



CAPTURING MALICIOUS DRONES IN A CIVILIAN ENVIRONMENT



NO DRONE ZONE

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Nomenclature

E – Youngs Modulus

K_{Ic} – Fracture toughness

σ_Y – Yield strength

ρ – Density

m – Mass

A – Area

$y_{\max}$ – Maximum deflection

Abbreviations

UAV – Unmanned Aerial Vehicle

UAS – Unmanned Aerial System

RF – Radiofrequency

FEA – Finite Element Analysis

CAD – Computer Aided Design

PID – Proportional Integral Derivative

FF – Feed-Forward Gain

MPI – Material Performance Index

AI – Artificial Intelligence

ML – Machine Learning

CNC – Computer Numerical Control

URDF – Unified Robot Description Format

Xacro – XML Macros for URDF

ROS 2 – Robot Operating System 2

IMU – Inertial Measurement Unit

FPV – First Person View

OSD – On-Screen Display

ESC – Electronic Speed Controllers

VTX – Video Transmitter

FC – Flight Code

PWM – Pulse Width Modulation

GNSS – Global Navigation Satellite Systems

UTM - Unmanned Aircraft System Traffic Management

ITS - Intelligent Transportation Systems

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Keywords

Drones, Capture mechanism, Computer Aided Design (CAD), Net, Finite Element Analysis (FEA), Simulink, ROS2, Gazebo, Physical Experiments

Use of AI

We acknowledge the use of ChatGPT 4.0 (<https://chatgpt.com/>) in proofing our report.

Continuations of Study

This project is a continuation of a MECH0020 project: "A Safe Approach to Defeating Malicious Drones in a Civilian Environment" by Benjamin Ashurov - 20093355.

Abstract

This project addresses a pressing need for a safe and effective drone interception system suitable for civilian environments. Existing counter-UAV technologies often rely on destructive or invasive methods, making them unsuitable for urban deployment. In response, a novel capture mechanism was developed, focusing on secure and reusable mid-air interception. The design draws inspiration from umbrella and trampoline structures to enable controlled net deployment and energy absorption. Structural elements were refined through CAD modelling, multi-objective optimisation and FEA, while system performance was evaluated using Simulink and Gazebo simulations. The final prototype was integrated into a commercial FPV drone, validated through physical testing and field trials and used to conduct a real life drone interception. This system demonstrates a feasible alternative to destructive counter-drone methods, supporting public safety and regulatory compliance in increasingly drone-dense civilian airspaces.

1. Introduction

1.1 Problem Background

The incorporation of drones and UAVs has dramatically transformed various sectors, including logistics, agriculture, and emergency response. In logistics, these UAVs have improved delivery efficiency, with research indicating potential delivery time reductions of up to 70% (Bouhria, Benhra, & Wissal, 2025). In agriculture, using drones for precision spraying has contributed to a 30% reduction in pesticide use and a 15% increase in crop yields (Borah, Padhiary, Sethi, Kumar, & Saikia, 2025). Additionally, UAVs have enhanced search-and-rescue operations, demonstrating a 50% improvement in response times compared to traditional ground methods (Lyu, Zhao, Huang, & Huang, 2023). These benefits arise from the inherent advantages of drone technology, including ease of use, speed, cost-effectiveness, and the ability to reach difficult or dangerous locations for ground vehicles (Shah, et al., 2024). The deployment of UAVs has decreased reliance on conventional transport, lowering emissions and conserving resources. By 2023, over 20,000 commercial drones were registered with the UK Civil Aviation Authority, with many additional units being used recreationally (Jackman, UK Commercial Drone Industry Professional, responsible and considerate drone use, 2023). Predictions suggest this number could soar, with estimates indicating that over 76,000 commercial drones could operate within UK airspace by 2030 (Baker, 2022).

However, the same characteristics that make drones beneficial, affordability, accessibility, and aerial capabilities, also leave them vulnerable to misuse, particularly in public environments (Clarke & Bennett Moses, 2014). The ease of drone access, often requiring minimal operational training, presents an increasing risk. In 2023, UK law enforcement documented over 6,200 drone-related incidents, marking a 63% increase since 2021, with roughly 11% connected to criminal activities such as contraband smuggling, unauthorised surveillance, and violating restricted airspace (Protect UK, 2024). High-profile incidents have highlighted the disruptive potential of malicious drone operations. In 2023, Frankfurt Airport faced multiple closures due to similar unauthorised drone activity (Bodell, 2023). Between 2021 and 2023, over 80 public event disruptions across Europe involving drones affected concerts, protests, and sporting events. Sometimes, drones spread propaganda or hazardous materials among large crowds (Cubukcu, 2024).

The transformation of consumer drones into improvised weapons has become evident in conflict zones as well. Iran-allied groups carried out more than 800 drone attacks from 2018 to 2024, resulting in over 1,300 deaths (Akbarzadeh, 2024). In Ukraine, commercial drones priced under £300 have been altered to carry explosive or incendiary payloads, effectively converting recreational devices into tools of war (Kunertova, 2023). Urban centres are particularly at risk. A notable incident in 2022 involved a drone near a residential tower in London, leading to a data privacy inquiry (Jackman & Hooper, Drone incidents and misuse: Legal considerations, 2023). With drones and UAVs becoming increasingly integrated into urban infrastructures, such as ITS and UTM networks, they pose systemic vulnerabilities (Tang., 2021). These vulnerabilities include interference with emergency services, unauthorised data collection, increased collision risks, and broader threats to public safety and critical infrastructure (Semenov, Krupska-Klimczak, Mazurek, Zhang, & Chernikh, 2025).

Current counter-drone technologies are primarily geared towards military use and are ill-equipped for civilian applications. Methods such as kinetic interceptors, high-energy lasers, radio frequency jamming, and anti-aircraft missiles can neutralise UAVs but also pose risks of collateral damage, communication disruptions, and legal consequences (Bruno Oliveira, Holland Michel, & Silkset, 2020). For instance, a 2021 trial of RF jamming in a European airport accidentally disrupted GPS services for nearby aircraft, highlighting the unintended outcomes of applying military-grade solutions in civilian airspaces (Eurocontrol, 2021). These escalating challenges underscore the pressing need for a safe, scalable, and legally compliant framework to address the threats posed by malicious drones. Such systems must effectively mitigate drone-related risks without introducing additional hazards to the public, endangering urban infrastructure, or undermining the substantial economic and operational benefits that drones and UAVs can deliver.

1.2 Problem Statement

The rapid proliferation of small drones and UAVs has significantly outstripped the advancement of safe and effective countermeasures for civilian airspace. Key locations, including airports, prisons, and urban areas, remain at risk of drone-related issues such as unauthorised surveillance, smuggling, and interruptions to essential services. Although military-grade counter-UAV technologies like jammers, lasers, and kinetic interceptors prove effective in combat situations, they present considerable dangers in civilian environments, such as uncontrolled debris, signal interference, and potential legal ramifications, rendering them unsuitable near densely populated areas or critical infrastructure (Wang, Liu, & Song, 2020). There is an urgent requirement for a non-invasive, real-time drone interception solution that can be safely deployed in high-traffic settings. This system should emphasise public safety, protect nearby infrastructure, adhere to civil aviation laws, and integrate smoothly with current security measures. This initiative

introduces a compact drone capture system designed specifically for urban settings. Rather than employing destructive methods, this solution captures drones or UAVs without inflicting any damage, allowing for safe recovery, and preserving forensic evidence for legal purposes. Comparative studies indicate that implementing this system could result in up to 70% of operational savings compared to traditional responses, such as lockdowns or multi-agency operations (Wendt, Voltes-Dorta, & Suau-Sanchez, 2020). By providing a scalable, legally compliant option, this project tackles a crucial engineering challenge: ensuring effective drone defence in civilian environments while maintaining safety, functionality, and conformity with regulatory requirements.

1.3 Project aims

The primary aim of this project is to design an innovative drone capture mechanism capable of safely capturing unauthorised drones in civilian environments. A key supporting aim is to validate the proposed solution through a combination of physical prototyping, analytical tools, and advanced simulation. This includes both structural analyses using FEA and control-oriented simulation using Simulink and ROS 2-based Gazebo environments. The simulation component specifically focuses on evaluating the mechanical feasibility, deployment dynamics, and control integration of the capture mechanism under realistic, physics-based conditions.

The project objectives are the following:

- Design and implement a robust capture mechanism deployment system, capable of mid-air activation and reliable mechanical operation
- Simulate predictions, in Simulink, of how integration of the capture mechanism on the drone affects flight performance
- Simulate mechanism behaviour and deployment sequences in Gazebo, using custom-built ROS 2 plugins and control nodes to validate actuation strategies and identify potential limitations before physical testing
- Ensure minimal collateral damage and risk to bystanders or property, through secure drone immobilisation and controlled descent post-capture
- Achieve high capture effectiveness through precise mechanism actuation and modular integration with an interception platform
- Ensure the mechanism's reusability and durability, with components operating effectively within the elastic regime and recoverable after deployment
- Rigorously adhere to regulatory and safety standards, ensuring ethical and legal compliance throughout system design and validation

1.4 Stakeholder Analysis

Key stakeholders for the proposed drone capture mechanism include critical infrastructure operators, emergency services, and government agencies. For instance, power grid substations require protection from drone sabotage, while emergency response teams need safe neutralisation methods during aerial operations. Early stakeholder engagement ensures the design meets real-world needs and simplifies regulatory alignment.

Airport operators are particularly vulnerable. In December 2018, two unauthorised drones shut down Gatwick Airport for three days, affecting over 10,000 passengers (Time, 2018). Similarly, Frankfurt Airport saw traffic restrictions for 10 days in 2023, with 67 drone-related air traffic interferences reported by Hesse police from May 2022 to February 2024, 11 breaching the control zone (Wolfsteller, 2024).

Government facilities also face threats from drone-borne munitions and contraband smuggling. A 2025 drone attack in Kursk caused a blackout impacting 1,500 households (Reuters, 2025). UK prisons like Manchester have reported coordinated drone drops, with authorities acknowledging a loss of airspace control to criminal gangs (Reuters, 2025). Other stakeholders include the agricultural and commercial sectors. Drones are increasingly used in large-scale farming for NDVI mapping and post-disaster assessments. Meanwhile, companies like Amazon are trialling drone deliveries (Nassi, Shabtai, Masuoka, & Elovici, 2019). Ensuring airspace safety is essential to support these beneficial uses while mitigating malicious threats.

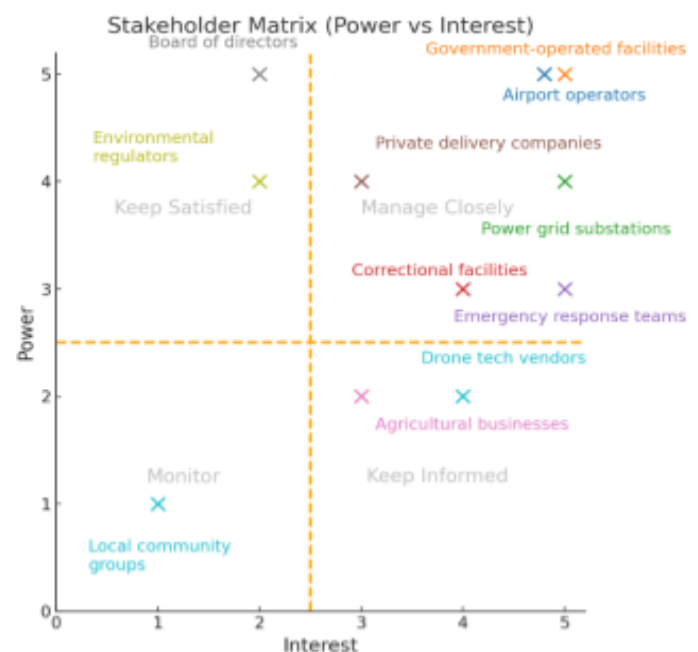


Figure SEQ Figure 1* ARABIC 1 – Stakeholder analysis matrix.

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2. UAV Technology Review

2.1 Threat Landscape & Scope

The swift incorporation of drones and UAVs into civilian settings has led to a range of unexpected vulnerabilities that traditional urban safety systems have failed to address. By 2021, there were over 870,000 drones registered in the United States (Nassi, Shabtai,

Masuoka, & Elovici, 2019), and the global drone market is projected to expand at more than 20% annually (Gharibi, Boutaba, & Waslander, 2016). This commercial and recreational drone usage surge is notable among hobbyists, private operators, and non-state actors. In the UK, drone incidents skyrocketed by 63% from 2021 to 2023, with over 6,200 reports in one year, covering issues from airspace violations to unauthorised surveillance and contraband delivery (EASA, 2024). This rapid growth has outstripped regulatory frameworks, revealing significant gaps in civilian airspace governance and response systems. The variety of drone models contributes to this continually evolving threat landscape. Consumer quadcopters, particularly those designed like DJI multirotors, are often involved in airspace breaches due to their stable flight, autonomy, and camera features. While these drones are commonly used for photography and surveying, they are increasingly being misused for unauthorised data gathering and entering restricted areas (Wei, Zhou, Leng, Maharjan, & Zhang, 2021). On the other hand, fixed-wing drones, such as the Parrot Disco, offer longer flight times and higher speeds, making them attractive for covert surveillance and smuggling activities (Al-Dosari, Hunaiti, & Balachandran, 2023). Heavy-lift drones, like those in the DJI Matrice series, can carry payloads over 5 kg, making them suitable for logistics and transporting dangerous or illegal items (Custers, 2016). Additionally, sub-250 g microdrones present a unique challenge for regulation due to their legal exemptions and ease of concealment, complicating enforcement efforts.

While different types of drones pose specific risks, FPV drones exemplify a distinct combination of speed, agility, and unpredictability. Unlike GPS-dependent models, these drones are manually operated at low altitudes and utilise live video feeds, often eluding traditional radar or radio frequency detection (Mekdad, et al., 2021). Frequently custom-built, FPVs enable operators to bypass safety limitations imposed by manufacturers and avoid no-fly zones. Their increasing presence in both military and civilian contexts, especially highlighted by their role in the Ukraine conflict, where they reportedly caused 70–80% of drone-related battlefield casualties (Austria, 2024), demonstrates their disruptive capabilities. In urban areas, these drones have been linked to incidents of smuggling, harassment of crowds, and surveillance of sensitive sites. Their agility poses significant challenges for existing countermeasures, which often struggle to respond in real-time or safely recover hostile drones. The emergence of autonomous drone swarms further complicates the situation, as these coordinated systems can easily overwhelm traditional defences (Gupta, Jain, & Vaszkun, 2016). Existing interception methods, whether leveraging RF jammers or kinetic and laser-based solutions, tend to be reactive, narrowly focused, or carry significant risks in urban environments. Many current approaches prioritise detection over safe recovery, limiting their effectiveness in populated areas. Thus, a comprehensive strategy for drone mitigation in civilian settings must be adaptable, engaging with the full range of UAV types while minimising collateral damage and aiding in evidence gathering. The growing complexity and variety of drone-related threats necessitate a flexible and precise response strategy rooted in practical operational realities.

2.2 Detection Technologies

Counter-drone detection rests on four main sensor families, each powerful yet flawed when used in isolation. RF sensing reliably tracks most hobby-class UAVs by recognising their control-link or telemetry signatures, but it fails against fully autonomous craft that emit no RF traffic [11]. Acoustic arrays identify the distinctive rotor noise of small multirotors and work well in quiet areas; however, urban soundscapes swamp those signatures, making the method unreliable where protection is most needed [12]. Electro-optical/infra-red cameras can classify and follow drones with high precision in good light and clear weather, yet their performance degrades sharply in fog, rain, low light or when heavy on-board processing is unavailable [13]. Radar operates day-night and all-weather, but small UAVs resemble birds or wind-borne debris on raw radar returns, generating high false-alarm rates [14].

Because no single modality delivers the low-false-alarm, all-weather coverage required in dense civilian airspace, recent work concentrates on multi-sensor fusion, combining radar's reach, RF's specificity, optics' classification power and acoustics' near-field cueing to raise detection probability while suppressing false tracks. Early fusion frameworks, such as the architecture demonstrated by Kim et al. [15], show marked improvements in urban test beds, suggesting that robust counter-drone defence will depend on tightly integrated, complementary sensors whose outputs feed real-time defeat systems able to act despite imperfect or partial cues.

2.3 Countermeasures Technologies

2.3.1 Electronic: Jamming & Spoofing

Jamming and spoofing methods have been extensively studied as non-kinetic techniques to disrupt rogue drone operations. According to Mathur (Mathur, 2022), jamming works by injecting disruptive noise into the radiofrequency (RF) channels used by drones for communication and navigation. This interference can sever the link between the drone and its controller or cause the drone's guidance systems to malfunction. Spoofing, on the other hand, involves transmitting counterfeit signals, most notably falsified GPS data, which mislead the drone about its true position, prompting it to deviate from intended flight path or even return to a predetermined safe zone. These techniques are particularly effective against drones that rely on conventional analogue RF communications. However, their operational success is constrained by regulations: the power output of jammers is limited to avoid collateral disruption, and the growing adoption of frequency-hopping protocols or fibre-optic links can render RF jamming and spoofing ineffective. Field trials with the Dedrone RF-300 jammer (Figure 2) disabled approximately 93 % of DJI Phantom 4 drones at a 1 km line-of-sight range using a 10 W burst (Dedrone by Axon, 2023).



Figure 2 – Dedrone RF-300 jammer

2.3.2 Directed Energy

Directed-energy systems, including lasers and microwaves, promise rapid-response neutralisation. Boeing's Compact Laser Weapon System (CLWS) has demonstrated success in operational tests, neutralising simultaneous drone threats, including swarms, across

considerable distances (Hein, 2025). Implementation challenges include precise targeting, significant energy demand, high operational costs and potential collateral damage from reflections or mis-targeting (Gonzalez-Jorge, et al., 2024). Furthermore, lasers have shown a diminished effectiveness in high moisture environments, limiting their inherent reliability (Chaari & Al-Maadeed, 2020). Nevertheless, CLWS neutralised a DJI Matrice at 1.2 km with a 4.8 s, 2 kW burn during 2024 live-fire trials (Boeing, 2024), showcasing its effectiveness in controlled conditions.

2.3.3 Acoustic Disruption

Prandtl Dynamics has demonstrated a non-lethal acoustic solution that focuses ultrasonic beams to upset drone gyroscopes and flight-control electronics. Trials in Canada's IDEaS sandbox programme proved practicality in dense civilian areas where collateral-damage tolerance is near zero (Hein, 2025) (The Economist, 2025). Current limitations are a 100 m effective range and susceptibility to wind and ambient noise. Correctional facilities show strong demand to halt contraband drops by small quadcopters. IDEaS trials reported an 84 % disable rate for quadcopters within the system's 100 m acoustic cone (Hein, 2025).

2.3.4 Net-Capture Solutions and Kinetic-Defeat Technologies

Net gun systems offer a non-destructive, kinetic approach to neutralising rogue drones by launching a deployable net that entangles the drone's propellers, leading to its descent. Their main benefits include ease of use and safety for civilian contexts, significantly lowering the risk of misuse and minimising the chance of detonating any onboard explosives thanks to the capture method. Demonstrations of equipment like the SkyWall 100 illustrate their effectiveness, showcasing a remarkable 92% capture rate on the first attempt from a distance of eight meters against quadcopters weighing up to six kilograms during controlled tests. However, a significant drawback is their limited effective range, typically between 50 and 100 meters, which requires operators to be closer to potential threats. This reduces reaction time and increases risk, especially in urban environments where obstacles often impede visibility. Trials have explored the possibility of mounting net guns on drones to improve mobility; however, this strategy has had limited success against fast and agile targets, such as FPV drones. Intercepting these drones is challenging due to the precision needed to aim and deploy the net mid-flight, resulting in net guns being most effective against slower, stationary drones like the DJI Phantom. Environmental factors like wind and obstacles can compromise reliability, leading to misfires or incomplete captures. Additionally, broader kinetic interception techniques, including net-based systems, do not depend on radio frequency vulnerabilities and can restrain drones while causing minimal collateral damage. Nonetheless, these systems require extremely high precision, relying on real-time tracking and instantaneous actuator response. Their effectiveness notably decreases when faced with multiple or coordinated threats, such as drone swarms.



Figure 3 – Image of a ground-based net gun, a drone with a net gun

2.3.5 Direct-Impact Interceptors

Modern counter-drone defence has advanced drastically with direct-impact technologies that collide with the target at high velocity, destroying or disabling it. This approach is much simpler than trying to manoeuvre a drone next to the target to fire a net gun at it, making it much more suitable for countering high speed FPVs. Anduril's Roadrunner (Figure 4) and Anvil interception systems are notable developments of this drone type. Roadrunner, an autonomous high-speed interceptor, engages rogue UAVs through rapid acceleration and controlled mid-air impact (Smith, The rise of rogue drones in civilian airspace: Implications for public safety, 2023). Anvil operates on a similar principle but is optimised for smaller-scale threats, making it suitable for engagements in urban environments where precision is paramount (Jones & Patel, 2024). Trials demonstrate the ability to intercept multiple threats in real time. Anduril's Anvil recorded an 89 % single-pass intercept success against 5 kg multirotors closing at 70 m s⁻¹ during U.S. SOCOM tests (Smith, Lee, & Williams, Autonomous kinetic drone interceptors: Performance analysis and operational effectiveness., 2023).



2.4 Regulatory, Ethical and Sustainability Considerations

The deployment of counter drone systems can raise several ethical and regulatory considerations. One example of this is with the public's right to privacy and data protection. Any utilisation of counter-drone sensors (such as RF scanners, acoustic arrays, or even optical cameras) leads to the collection of extensive metadata, which can inadvertently record personal information. Furthermore, an incorrect capture of an innocuous drone may lead to unauthorised capture of private data and imagery, leading to a potential violation of GDPR stipulations on lawful processing and data minimisation. The Information Commissioner's Office (ICO) emphasises the performance of Data Protection Impact Assessments (DPIAs) to implement anonymisation techniques and minimise the risk of GDPR violations (Information Commissioner's Office, 2018).

Due to the current environment around unmanned aerial vehicles within the UK, a kinetic interceptor must be maintained within visual line-of-sight and fly below 400 feet. Additionally, cooperation through Notices to Air Missions (NOTAMs) alongside the fitting of electronic conspicuity devices, which continuously broadcast the UAV's identity and altitude, preventing conflicts with manned

aircraft, is necessary to safely and legally operate such a system (Civil Aviation Authority, 2018). Finally, the current and future increased proliferation of legitimate commercial and emergency drones will require the advent of an authentication framework. This is so that the vast number of compliant, legal drones are not targeted by the interceptor incorrectly, potentially causing GDPR and legality issues and wasting the drone's valuable use on non-malicious targets. Sustainability is another concern with current technologies. Many rely on single-use components or destructive capture methods, leading to inflated operational costs and greater environmental impacts with each deployment (Ryou, Beyer, & Karaman, 2025). Furthermore, most systems are optimised for open environments and struggle in dense, GPS-denied, or signal-cluttered areas, where real-world drone threats frequently arise (Vinogradov & Pollin, 2021) (Kim, Kim, Shin, Wang, & Matson, 2023). Few solutions emphasise intact retrieval and controlled descent, which are increasingly vital for legal accountability, evidence retention, and public safety. Studies (Saranovic, Pavlovski, Power, Stojkovic, & Obradovic, 2021) point out that interception systems in civilian settings must be able to recover drones without introducing new hazards. Unfortunately, most available designs destroy the drone or leave its descent uncontrolled, making them unsuitable for sensitive areas. These challenges underscore the requirement for an interception system that allows for safe mid-air capture, controlled descent, low-cost reuse, and adaptability to complex civilian environments, features not currently integrated in any existing solution. This project directly addresses these gaps by introducing a design focused on controlled capture, modularity, cost-efficiency, and suitability for urban conditions, providing a viable alternative to destructive and infrastructure-heavy systems.

(Civil Aviation Authority, 2018) (Ryou, Beyer, & Karaman, 2025) (Vinogradov & Pollin, 2021) (Kim, Kim, Shin, Wang, & Matson, 2023) (Saranovic, Pavlovski, Power, Stojkovic, & Obradovic, 2021)

2.5 Limitations of Current Interception Methods for Civilian Environment Use

Many of these systems were developed for military applications, where the primary focus is neutralising threats without concern for recovery or collateral damage. In civilian environments however, there is a critical need for solutions that are safe, reusable, cost-effective, and capable of capturing drones without endangering people, property, or infrastructure (Vinogradov & Pollin, 2021). Existing electronic, laser and microwave effectors, even when successfully used, will cause uncontrolled crashes, complicating recovery efforts and posing unacceptable risk of collateral damage (Zhang, et al., 2024). In addition, a malicious actor could rather easily adapt a drone to attack a target which is protected by these technologies if they know that they will be used. If the drone is connected to the control station via a tethered fibre optic cable, jamming will have no effect (Hambling, n.d.). Lasers can largely be overcome by surrounding the drone in reflective material, while microwave effectors can be countered by surrounding sensitive electronics, such as the flight controller, in a Faraday cage. Similarly, acoustic disruption systems can also be mitigated by wrapping the affected electronics in sound-absorbing materials. Therefore, there is a fundamental reliability concern for these systems, which is not acceptable in a civilian environment where a near 100% guarantee of success is required to match existing health and safety standards (Safety and Reliability Society, n.d.).

Net-based kinetic systems offer a non-lethal approach to physically capturing drones but have substantial operational constraints. Ground-based net guns typically have a maximum range of under 100 meters and require manual reloading, which limits their effectiveness for quick or repeated deployments (Brust, et al., 2018), once again posing a reliability constraint. While drone-mounted net systems enhance mobility, they are only able to reliably engage relatively slow, non-maneuvring targets. Moreover, these systems cannot guide a captured drone to a safe landing zone, so the malicious drone can still land (and deliver its payload) to a protected area. Therefore, this approach can't reliably eliminate the risk of accidental damage or injury (Pliska, Vrba, Tomáš, & Saska, 2024).

Advanced autonomous interceptors, such as Anduril's Anvil, are in many ways the best existing solution to countering drones in a civilian environment. They utilise automated high-speed tracking, which is paired with external data provided to the interceptor drone by an advanced sensor suite, ensuring high interception accuracy. Therefore, this approach can be seen as highly reliable, which makes it suited for civilian defence uses. Nevertheless, these systems are destructive, prohibitively expensive, and can feature classified technology, making their widespread civilian use unlikely (Weisgerber, 2023). Furthermore, it is designed such that a successful interception results in one or both drones falling out of the sky. This poses a near certainty of collateral damage or even bystander casualties if used in a dense urban environment.

2.6 Identified the Gaps in Capabilities

An ideal solution for defeating drones in a civilian environment must have high levels of reliability and a long range, akin to that of kinetic interceptors. At the same time, to fulfil cost and sustainability goals which are inherent to any commercially viable product, this solution should be highly reusable, just like non-kinetic solutions, such as jammers and lasers. Crucially, unlike any existing technology, the targeted drone should be defeated in a controllable manner. Upon successful interception, the drone must not simply fall from the sky, risking collateral damage. Instead, it should be disabled in mid-flight and guided to land safely in a designated area.

3. Initial Design

3.1 Concept Formulation

Based on the existing technology overview, it was concluded that kinetic interception – physically impacting a drone with a projectile, remains the most reliable method for defeating drones. This insight led to an idea that a system that can be designed such that upon an interceptor's collision with its target, the malicious drone can be captured instead of knocking it out of the sky. Specifically, this capture mechanism can deploy a net that ensnares the targeted drone upon contact with the kinetic interceptor. This design guarantees that the captured drone is securely connected to the interceptor, facilitating controlled descent, with no inherent risk of collateral damage.

To develop this design, the envisioned system was divided into two primary components:

1. The Effector System: The capture mechanism featuring the net that would ensnare the target and absorb the impact.
2. The Carrier Platform: The quadcopter, VTOL or missile platform responsible for delivering the effector to the target.

Given the variety of existing carrier systems, the aim of this project is to design the capture mechanism that would be highly compatible with a variety of carrier platforms.

3.2 Functions, Objectives and Constraints

Based on the above problem background and statement, the following functions, objectives and constraints (FOC) were identified. Certain quantifiable targets were derived later, as indicated by the references of the report. The manner in which the various elements of the FOC have been fulfilled is detailed in the validation (Section 5) and testing (Section 7) sections of the report.

Table 1 – Functions, objectives, and constraints of this project.

Item	Target	Rationale
Functions		
Net Deployment	Cross-sectional area of net $\geq 0.5 \text{ m}^2$	Covers the surface area of the carrier platform with high certainty (Arora, 2022)
Collapsibility	The mechanism's width may not exceed that of the drone's base when in a collapsed configuration	The mechanism must be able to become compact to ensure it can be integrated into a variety of carrier platforms, such as rockets
Drone capture	The mechanism must be able to catch and secure the drone upon impact with the interceptor	The net wraps around the rotors, and spring-loaded claws then clamp the drone in place
Impact Energy Absorption	No plastic deformation occurs when the peak force due to impact (500 N) is applied to the system	Based on the force absorption model, as seen in Section 4.1
Objectives		
Safety	Safety factor ≥ 1.5	Meets aerospace safety factor standards (Zobeiry, 2021)
Mass Minimisation	Total weight $\leq 8 \text{ kg}$	High masses would decrease the manoeuvrability and speed of the carrier platform
Deflection Control	Arm tip deflection $\leq 10 \text{ cm}$	Preserves clearance from the carrier platform
Capture Test	Valid interception carried out in realistic conditions	Validated in field tests under moderate wind
Sustainable construction	Mechanism made of sustainable material	Prototype must be able to be disassembled and recycled/reused after project timeline
Constraints		
Standards Compliance	CE Machinery Directive; UK CAA UAV regs	Ensures lawful use for testing purposes
Environmental settings	Wind $\leq 6 \text{ m/s}$; $-10 \text{ }^\circ\text{C}$ – $40 \text{ }^\circ\text{C}$; Relative humidity $\leq 80 \%$	Guarantees reliable operation in civilian settings
Manufacturability & Cost	Prototype cost $\leq \text{£}3 \text{ 000}$; lead time $\leq 4 \text{ weeks}$	Per the assigned budget and based on the initial timeline assessment.
Malicious drone	The one used for testing will have a mass equal to or below 1 kg	Representative of the small drone class (Appendix A.1) that the mechanism is designed to defeat

3.3 Design Inspiration

The design of the capture mechanism must satisfy two essential aims: firstly, it needed to endure high-speed collisions between the carrier platform and the targeted drone without any permanent deformation, ensuring structural integrity and allowing for the system

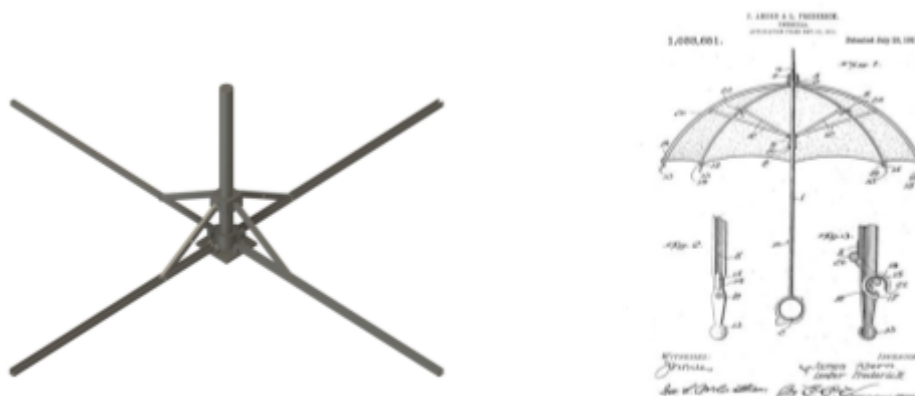


Figure 5 – Umbrella based initial design (left), umbrella structure engineering drawing (right) CITATION Ure \1 2057 (Uretimpark, n.d.)

to be reused. Secondly, it must be collapsable which is crucial for integration into a variety of exiting carrier platforms, such as rockets.

The initial round of brainstorming led to examining the design of an umbrellas, which shares many designs aims with the envisioned capture mechanism – it needs to be compact, to allow for storage and are designed to support large, distributed loads. To achieve this, umbrellas utilise braces that are connected to a slider. This feature allows them to support loads effectively, since the braces fix the umbrella arms in place, and allow for it to open and close as the slider moves up and down the central mass. Despite showing great promise, further research revealed that attaching a net to the arms of an umbrella inspired structure, (Figure 5) would result in excessive deformation of the supporting arms under impact of a drone (Wang, Contreras-Jiménez, Jorquera-Lucerga, & Garlock).

Upon further brainstorming, it was determined that certain design decisions can be adapted from a system that was specifically designed to repeatedly absorb high-energy impacts – a trampoline. This mechanism type ensures structural integrity by incorporating springs that increase stopping distance of the impacted object, thereby minimising the force applied to its frame. Building on this principle, the design concept utilised custom “end caps” at the ends of the arms (as seen in Figure 6) which in turn connect to springs that support a net suspended above the arms of the mechanism. As showcased in Section 4.6 this modification allows for an arm with a relatively small mass to remain undeformed even when high stresses would be applied to the net during an impact with a drone.

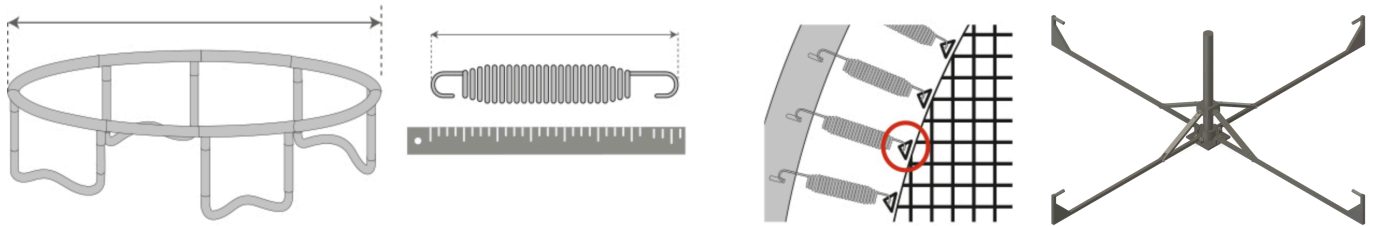


Figure 6 – Diagram of springs in a trampoline (left), springs integrated into the umbrella structure (right).

3.4 Refinements and Prototyping

To validate this concept, several basic prototypes were constructed to test the various aspects of the design. Firstly, a small-scale, 3D-printed version of the umbrella design was constructed. This was used to validate the idea that the length of the brace and its position on the arm (as defined in Section 4.4) can vary without compromising the ability of the mechanism to open and close. Next, a full-scale prototype constructed was constructed out of readily available, sustainable materials – wood, cardboard and recycled PLA (Replique, n.d.), which incorporated the above-mentioned end caps to support a net. Crucially, this prototype revealed a significant challenge of integrating end caps into an umbrella design – the end caps collide with each other as well as the central column upon deployment.



Figure 7 – Initial 3D-printed prototypes of capture mechanism.



Figure 8 – Single end cap (left), double end cap (middle), zoomed-in image of the endcaps in the closed position (right)

To resolve this fundamental issue, two key design solutions were implemented. FOpposite end caps were redesigned to interlock. The single end cap consists of a right-angled triangle formed by a vertical beam and a support beam set at a 30° angle, connected to two of the arms via bolts. On the opposing arms, double end caps were used. These replicate the same triangular structure but are offset laterally by 6 mm, creating a central slot, as seen in Figure 8. This slot allows the single end cap to nest within the double end cap,

preventing a collision when closing the mechanism. The second solution involved shortening one pair of opposing arms by 100 mm to create a staggered structure. This ensured that the longer arms would fold over the shorter arms without interference, as shown in Figure 8, thereby eliminating the clashing issue observed in the previous prototype. The final design, as shown in Figure 9 effectively allowed the capture mechanism to open and close as intended.



Figure 9 Final design concept in the opened and deployed state

3.5 Testing platforms design



Figure 10 – The SG 15L by SG UAV

To test the final design, a carrier platform was required to transport the capture mechanism to its target and execute an interception. Several options were evaluated, including a heavy lift hexacopter [61], known for their high payload capacity; a large model rocket, which could facilitate testing of very high-speed interceptions, and a large FPV drone, offering a balance between speed and payload capacity. Ultimately, the large FPV drone was selected due to its ease of operation, adaptability for modifications, and significantly lower cost compared to the other alternatives.



Figure 11 – Image of drone used as target

While constructing a new drone from scratch was considered, it was instead decided to buy a commercial 15-inch quadcopter with known payload capacity and pre-tuned PID values for large payloads. Therefore, the SG 15L by SG UAV, shown in Figure 10, was purchased (SG UAV, 2024). During testing, it was found that the maximum velocity of this drone is 30 ms^{-1} .

Furthermore, it was also necessary to construct a target drone to be captured by the designed capture mechanism, as well as to practice drone piloting skills. Therefore, a standard 5-inch FPV drone (Figure 11) was constructed and used for the final testing stage.

4. Detailed Design

4.1 Force Calculations

The impact between the capture mechanism and the targeted drone is a complex physical phenomenon. Therefore, a simplified model was created to determine the maximum forces applied to the system. Specifically, this model reflects what was determined to be the worst-case scenario where the targeted drone hits the net along the length of one of the arms. This way, the impact force is entirely absorbed by a single arm, rather than being distributed amongst multiple arms.

Therefore, this case can be reduced to a 2D problem, as seen in Figure 12.

In this model, the net is assumed to behave as an inextensible cord which is attached to the spring at one end and to a fixed support - the central column, at the other. Therefore, when an impact occurs at any point along the “cord”, the deflection in its shape arises from the extension of the spring.

The maximum extension is derived to be (see Appendix A.2):

$$x = L - \frac{y_{\max}}{\sin(\theta)} - \sqrt{\frac{y_{\max}^2}{\sin^2(\theta)} + L^2 - 2 \frac{y_{\max}}{\sin(\theta)} L \cos(\theta)}$$

Where θ is the maximum angle of the spring, $y_{\max}$ is the distance between the net and the arm and L is the length of the arms. These parameter values were estimated based on the initial CAD design (Figure 6). Therefore, the maximum extension of the spring for this setup was calculated to be 0.3 m.

To determine the appropriate spring constant, it was surmised that the impact energy acting on the mechanism during the collision depends on the mass and relative velocity of the interceptor and the target. Bearing in mind that the final interception testing was carried out on a near stationary drone, its velocity can be disregarded. As detailed in Section 7.3 the mass of the target drone was equal to 1kg (1sf) and maximum velocity of the interceptor is 30m/s. Therefore, based on the kinetic energy equation, the total energy applied to the system during the collision would be equal to 450 J.

Since the design is made to function as a trampoline, in the worst-case scenario, a single spring would have to absorb this energy. By equating the kinetic energy to the potential energy of the spring, based on the previously derived maximum possible extension of the spring in the system, the spring constant (k) was found to be 10 000 N/m.

Finally, using Hooke’s law, the force acting on the system was calculated as being 500N (Appendix A.2).

The fundamental limitation of this model is the approximate nature of the geometric constants used to calculate the maximum possible extension of the spring. Therefore, the derived force is a general case approximation, and the actual force acting on the system will vary based on the specific dimensions of the prototype. Another assumption is the fact that no strain would occur in the net, which

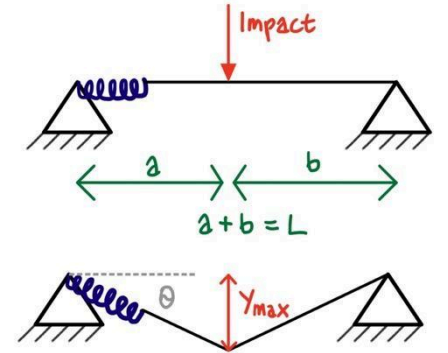


Figure 12 – 2D Diagram showing relevant forces

requires it to be perfectly rigid and taut during impact. In fact, in the final configuration the net was purposely set in a semi-taut configuration, to increase the probability of capture.

4.2 Material Selection Analysis

4.2.1 Introduction to Material Requirements

Optimal material selection is key to creating a successful capture mechanism design. Based on the FOC, the key requirements for the mechanism include resistance to elastic deformation, to ensure structural integrity during impact, as well as resistance to plastic deformation and fracture, to ensure reusability. Furthermore, the design must be lightweight so that the carrier platform can take off and effectively pursue the malicious drone. Finally, good machinability is crucial for ease of manufacturing, while sustainability considerations make recyclability a key selection criterion.

4.2.2 Material Selection Criteria

The key functions, objectives, and constraints that guided the material selection process are summarised below:

Table 2 – Material selection criteria.

Category	Description	Quantification / Justification
Yield Strength	Resistance to plastic deformation under load	High yield strength is required to avoid permanent deformation
Fracture Toughness	Resistance to crack propagation and sudden failure	High fracture toughness to ensure durability
Stiffness (Young's Modulus)	Minimising elastic deformation and bending during use	High Young's Modulus
Low Density	Reducing weight to enhance drone flight performance	Preferably < 3000 kg/m ³
Machinability	Ease of manufacturing complex geometries	Rated "Good" or better in machinability databases
Corrosion Resistance	Durability and reliability in outdoor environments	Must be corrosion-resistant or easily protected
Sustainability	Promoting recyclability and reduced environmental footprint	Fully recyclable; low embodied CO ₂
Economic Feasibility	Affordability and ease of procurement	Cost < £50 per kg
Operational Temperature	Reliable operation across typical environmental conditions	Functional range: -20°C to +60°C

4.2.3 Material Selection Process (ANSYS Granta EduPack 2020 Analysis)

The material selection process was guided by key selection stages used to find the optimal material for this use case. At first, materials were filtered in an initial screening stage. Next, constraints were rigorously defined and applied to eliminate unsuitable candidates:

- Corrosion Resistance: Materials susceptible to corrosion under typical operational conditions are excluded
- Price: Limited to economically feasible materials priced under £50/kg
- Operational Temperature Range: Exclude materials unable to sustain mechanical integrity between -20°C to 60°C

Next, ranking was carried out based on the following performance indices:

Key Material Performance Indices (MPIs) were methodically identified and applied to rank materials based on their capability to satisfy design objectives. All MPI derivations are in Appendix A.3.

Yield Strength vs Density: This MPI evaluates the strength-to-weight ratio, essential for ensuring structural integrity while minimising weight.

MPI:

$$\log \log (\sigma_y) = C + \log (\rho)$$

Therefore, the MPI line has a slope equal to 1.

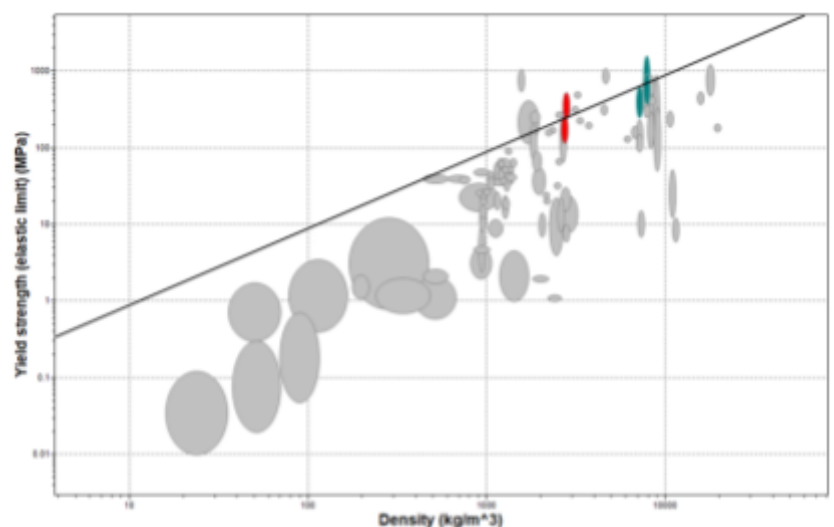
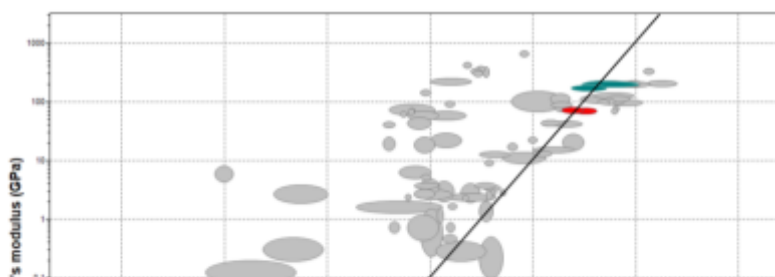


Figure 13 – Yield strength against density plot



Young's Modulus vs Fracture Toughness: This MPI assesses resistance to elastic deformation and fracture, crucial for durability under repeated load cycles.

MPI:

$$\log(E) = C - 2\log(K_{Ic})$$

Therefore, the MPI line has a slope equal to 2.

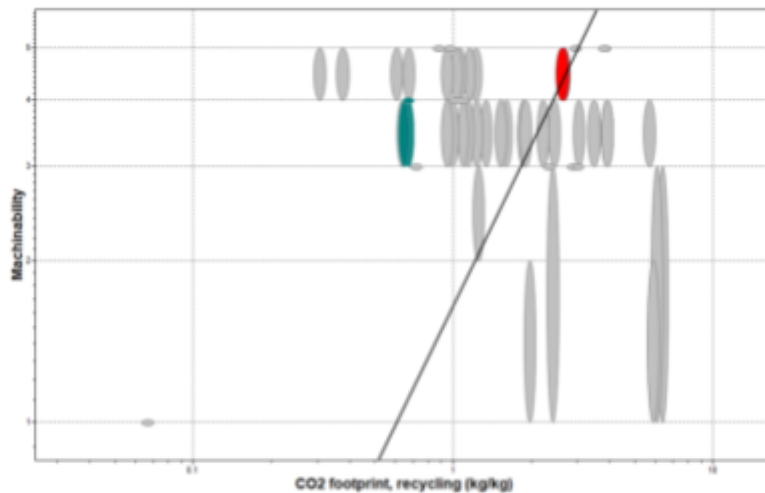


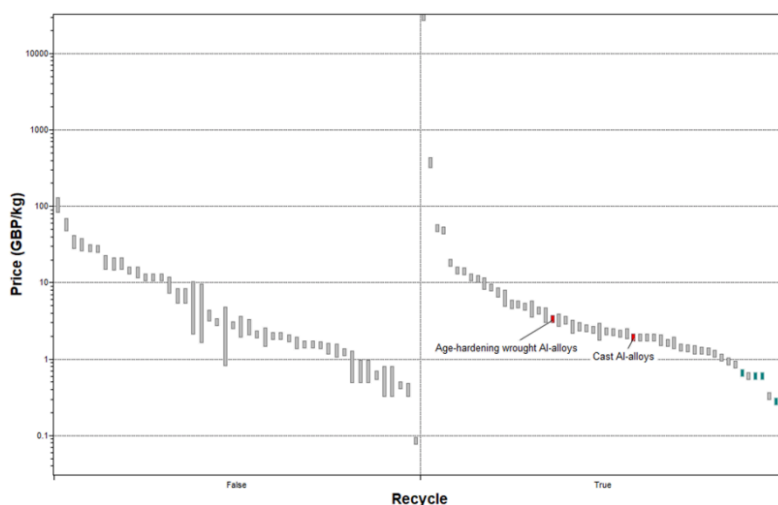
Figure 15 Machinability against CO2 footprint plot

Machinability vs CO2 Footprint: Considering sustainability and manufacturing practicality simultaneously.

MPI:

$$\log(\text{Machinability}) = C + \log(\text{CO}_2 \text{ footprint})$$

Therefore, the MPI line has a slope equal to 1.



Recyclability vs Price: Assessing environmental and economic practicality.

MPI is not needed as a material is either recyclable or not.

- Age-hardening wrought Al-alloys
- Cast Al-alloys
- Cast iron, ductile (nodular)
- High carbon steel
- Low alloy steel
- Medium carbon steel

Figure 16 (Left) Price against Recycle plot, (right) shows what material each colour is in the plot.

4.2.4 Evaluation of Finalist Materials

After applying the defined constraints and evaluating relevant MPIs, six material categories emerged as finalists. Whilst all the materials met baseline requirements, a comparative analysis led to the selection of age-hardening wrought aluminium alloys (e.g. Aluminium 6061 T6) over the others.

Table 3 – Summary of reasons for rejection for materials that met selection criteria.

Material	Reason for Rejection
Cast Aluminium Alloys	Good recyclability and corrosion resistance, but inferior mechanical properties (lower strength and toughness) compared to age-hardened wrought Al-alloys.
Cast Iron (Ductile/Nodular)	Strong in compression and durable, but too dense, making it unsuitable for lightweight drone applications.

High Carbon Steel	Excellent strength, but very high-density compromises flight efficiency; also, more difficult to machine.
Low Alloy Steel	Better strength-to-weight ratio than high carbon steel, but still significantly denser than aluminium and more prone to corrosion.
Medium Carbon Steel	Provides balanced strength and toughness but is still relatively heavy and less corrosion-resistant compared to aluminium.

4.2.5 Justification for Aluminium 6061 T6

The selection process evaluated and addressed competing objectives, including weight versus cost, strength versus machinability, and sustainability versus economic feasibility. Aluminium 6061 T6 emerged as the balanced optimal solution, expertly navigating these critical trade-offs. Table 4 on the following page summarises how Aluminium 6061 T6 satisfies each critical material requirement (ASM Aerospace, 2024).

Table 4 – Table summarising how the material chosen meets the criteria assigned.

Material Property	Performance	Justification
Yield Strength	276 MPa	Sufficient to resist plastic deformation under loads
Young’s Modulus	69 GPa	Ensures stiffness and minimal elastic deflection
Density	2700 kg/m³	Lightweight for drone payload efficiency
Machinability	Rated as "Excellent"	Suitable for manufacturing intricate parts
Corrosion Resistance	Naturally corrosion-resistant	Provides durability and reliability in outdoor environments
Recyclability	Fully recyclable with minimal property loss	Reduces environmental impact and supports circular economy principles
Cost	Typically < £5 per kg	Well below the £50/kg constraint
Manufacturing Compatibility	Easily machined, welded, and formed	Facilitates prototyping and mass production

4.2.6 Applications of Aluminium 6061 T6 in Similar Use

Aluminium 6061 T6 has a well-established use profile in similar demanding applications requiring optimal strength-to-weight ratios, fatigue resistance, and environmental durability. These include aerospace components, drone parts, bicycle frames, and automotive and marine components. This vindicates the fact that its comprehensive properties uniquely suits it to the dynamic, structural demands of drone-capture mechanisms (Kumar , 2023).

4.2.7 Sustainability Profile of Aluminium 6061 T6

Aluminium 6061 T6 exhibits significant sustainability benefits, primarily due to its high recyclability and efficient life-cycle performance. Unlike many other structural materials, aluminium can be recycled repeatedly without significant degradation of properties, preserving its mechanical integrity and ensuring a lower environmental footprint. The recycling process for aluminium consumes substantially less energy compared to primary production, reducing CO₂ emissions dramatically, up to 95% lower than virgin aluminium production (itrecycle, 2024). Additionally, the long service life, corrosion resistance, and ease of maintenance contribute to reduced environmental impacts during operation (Dean, 2024). Its widespread availability and ease of recycling further enhance overall sustainability, making Aluminium 6061 T6 an environmentally responsible choice for structural engineering applications (Edwards, 2024).

4.3 Cross-section analysis

4.3.1 Comparing different beam types

Various cross-section beam geometries result in varying stiffness-to-weight ratios of a given beam. Therefore, it is necessary to determine the ideal cross section type for the various components the capture mechanism. Therefore, the deflection formula was calculated for solid rectangular, I-section, and hollow-rectangular beam profiles to find the minimum area needed to achieve a given free end deflection. Full derivation can be found in Appendix A.4.2.

$$y_{max} = \left[\frac{1}{EI} \left[\frac{F_1(a-L)+M_k}{6} \frac{L^3}{6} + \frac{F_1L-M_k}{6} \frac{(L-a)^3}{6} - \frac{F_1(a-L)+M_k}{6} \frac{a^3L}{6} \right] \right]$$

For the different beam types, only *I* changes values, and the rest of the formula stays the same. The solid rectangle used a fixed depth-to-width ratio; the I-section used standard flange/web proportions; and the hollow tube used a prescribed outer aspect ratio and wall-thickness fraction. Figure 17 shows that, for any given deflection, the hollow tube always requires the smallest cross-sectional area, confirming its better mass efficiency.

4.3.2 Comparing various aspect ratios

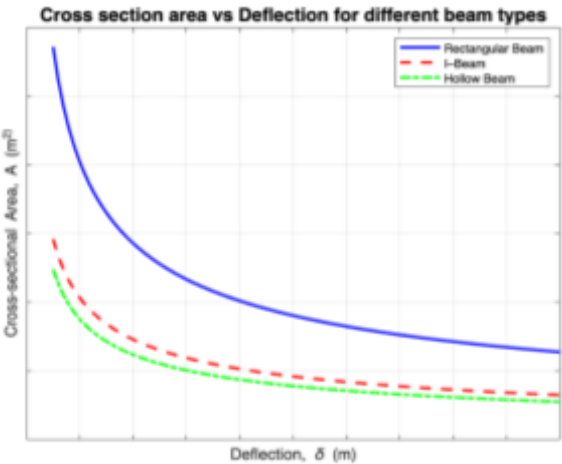


Figure 17 – Plot shows the trend between deflection and different beam types.

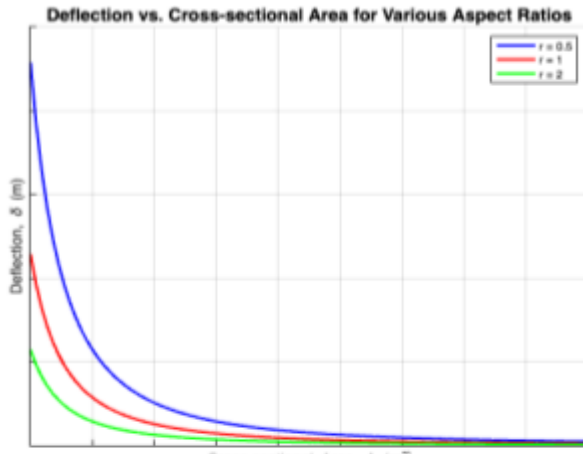


Figure 18 Plot shows the trend between deflection and different aspect ratios.

A further parametric study isolated the depth-to-width ratio for a non-hollow beam using the same deflection formula used in the previous section. Keeping beam length and loading constant, aspect ratios of 0.5, 1.0, and 2.0 were evaluated, and deflection was plotted against area. As shown in Figure 18, taller, narrower sections (higher aspect ratios) produce much lower deflections for a given beam area. Above an aspect ratio of two, the curves begin to converge, indicating diminishing returns for higher aspect ratios. Therefore, it was determined that an aspect ratio of 2.5 would achieve a high stiffness without unnecessarily increasing the system mass.

4.4 Arms & Brace Dimensions Derivation

4.4.1 Problem Definition

Optimisation is necessary to determine the ideal dimensions of the net arms and brace as part of the wider capture mechanism. The primary components under investigation are structured as four distinct beams, labelled as Beams 1, 2, 3, and 4, each designed with specific geometric and mechanical characteristics, shown in Figure 19. The lengths and sizes of each of these parts have multiple competing functional objectives, requiring a method to correctly allocate dimensions.

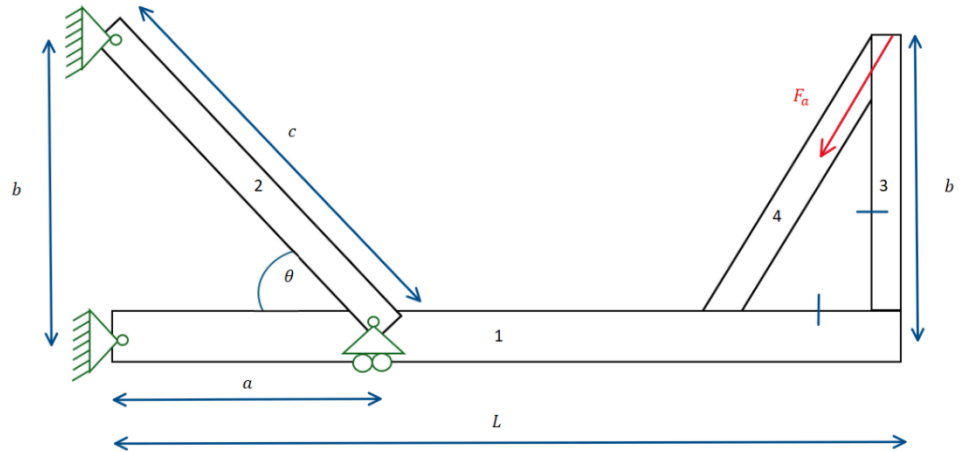


Figure 19 – Free body diagram of the net arm-brace assembly showing applied load, geometric parameters (arm length L , brace dimensions and length), support conditions, and internal beam numbering (1–4).

Beam 1 serves as the primary horizontal arm, Beam 2 functions as a diagonal brace, Beam 3 acts as the vertical support structure at the ends of the horizontal arms, to which the net is tied to, and Beam 4 is the connecting diagonal arm. Beams 1, 2 and 4 feature hollow rectangular cross-sections, justified in Section 4.3, while Beam 3 possesses a solid rectangular cross-section, reflecting its function to withstand greater direct loads.

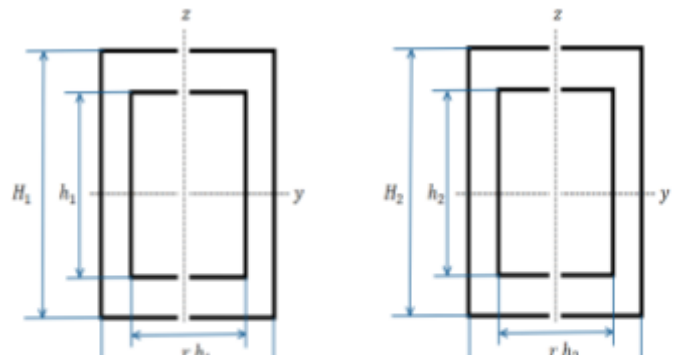
The optimisation problem considers four distinct objectives to ensure functionality and reliability:

1. Minimisation of Mass (m): Reducing the mass of each component enhances the carrier drones manoeuvrability and efficiency, which is crucial for agile pursuit and capture of target drones.
2. Minimisation of Stress (σ_{max}): Limiting the maximum stress experienced by the arms is critical to prevent structural failure, especially under collision loads during interception.
3. Minimisation of Deflection (y_{max}): Keeping deflection to a minimum ensures the structural integrity and precise positioning of the net during high-speed manoeuvres and impacts.
4. Maximisation of Arm Length (L): Increasing the length of the net arms directly enlarges the capture area, facilitating the interception of larger or evasively manoeuvring drones.

4.4.2 Requirements & Constraints

Apparatus Setup

Beam 1 (Net Arm): This beam is cantilevered, with one end rigidly fixed (screwed) to a central column, while the opposite end is free. This net arm must withstand bending stresses



resulting from collisions with intercepted drones, hence the need to minimise deflection and stress.

Beam 2 (Brace): This beam supports the net arm, with one end attached via a roller support to Beam 1, allowing it to move along the net arm without inducing additional bending moments. The other end is rigidly fixed (screwed) to a slider positioned along the central vertical column, ensuring structural integrity and stability.

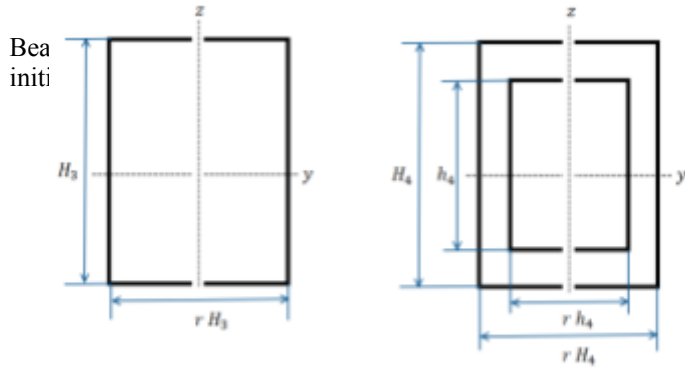


Figure 21 Cross-sectional geometries of Beam 3 (left) and Beam 4 (right), both featuring hollow rectangular profiles.

ports. Note that due to high deformations in this beam, as observed in

13. This is the beam through which the primary force from the net is

All beams are fabricated from Aluminium 6061.

Free Design Variables

Free design variables were constricted by realistic bounds to ensure manufacturability and so a sizeable drone could be captured within the net:

- $0.1 \leq a \leq 1.8 \text{ m}$: Position of the roller support attachment along Beam 1
- $0.1 \leq b \leq 1.2 \text{ m}$: Vertical position of the brace slider along the central column when the arms are open completely
- $0.005 \leq H_1 \leq 0.1 \text{ m}$, $0.003 \leq h_1 \leq 0.06 \text{ m}$: Outer and inner heights of Beam 1 (net arm, hollow rectangular cross-section)
- $0.004 \leq H_2 \leq 0.08 \text{ m}$, $0.002 \leq h_2 \leq 0.04 \text{ m}$: Outer and inner heights of Beam 2 (brace, hollow rectangular cross-section)

Fixed Design Variables

- $H_3 = 0.01 \text{ m}$: Outer height of Beam 3 (vertical net support, solid rectangular cross-section)
- $H_4 = 0.01 \text{ m}$: Outer height of Beam 4 (diagonal net support, solid rectangular cross-section)
- $h_4 = 0.007 \text{ m}$: Inner height of Beam 4 (diagonal net support, solid rectangular cross-section)
- $c = \sqrt{a^2 + b^2}$: Length of connecting brace
- $F_a = 500 \text{ N}$: Overestimated impact load applied to the net arm during collision with drone (see Section 4.1)

Constraints

Table 5 – Design constraints of net arms/brace.

Constraint	Expression	Purpose
Mass	$m \geq 0.2 \text{ kg}$	Ensures structural functionality
	$m \leq 1.5 \text{ kg}$	Limits payload
Beam wall thickness	$H_i - h_i \geq 1 \text{ mm}$	Ensures manufacturability
Support position	$a < L$	The roller support must always be positioned within the length of the net arm
	$b < L$	Brace attachment point along the vertical central column must not exceed net arm length
Height-to-width ratio (r):	2.5	Cross-sectional proportion.

4.4.3 Assumptions

To simplify the optimisation problem, the following key assumptions were made:

- **Idealised Geometric Model:** The geometry was simplified into an idealised 2D beam model, neglecting the 3D effects and any minor geometrical irregularities that may arise during manufacturing.
- **Rigid Supports:** Supports were assumed idealised. A screw connection, at the base, was modelled as a fixed support, providing full constraint against translation, but realistically, screws may permit slight rotation and displacement.
- **Roller Support Simplification:** The roller support, where the brace connects to the arm, was modelled to allow free movement along Beam 1 without friction or resistance, ignoring possible frictional forces and moments that could arise in a realistic sliding connection.
- **Static Loading Conditions:** Only static loading was considered in this analysis, neglecting dynamic, fatigue, and aerodynamic effects during drone operation and drone-to-drone collisions.

- **Material Homogeneity & Isotropy:** Aluminium 6061-T6 was assumed perfectly homogeneous and isotropic throughout, disregarding any material imperfections, anisotropy, or variations in mechanical properties from manufacturing processes.
- **Neglecting Minor Loads:** Secondary effects such as wind loads, vibrations from drone motors, and impacts aside from the primary collision load were not considered in the analysis.
- **Linear Elastic Material Behaviour:** Material behaviour was assumed linear elastic within the analysis range, neglecting plastic deformation and any non-linear effects.

4.4.4 Formulation of Objective Functions

Table 6 – Summary of objective functions.

Objective	Equation	Purpose
Mass Minimisation	$m = \rho r \left[L(H_1^2 - h_1^2) + \sqrt{a^2 + b^2}(H_2^2 - h_2^2) + bH_3^2 + \frac{b}{\cos \cos(45^\circ)}(H_1^2 - h_1^2) \right]$	Minimise payload
Stress Minimisation (Max Stress)	$\sigma_{max} = (\sigma_{arm} , \sigma_{brace})$ $\text{Beam 1: } \sigma_{arm} = \frac{6M_0 H_1}{r(H_1^4 - h_1^4)} \quad \text{Beam 2:}$ $\sigma_{brace} = \frac{F_L L - M_0}{a \sin \sin(\theta) r(H_2^2 - h_2^2)}$ Where: $F_L = F_a \cos \cos(45^\circ)$ (Vertical component of applied load) $M_0 = F_a b \sin \sin(45^\circ)$ (Moment experienced by Beam 1 as a result of horizontal component of applied load) $\theta = \arctan \arctan\left(\frac{b}{a}\right)$	Avoid structural failure
Deflection Minimisation (Max Deflection)	$y_{max} = \left \frac{1}{EI} \left[\frac{F_L(a-L)+M_0}{a} \frac{L^3}{6} + \frac{F_L L - M_0}{a} \frac{(L-a)^3}{6} - \frac{F_L(a-L)+M_0}{a} \frac{a^2 L}{6} \right] \right $ Where: $I = \frac{r(H_1^4 - h_1^4)}{12}$ (Second moment of area)	Maintain stiffness and precision
Arm Length Maximisation	$(-L) \Leftrightarrow (L)$	Maximise capture range

Derivations of equations are in Appendix A.4.

4.4.5 Methodology & Implementation

The optimisation employed the NSGA-II algorithm within the Python library Pymoo (Version: 0.6.1.3) (Blank & Deb, '2020), providing multi-objective constrained optimisation by evaluating trade-offs between minimising mass, stress, and deflection. The key variables optimised include arm lengths and cross-sectional dimensions, constrained by material strength, geometric compatibility, and manufacturing practicality. This approach ensures a robust design solution, balancing performance efficiency, structural integrity, and operational effectiveness. The custom problem was encoded programmatically and explored with a population size of 100 individuals over 200 generations. NSGA-II's evolutionary operators: selection, crossover, mutation and Pymoo's constraint handling facilitated the identification of optimal, with practical solutions forming a balanced Pareto front.

4.4.6 Results & Pareto Analysis

Top 3 Solutions

The optimisation produced a set of Pareto-optimal designs. To select the top three solutions, a Euclidean norm was calculated for each design solution to measure its proximity to the ideal solution (the component-wise minimum). The top three closest-to-ideal solutions are presented in Table 7.

Table 7 – Top 3 Pareto-optimal solutions. Objective functions are highlighted in blue.

Rank	L (m)	a (m)	b (m)	H ₁ (m)	h ₁ (m)	H ₂ (m)	h ₂ (m)	m (kg)	Max σ_{arm} (MPa)	Max σ_{brace} (MPa)	y _{max} (m)
1	0.757	0.238	0.100	0.024	0.017	0.013	0.007	1.443	8.399	8.146	0.005
2	0.756	0.238	0.103	0.024	0.017	0.013	0.008	1.433	8.633	8.493	0.005
3	0.757	0.278	0.122	0.024	0.017	0.013	0.007	1.500	10.20	6.490	0.004

All top solutions exhibit higher stresses in the net arm compared to the brace, indicating the net arm predominantly drives the structural integrity and sizing decisions.

Pareto Analysis & Observations

Pareto plots provide visual insights into the trade-offs between key objectives:

Pareto Front: Mass vs Stress vs Deflection



Mass vs Stress vs Deflection:

The Pareto front (Figure 22) clearly illustrates that lower masses (< 0.5 kg) generally correspond to higher deflections (> 50 mm). Conversely, designs that exhibit lower deflections typically incur increased mass, indicating a strong trade-off between these two objectives. The top three designs identified effectively balance moderate mass (~ 1.4 - 1.5 kg) with relatively low deflection (~ 4 - 5 mm) and acceptable stress levels (< 11 MPa).

Arm Length vs Stress vs Mass:

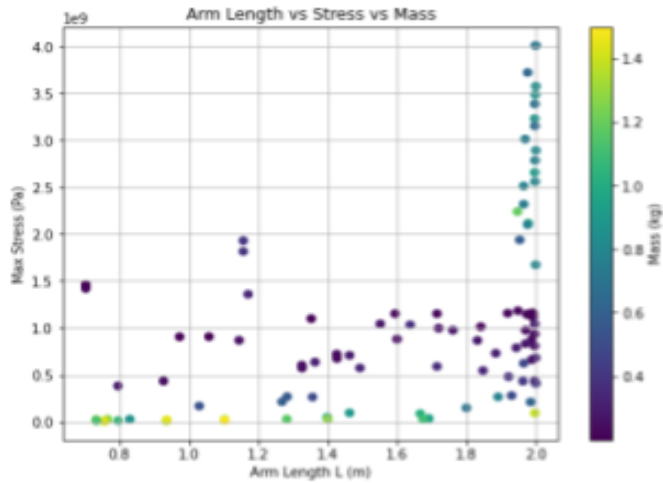


Figure 23 – 2D plot showing the relationship between arm length (L) and maximum stress, with point colour indicating mass. Figure 23 – 2D plot showing the relationship between arm length (L) and maximum stress, with point colour indicating mass.

Figure 23 shows that designs with longer arms approaching the upper bound (2 m) generally experience significantly elevated stress levels, often above 2 GPa, and tend to correlate with a wider mass variability. The chosen top designs maintain relatively short arm lengths (approximately 0.75 m) to effectively limit stress levels below approximately 10 MPa without excessively increasing mass.

Constraint Activity

The optimisation constraints significantly influenced the resulting Pareto-optimal set. Solutions closely approached the upper mass limit, suggesting that increased structural integrity strongly correlates with approaching this upper bound. The thickness constraints for the beams were frequently close to the minimum required difference, indicating structural sizing was directly driven by these limits.

4.4.7 Discussion

Mass vs Performance

Increasing the net arm length generally led to higher mass and greater stresses due to larger bending moments. The chosen Rank 1 solution provided an optimal balance with a manageable mass (1.44 kg per arm/brace), while comfortably meeting structural requirements (8.40 MPa arm stress, 8.15 MPa brace stress), well below the yield strength of Aluminium 6061-T6 (276 MPa). With four arms, the total mass (5.76 kg) remains within the drone's recommended payload limit of 8 kg, providing sufficient margin for additional components (e.g., net, actuators, electronics). Despite the relatively high mass, the net arms and braces serve as critical functional components in the capture mechanism, justifying this compromise.

Brace Design Role

The brace significantly controlled net arm deflection and reduced stresses, evident from the minimal deflection (5.18 mm in the Rank 1 solution). The optimisation consistently converged the brace position to its lowest feasible point (b at its lower bound in Table 7), advantageous both for reducing bending moments in the net arm and minimising stress on the central column.

Hollow Cross-Section Efficiency

Generally, the chosen optimisation process showed a preference towards thinner-walled hollow sections, as these provide high structural efficiency by maximising the second moment of area, thus reducing stress and deflection while minimising mass. The relatively thin walls of the Rank 1 solution (e.g., $H_1 - h_1 = 7.11$ mm, $H_2 - h_2 = 6.15$ mm) reflect this efficiency-driven approach, offering the best structural performance-to-weight ratio.

Manufacturing Impact

From a manufacturing perspective, the Rank 1 solution provides a practical compromise. The optimised dimensions are within feasible manufacturing tolerances for standard aluminium extrusion and machining processes. Although the wall thicknesses selected are relatively thin, they remain comfortably within achievable tolerances for custom aluminium profiles. This ensures the solution does not necessitate specialised production techniques.

Limitations

Despite providing valuable insights, the optimisation model incorporates certain simplifications which limit its applicability:

- **Large Deflection Effects:** The analysis relied on small bending assumptions inherent in simple beam theory. The relatively large deflections observed; however, larger real-world deflections may necessitate nonlinear, large-deformation analyses for accuracy.
- **Evolving Design:** Further design modifications during later development phases could affect the optimality of the current solutions. This optimisation provides initial guidance rather than definitive final dimensions.
- **Manufacturing Constraints:** The analysis assumed beams could be manufactured and assembled as separate idealised segments. However, practical manufacturing constraints might require certain elements, such as Beam 4, to be fabricated as one integral component, potentially impacting structural properties and the practicality of the optimal solutions.
- **Simplified Brace Representation:** The brace was modelled as a single component, rather than two separate parts positioned on opposite sides of Beam 1 to allow the umbrella mechanism to fold properly. This simplification could underestimate complexity in assembly, increase the actual mass slightly, and affect structural load distributions in practice.

These limitations highlight the need for additional analyses, such as large deflection studies, nonlinear structural simulations, and consideration of detailed manufacturing constraints.

4.4.8 Summary

The optimisation successfully identified a balanced design for the drone interception net arm and brace, achieving an optimal compromise between mass, stress, deflection, and arm length. The selected Rank 1 solution provided a structurally sound and relatively lightweight design at 1.44 kg per arm/brace set (5.76 kg total for all four sets), well within the drone's recommended maximum payload of 8 kg, allowing sufficient capacity for additional components. The design's stress (approximately 8.40 MPa for the arm and 8.15 MPa for the brace) and deflection (approximately 5.18 mm) were comfortably within acceptable limits, ensuring reliable operation and structural integrity. However, this study involved simplifications such as neglecting dynamic loading, aerodynamic effects, and fatigue considerations. Future studies should focus on incorporating these aspects, including dynamic analyses through FEA, fatigue assessments, aerodynamic investigations via CFD, and experimental validations to confirm performance in realistic operational scenarios.

4.5 Central Column

4.5.1 Problem Definition

The central column is a critical structural component of the capture mechanism. Mounted vertically atop the drone's upper surface, it provides essential support for the slider mechanism, which facilitates the controlled opening and closing of the net arms. Due to its critical position and functional responsibilities, the central column directly impacts the structural integrity and operational reliability of the capture system.

Optimisation of the central column dimensions is crucial for several reasons. Primarily, minimising its mass reduces the overall system weight, which directly improves the drone's manoeuvrability, flight endurance, and battery efficiency. Additionally, ensuring the column can withstand bending and shear stresses, resulting from operational loads such as the forces transmitted through supporting braces during capture events, is essential for maintaining structural safety and reliability.

A linear ball bearing is attached on the outer surface of the central column. The slider is then mounted directly onto the outer surface of the bearing, facilitating smooth, low-friction translational movement along the vertical axis. This design ensures precise and stable operation of the slider during deployment. Hence, establishing optimal dimensions that balance minimal mass with maximal structural performance under defined constraints represents a key engineering challenge addressed in this analysis.

4.5.2 Objectives

The optimisation of the central column was guided by clearly defined engineering objectives:

- Minimise mass (m): Crucial for enhancing drone performance by reducing the overall system weight, thereby improving flight endurance, agility, and efficiency.
- Minimise bending stress (σ_{max}): Ensuring structural integrity under operational loading, specifically from the forces transmitted via the supporting braces during the capture mechanism's operation.
- Minimise shear stress (τ_{max}): Vital for structural safety and reliability, specifically to withstand the longitudinal component of forces applied by the brace against the column's surface without failure.

Initially, friction and interface pressure reduction were prominent objectives. However, upon implementing linear ball bearings, frictional forces were dramatically reduced. This innovation effectively eliminated friction as a design concern, allowing for a simplified optimisation focus on mass, deflection, and shear stress. This iterative refinement process highlights the importance of continually revising design objectives based on practical engineering innovations and advancements.

4.5.3 Design Variables & Constraints

Design Variables

- r_o : Outer radius of central column
- r_i : Inner radius of central column
- L : Length of central column

Constraints

Several critical constraints were defined, addressing practical considerations, structural safety, and geometric compatibility with existing components:

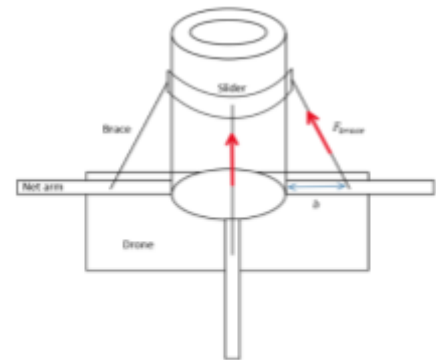


Figure 24 – Free body diagram of the central column assembly with slider, braces, and net arms.

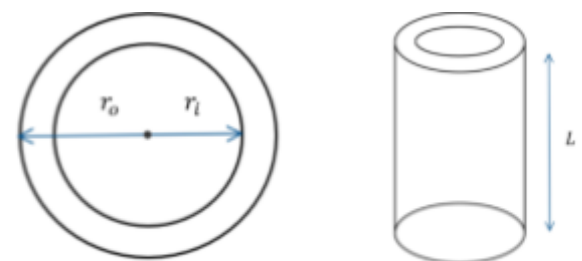


Figure 25 – Geometry of the central column, showing internal radius, external radius, and length.

- Length ($L \geq d_s$): The column length must be equal to or greater than the required sliding distance (d_s) for the slider mechanism to fully open and close the capture net.
- Outer diameter ($r_o < r_{o_{max}}$): The outer diameter of the column ($2r_o$) cannot exceed the width of the drone surface ($2r_{o_{max}} = 123 \text{ mm}$) to which it will be mounted.
- Outer diameter for bearing compatibility ($r_o \leq r_{i_{bearing}}$): The outer diameter of the column must be compatible with commercially available linear ball bearings ($r_{i_{bearing(max)}} = 25 \text{ mm}$).
- Material: Aluminium 6061 T6 was selected, characterised by a yield strength $\sigma_y = 276 \text{ MPa}$ and a shear limit $\tau_y = \frac{\sigma_y}{2} \Rightarrow \tau_y = 140 \text{ MPa}$, calculated according to maximum shear stress theory as half of the yield strength.
- A safety factor (s.f.) of 1.5 was implemented for both bending and shear stresses to ensure operational safety and reliability.
- Bending stress (with s.f. of 1.5): $\sigma_{max} \leq \frac{\sigma_y}{1.5} \approx 186.7 \text{ MPa}$
- Shear stress (with s.f. of 1.5): $\tau_{max} \leq \frac{\tau_y}{1.5} \approx 93.3 \text{ MPa}$

4.5.4 Loading Conditions

The central column experiences both bending and shear because of the force applied by the brace-slider assembly, which is connected to four net arms arranged symmetrically around the column's circumference.

During capture, a drone colliding into a net quadrant will exert a force that is transmitted through the affected arms and into the brace. This, in turn, applies a combination of forces to the central column via the slider. The brace is diagonally aligned with respect to the column and slider, meaning that the force transmitted through it has both vertical and horizontal components. As a result, in addition to inducing bending, the force also contributes a shear force on the central column's surface along its longitudinal axis.

For simplicity, two loading cases were analysed:

1. Single Arm Loading (Worst Case): The full weight or impact of the captured drone is assumed to act through a single net arm, exerting a force on one side of the brace. This produces a concentrated bending in the direction of the affected net arm and shear force along the column at the side of the activated arm.
2. Dual Arm Superposition (Quadrant Capture): In a more realistic case, the drone is captured within a net quadrant, and two net arms are activated. The forces from both arms are assumed to be symmetric and superimposed to evaluate combined loading effects on the column. This captures the distributed but directional bending and shear expected in typical operation.

The net-arm forces act at the end of the brace at a fixed distance from the central column's centroid. The corresponding reaction is modelled as a point load at this distance, resolved into its vertical and horizontal components to evaluate the resulting bending moments and shear forces along the central column. The column is thus treated as a vertical cantilever beam fixed to the drone base, subject to lateral point loading at its free end.

4.5.5 Assumptions

To simplify analysis and allow the use of closed-form solutions, several assumptions were adopted in modelling the structural behaviour of the central column:

For Pure Bending Stress Analysis:

- The beam (column) is initially straight and free from residual stresses.
- The material (Aluminium 6061 T6) is perfectly homogeneous and isotropic.
- Plane cross-sections remain plane before and after bending.
- All cross-sections are symmetric about the plane of bending.
- Young's Modulus is constant and the same in both tension and compression.
- The elastic limit is not exceeded; material behaves linearly elastically

For Shear Stress (Uniform Shear Flow) Analysis:

- The shear force is uniformly distributed along the wall thickness of the hollow cylinder.
- The wall thickness is sufficiently small relative to the outer radius to assume a thin-walled approximation where required.
- Shear flow is constant through the thickness at a given cross-section.
- The central column's cross-section does not distort significantly under loading.
- Torsional shear and direct shear are considered additive and act independently.

4.5.6 Optimisation Methodology

To determine the optimal dimensions of the central column, a MATLAB-based (MathWorks, 2023) parametric analysis was conducted. Three performance metrics were defined as objective functions: total mass m , maximum bending stress σ_{max} , and maximum shear stress τ_{max} . The analytical expressions for each objective were derived, and their detailed derivation steps are documented in Appendix A.5.

The mass objective was calculated using the hollow cylinder volume equation: $m = \rho \pi L (r_o^2 - r_i^2)$ (see Appendix A.5.1).

The bending stress was calculated based on beam theory and superposition of forces from two arms during drone capture: $\sigma_{max} = -\frac{4F \cos(\theta) b r_o}{\pi(r_o^4 - r_i^4)}$ (see Appendix A.5.2). When a target drone is captured within a net quadrant, its weight and impact will be distributed between two net arms, meaning that the stress on the central column in a singular direction will be less than if the drone's impact were to be absorbed solely by one arm. Therefore, the maximum bending stress that the central column will experience is in a scenario when a drone directly impacts an isolated net arm.

The shear stress was derived using shear flow theory for thin-walled sections: $\tau_{max} = \frac{F \sin(\theta)}{2\pi r_o (r_o - r_i)}$ (see Appendix A.5.3).

The design process proceeded by fixing the length of the central column to $L = 0.5 \text{ m}$, a practical value satisfying inertia and mounting constraints. Commercially available outer diameters (20 mm to 50 mm) compatible with linear bearings were used as discrete cases. For each fixed r_o , a sweep of inner radius r_i was performed to evaluate performance across the design space.

MATLAB was used to compute and plot the relationships between the inner radius r_i , mass m , and the two stress metrics (σ_{max}/τ_{max}) for each case. Overlaid on these plots were safety factor limits for both bending and shear, allowing for quick identification of feasible regions. The graphs enabled visual filtering of infeasible geometries and selection of configurations that met all constraints while minimising mass. This method ascertains that design decisions were driven by analytical evaluation and visual interpretation of performance trends, resulting in a practical, lightweight, and structurally sound central column design.

For all graphical analyses, the allowable inner diameter was constrained to integer values to ensure manufacturability, as most standard manufacturing processes do not support tolerances as fine as 0.1 mm. The selected inner radius must also facilitate maximum bending and shear stresses below the safety factor. This discrete approach aligns with commercially feasible machining resolutions and simplifies procurement.

4.5.7 Results & Analysis

Outer diameter $D_o = 50 \text{ mm}$:

When $D_o = 50 \text{ mm}$, the optimal internal radius under the safety factor constraint was $r_i = 24 \text{ mm}$. At this point, mass reduced to

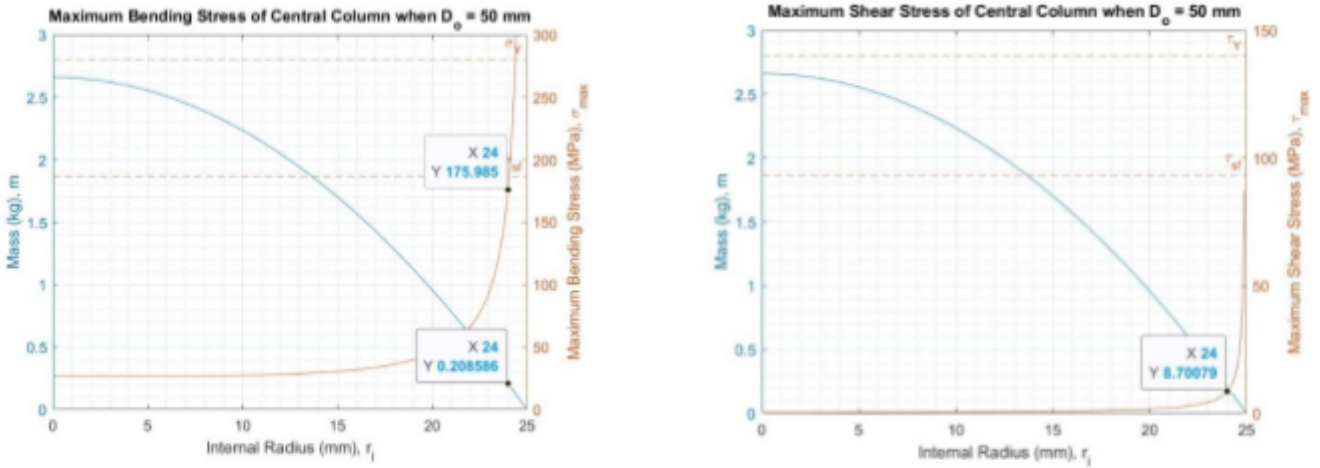


Figure 26 – Variation of mass, maximum bending (left) and shear stress (right) with internal radius for a central column of outer diameter 50 mm.

$m = 0.209 \text{ kg}$, $\sigma_{max} = 175.985 \text{ MPa}$ and $\tau_{max} = 8.701 \text{ MPa}$.

The same approach used to select the inner diameter for an outer diameter of 50 mm was used to select an appropriate inner diameter for outer diameters of 40 mm, 30 mm and 20 mm (see Appendix A.5.4).

All of the graphs display an exponential relationship between inner radius r_i and both maximum stresses, σ_{max} and τ_{max} as well as asymptomatic behaviour around the maximum possible inner radius r_i (where $r_i < r_o$).

A summary table was constructed to compare the performance of the three feasible configurations:

Table 8 – Summary of central column configurations and their corresponding properties.

Outer Diameter (mm)	Inner Diameter (mm) ~ mm that is under sf	Mass (kg) ~ g	σ_{max} (MPa)	τ_{max} (MPa)
50	48	0.209	175.985	8.701

40	36	0.324	150.575	5.438
30	22	0.443	172.686	3.625
20	19	0.042	N/A	43.504

Based on this table, the configuration $D_o = 50 \text{ mm}$, $D_i = 48 \text{ mm}$ was identified as the optimal solution, offering the lowest mass ($m = 0.209 \text{ kg}$) while satisfying the safety-factor adjusted structural limits for both σ_{max} and τ_{max} . A clear trend is also observed where decreasing the outer diameter necessitates an increase in material mass to satisfy the same structural stress constraints, making smaller-diameter columns less efficient for this design.

4.5.8 Discussion

The optimised central column dimensions (50 mm outer diameter, 48 mm inner diameter) effectively balance structural integrity and minimal mass. These dimensions ensure bending, and shear stresses remain safely within limits, adhering to the 1.5 safety factor. The reduced mass directly enhances drone agility, flight endurance, and energy efficiency, crucial for extended operational performance.

Selecting a relatively large outer diameter provided key practical advantages, simplifying attachment points for net arms and allowing adequate space for mechanical components, thus reducing potential interference during net deployment. However, the commercially available linear bearings posed a limitation by capping the maximum inner diameter at 50 mm, restricting further mass optimisation. Despite this constraint, the resultant 2 mm wall thickness proved sufficient for structural requirements.

To address frictional concerns, a KH50 Relubricatable Closed Linear Ball Bushing (50x62x70 mm) (Simply Bearings, n.d.) was integrated. This bearing ensures smooth and low-friction linear movement for the slider, simplifies assembly, and enhances overall mechanism reliability.

A notable consideration arises from the thin wall thickness (2 mm), which could lead to unintended local thinning during manufacture and increased stress concentrations. The applied safety factor and conservative load estimations mitigate these potential risks.

Overall, the final dimensions represent an effective compromise between manufacturability, structural reliability, and performance, demonstrating thoughtful integration of commercially available components and practical engineering constraints.

4.5.9 Limitations

While the design meets current performance targets, several caveats should be acknowledged for this design process:

- Static, linear analysis – dynamic impact loading, rotor-induced vibration and fatigue life were not assessed.
- Idealised load distribution – real collisions may impose distributed, time-varying forces beyond the two-point-load scenarios modelled.
- Environmental durability – corrosion and long-term weathering effects were not investigated.

4.5.10 Validation

To verify the structural reliability of the chosen central column configuration, FEA simulations were carried out for the two critical operational scenarios:

- Single Arm Loading: A single point force was applied to one side of the central column, resulting in a concentrated and asymmetrical load.
- Dual Arm Superposition: Two-point forces, each half the magnitude of the original load, were applied to two perpendicular sides of the column.

The aim of these FEA simulations was to ascertain whether the worst-case loading condition (Scenario 1) led to the highest stress concentrations and deformations, validating the conservative assumptions made during hand calculations. The results confirmed this hypothesis.

FEA results summary:

Table 9 – Summary of FEA results comparing one-arm and two-arm loading scenarios on the central column.

Scenario	Max. Stress (MPa)	Max. Deflection (mm)	Min. Safety Factor
One-arm (worst case)	150.0	2.734	1.84
Two-arm (equally balanced)	207.52	1.933	1.33

In Single Arm Loading, the central column experienced higher von Mises stress, greater tip deflection, and a slightly reduced minimum safety factor compared to Dual Arm Superposition. Nonetheless, in both cases, the structural stresses remained within the yield limits defined by the material (Aluminium 6061 T6) and the design safety factor of 1.5. These findings validate the robustness of the analytical assumptions made during optimisation and confirm that the selected geometry (50 mm outer diameter, 48 mm inner diameter) offers sufficient structural performance under both balanced and unbalanced load conditions.

4.5.11 Summary

The optimisation delivered a 50 mm OD × 48 mm ID Aluminium 6061-T6 central column that satisfies a 1.5 safety factor for both bending and shear while adding only 0.209 kg to the aircraft. The geometry mates directly with a KH50 linear ball bushing, minimises

friction for the slider, and leaves ample surface area for net-arm attachments and the retractable drive system. Finite-element simulations under both balanced two-arm and worst-case single-arm impacts confirmed that stresses and deflections remain safely below yield limits, validating the analytical sizing approach. In future work, limitations can be addressed through dynamic testing, fatigue evaluation and exploration of custom bearing or composite options. These will be key steps in a later iteration of the capture mechanism.

4.6 FEA Validation

A FEA was performed using the ANSYS Workbench 2024 R1 Student Version (ANSYS, 2024) (Ansys, 2024) to assess the structural integrity of the capture mechanism under extreme loading conditions. The main goal of this analysis was to ensure that the design could endure a 500 N impact force (from Section 4.1) without surpassing the yield stress of Aluminium 6061 T6 (276 MPa) while achieving a minimum safety factor of 1.5. This safety factor is consistent with aerospace engineering standards mentioned in the objectives in Section 1.3. Additionally, it was crucial to minimise the deflection of the arms to maintain precision and functionality, especially considering the dynamic environment in which the mechanism operates.

4.6.1 Assigning material

Each part of the capture mechanism was subjected to FEA, including the arm brace, base, slider, central column, bearings, and end caps. These components were first designed in FUSION 360 due to its strong parametric modelling features, and were subsequently imported into ANSYS Workbench. Aluminium 6061 was assigned as justified in the material selection part. Material properties were defined, including Young's modulus (69 GPa), Poisson's ratio (0.33), and yield strength (276 MPa), to represent the material accurately. However, idealised assumptions used in the setup regarding isotropic and homogeneous material properties pose constraints. This is as they overlook real-world imperfections such as residual stresses and microstructural inconsistencies that could affect actual performance.

4.6.2 Mesh convergence study

Meshing plays a crucial role in the FEA process, significantly impacting simulations' accuracy and computational efficiency (Mistry, et al., 2014). The use of automatic meshing was advantageous due to its ability to effectively manage complex geometries, employing adaptive algorithms that modify element size and distribution in response to local geometric features (Li & Robinson). This method ensured a high-quality mesh throughout the model while minimising human error during setup. (Jien & Nishiura, 2018) However, one drawback of automatic meshing is its potential inability to provide adequate refinement in areas of high stress, leading to inaccuracies in local stress predictions (Mistry, et al., 2014). On the other hand, while manual meshing allows for focused refinement, it is more labour-intensive and can introduce inconsistencies, particularly in larger assemblies (Li & Robinson). Consequently, the study utilised automatic meshing as the baseline, carefully monitoring mesh refinement through convergence analysis to ensure its effectiveness (Jien & Nishiura, 2018).

A mesh convergence study was performed to confirm the stability and reliability of the simulation results. Two key parameters were evaluated: total deformation and von Mises stress. These metrics were selected to provide a comprehensive assessment of the structural performance under operational loads, with total deformation reflecting global stiffness and von Mises stress indicating peak localised stresses. Total deformation was established as the convergence criterion due to its stable representation of global stiffness and because the values converged. In contrast to von Mises stress, which exhibits high sensitivity to mesh size near geometric discontinuities like corners and joints, total deformation represents the overall mechanical response of the structure. As shown in Figure 27, von Mises stress displayed fluctuating trends with mesh refinements, decreasing nonlinearly from 253.8 MPa at a 10 mm mesh size to 100.7 MPa at 1 mm, without reaching a definitive plateau. In comparison, total deformation demonstrated a stabilising curve which converged at around 9.57 cm (using a 1% criterion) (Patil & Jeyakarhikeyan, 2018), which is at a mesh size of 3 mm, shown in Figure 28.



Figure 27 – Plot showing element number against von Mises stress.



Figure 28 – Plot showing element number vs total deformation at the free end.

This mesh size was uniformly applied in subsequent design iterations, which involved geometric modifications, including adjustments to joint hole diameters, reinforcement of critical areas, and enhancements to arm profiles. Following the iteration strategy outlined in the FEA documentation, these modifications maintained the fundamental topology, material properties, and boundary conditions. Consequently, the main deformation mechanisms, bending and axial stiffness, remained unchanged, confirming the continued appropriateness of the 3 mm mesh size for future simulations unless significant geometric or material alterations were made.

4.6.3 Element Type

The mesh included quadrilateral (hexahedral) and triangular (tetrahedral) elements tailored to the geometry of the components. Hexahedral elements (SOLID186), characterised by a 20-node solid formulation with quadratic displacement behaviour, were applied to straightforward, prismatic sections like the arm (Rohan, Lobos, Nazari, Perrier, & Payan, 2014). These elements provide high stress and strain resolution precision while being computationally efficient (SOLID186, n.d.). Conversely, tetrahedral elements (labelled as SOLID187), comprising 10-node solids, were incorporated for more complex geometries, such as braces and end caps, as they adapt better to irregular shapes while delivering reliable performance (Gowtham & Srivatsa, 2018). Elements such as SOLID186 are optimal for use with structured meshes and simple, regular shapes, which makes them well-suited for straightforward geometries (SOLID186, n.d.). In contrast, SOLID187 elements excel in handling complex or irregular meshes, particularly when it comes to accurately capturing stress distributions (SOLID187, n.d.).

4.6.4 Boundary Conditions and Modelling Assumptions

To accurately represent the behaviour of the capture mechanism under load, several assumptions were made to ensure both the precision of the simulation and its computational efficiency. The arms' motion was modelled using revolute joints, which permitted rotation solely around the z-axis while constraining all other movements (ANSYS, 2025). This approach effectively captured the hinge-like action of the arms without introducing unnecessary complexity or issues such as artificial stiffness or redundant constraints that could distort stress predictions (ANSYS, 2025).

It was assumed that contacting components were fully bonded, with no slipping or separation allowed. Although this simplification ignores minor gaps or friction that might be present, it reflects a commonly accepted practice for components that are tightly secured in assembly (He, 2011). For external loading simulations, a force of 500 N was applied at a 45-degree angle to the arm's axis at the tip of the end cap. This scenario was designed to represent a worst-case impact, like a drone becoming trapped in a flexible net, resulting in both bending and shear stresses. While such impacts can be dynamic, they were modelled as static loads to focus on peak stress and deformation, an appropriate approach in the early design phase when detailed dynamic analysis is impractical. The model deliberately excluded time-dependent factors like vibrations or fatigue, concentrating instead on the immediate structural response to a singular, extreme load. This focus is particularly useful when cyclic or long-term effects are not the primary concern.

4.6.5 Outputs from Simulations

The ANSYS Workbench Student Version proved adequate for this project, though it has certain academic limitations. These include a maximum of 32,000 nodes and elements, and the lack of advanced solver capabilities like nonlinear analysis and explicit dynamics (ANSYS, 2023). While these restrictions may confine the complexity and detail of models, they did not significantly affect the goals of this study. The software still provided valuable results, such as total deformation, equivalent von Mises stress, and safety factors, which were essential for assessing the structural performance of the capture mechanism. Total deformation measurements indicated that deflection remained within acceptable limits, while the von Mises stress analysis revealed critical areas below the yield strength of Aluminium 6061. The safety factor, aimed at 1.5, confirmed that the design had a sufficient margin before reaching failure. Although simplified modelling and idealised assumptions were necessary due to software constraints, the simulations effectively identified key stress concentrations and guided iterative enhancements. In this regard, the limitations commonly found in student software did not impede the derivation of meaningful insights or the achievement of performance and safety objectives. Future analyses could expand on this foundational work by introducing dynamic loading, improved contact conditions, and advanced material models to increase accuracy and realism.

4.6.6 Topology Optimisation Setup

FUSION 360 was employed to perform topology optimisation on essential components of the capture mechanism, primarily aiming to reduce structural weight while ensuring mechanical performance under static loading conditions. This was particularly crucial because of the drone's payload limit of 10 kg. The optimisation sought to achieve a safety factor of around 1.5 when subjected to a 500 N load applied at a 45-degree angle, reflecting the worst-case impact scenarios. To accurately simulate assembly constraints, the base of the central column was fixed in all three translational directions, and pin constraints were placed at bolt hole locations to maintain critical geometrical features during the optimisation. Although various components were considered for optimisation, including the arm, brace, slider, base, and central column, FUSION 360's restriction to one target body per simulation meant that each part had to be handled separately. The shape optimisation engine within FUSION 360 employs established material distribution techniques to eliminate non-essential material and progressively arrive at an optimal design. The experience with ANSYS for topology optimisation showed that the solve times were excessively long, with each simulation for the complete model exceeding one hour. Therefore, FUSION 360 was used due to its streamlined case studies and fast computation times, enabling multiple simulations. FUSION 360 effectively generated lightweight design options that satisfied the required stress and safety under the specified static load. Nevertheless, while FUSION 360 excels in conceptual lightweight designs under simple static conditions, it has limitations in accurately capturing localised stress concentrations and transient dynamic responses.

4.6.7 Design Iterations through FEA and Topological Optimisation

This section details the application of FEA using ANSYS and topological optimisation using FUSION 360 to improve several components of the capture mechanism iteratively. The aim was to systematically evaluate mechanical performance, reduce mass where structurally viable, and enhance the overall design. Each component underwent multiple iterations, with results from FEA and TOPO used to inform further design changes. Where relevant, results are visually supported with representative images.

Arms of the capture mechanism

The original design of the arms featured a solid, non-hollow cross-section aimed at maximising strength while simplifying manufacturing. Although this approach provided a high average safety factor of around 2 and an average von Mises stress of 137.8 MPa, significant stress concentrations were detected around the joint holes, as shown in Figure 29. These stress concentrations occurred from geometric discontinuities that amplified the load in these areas, causing maximum stress levels to exceed the yield strength of 276 MPa. This raised concerns about the potential for plastic deformation and compromised the reusability of the arms, underscoring the necessity for optimisation to alleviate localised stresses and reduce weight while preserving structural integrity. Furthermore, the deflection was 9.52 cm, which can be further reduced.

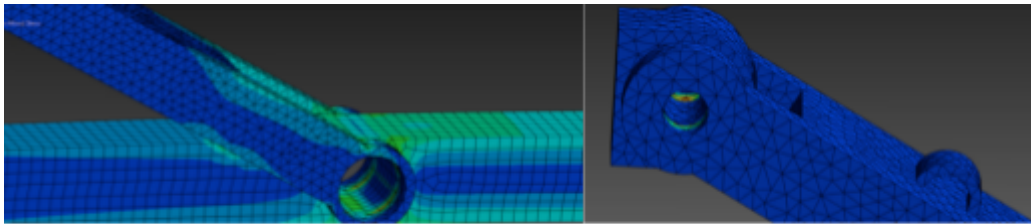


Figure 29 – Stress distribution at the joint where the arm and brace connect (left), and stress distribution at the joint between the brace and slider (right).

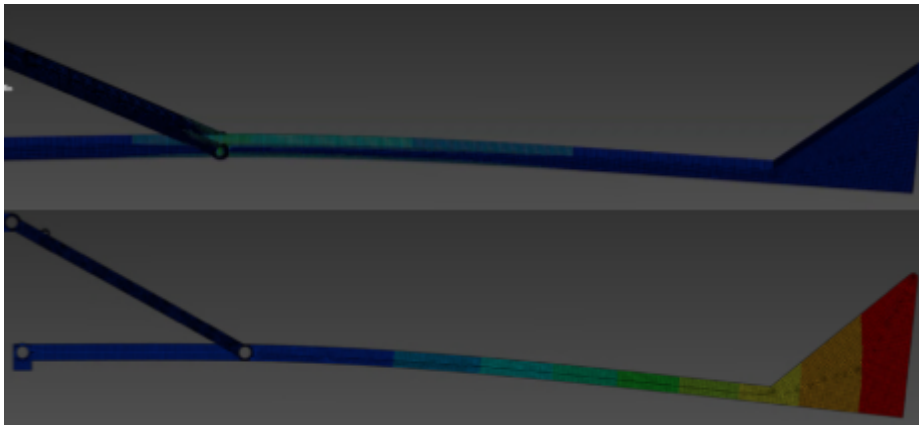


Figure 30 – Stress distribution (top) and total deflection distribution (bottom).

In the first iteration, two significant enhancements were made: the size of the joint holes was reduced, and an end cap was added. By minimising the joint hole diameter, stress amplification was mitigated, facilitating a more even distribution of forces. Adding the end cap improved the arm's stiffness by enhancing its moment of inertia, which is critical for resisting bending forces. These changes substantially decreased deflection by 52% (4.57 cm), improving the arm's performance and stability. The end cap was also important as it was used to attach the springs to the arms, as described in Section 3.

The end cap size was further diminished to the one displayed in Figure 31, for the second iteration to achieve additional weight savings. Although this change introduced a slight increase in deflection by 35.56% (to 6.2 cm) due to the reduced stiffness, the overall structural performance remained satisfactory. Peak stress levels remained below the material yield strength at approximately 120 MPa. This iteration demonstrated effective material utilisation, concentrating it where necessary and eliminating superfluous mass, thus balancing reliability and weight efficiency.

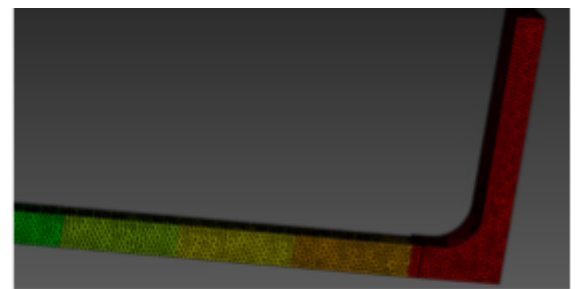


Figure 31 – The total deflection at the end cap is shown

The third iteration pursued weight reduction by implementing a hollow cross-section with a reduced area. While this design decision significantly lowered the arm's weight, the thinner material at the brace-arm joint resulted in stress levels surpassing the material limit, with maximum stresses reaching around 420 MPa, visible in the red region in Figure 33. This indicated a need for reinforcement at the joint interface to uphold structural integrity. Concurrently, topological optimisation (Figure 32) revealed that certain low-stress zones could be further reduced to enhance material distribution, emphasising the delicate balance between minimising weight and managing critical stress areas.



In the fourth iteration, the arm's aspect ratio was increased from 2.5 to 3.3, resulting in an elongated geometry. This modification improved the arm's capability to distribute forces evenly along its length, thereby reducing stress concentrations and enhancing stability. The slider position was also adjusted along the arm to optimise force transfer and ensure a more balanced load distribution. These modifications increased the average safety factor to 1.47 and minimised deflection, contributing significantly to the arm's durability and suitability for repeated use. However, a small area near the brace-arm junction continued to exhibit stress levels above yield strength, visible in Figure 34. Despite this, topological optimisation indicated that the overall design was nearly optimal, achieving a commendable balance between weight efficiency and structural performance.

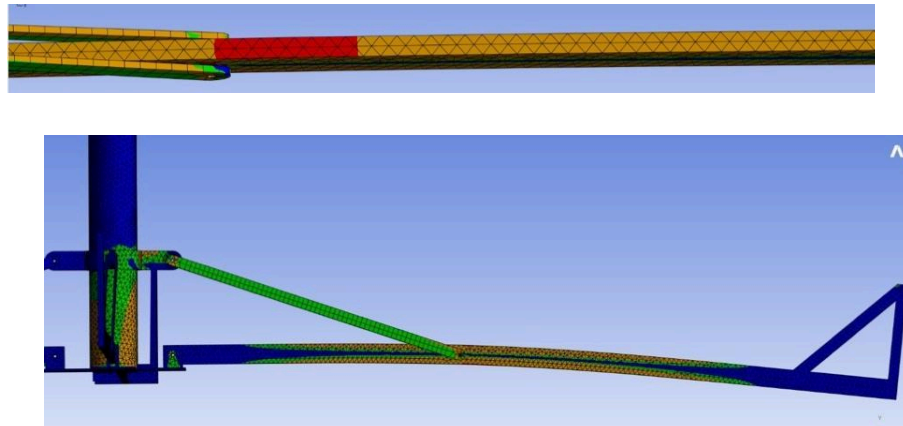


Figure 34 – Severe safety

critical component (top) and safety factor distribution of complete structure (bottom).

factor distribution in

The most recent iteration introduced a sleeve to reinforce the arm section at the brace joint, where persistent high stress concentrations had been identified, particularly at the brace-arm junction. Nylon PA6 was selected as the material due to its lower Young's modulus compared to that of aluminium 6061, which helps distribute stress better, as well as its ease of manufacturing. The introduction of this component ensured that the arm maintained elastic characteristics under cyclic loading, a vital requirement for reusability. This enhancement elevated the safety factor to 1.5 as shown in Figure 35, lowered stress levels to well below the yield threshold of 180 MPa, leading to a 50% decrease in stress from the previous iteration and reduced the maximum deflection by 4cm. By addressing critical stress zones and enhancing stability, the sleeve ensured that the arm met operational requirements while surpassing expectations for durability and resilience.

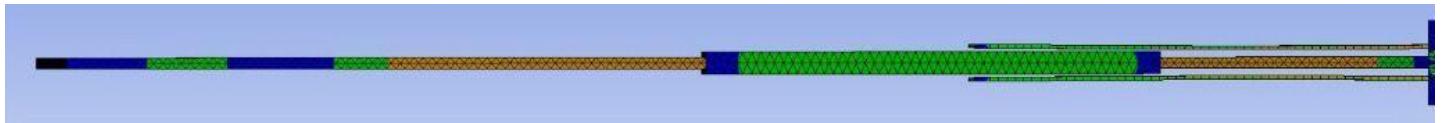


Figure 35 – Final iteration safety factor distribution.

Brace

The original brace design, shown in Figure 36, featured a substantial and sturdy structure, prioritising strength, and minimal deformation under load. However, localised stress concentrations were identified near the arm-brace joints, with stress peaking at approximately 200 MPa. Coupled with the brace's excessive weight, this underscored inefficiencies in material distribution and highlighted the need for optimisation.

The first iteration involved reducing the brace's cross-section and joint thickness, improving material efficiency. While this maintained structural integrity, stress levels around the arm-brace joint increased to 300 MPa, due to the reduced material at crucial connection points. A topological analysis revealed potential for further mass reduction, particularly in the mid-to-upper sections of the brace, shown in Figure 37.

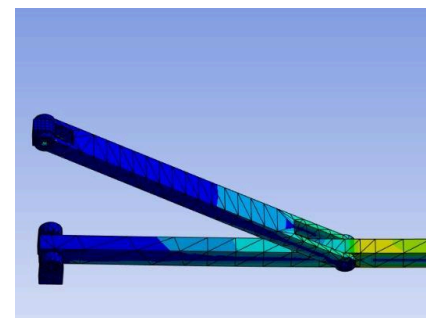
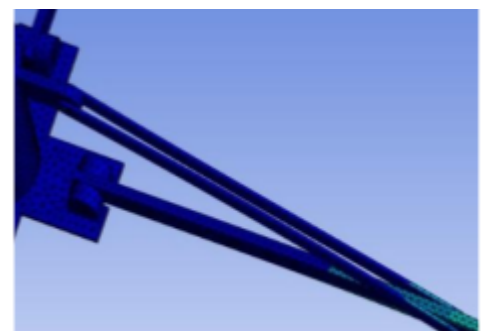


Figure 36 – This schematic shows the stress distribution at the brace region.



The second iteration introduced a hollow central span with reinforced ends to redistribute material more effectively, using the optimised parameters from Section 4.4. By relocating mass towards the brace tips, which encountered higher loads, this design reduced stress levels to approximately 150 MPa while achieving significant weight savings. This iteration ensured a more efficient and balanced stress distribution throughout the brace. Figure 38 shows that in the



final iteration, the brace was extended to accommodate the longer arm design from the latest arm iteration. This elongation improved load distribution across the arm-brace connection points, promoting smoother force transfer and ensuring compatibility with the overall assembly. Topology optimisation indicated minimal opportunities for further enhancement, thereby validating the brace as lightweight, reliable, and structurally sound.

Base and Arm Supports

The original base and arm support employed a thick, box-like structure prioritising structural robustness. While this approach ensured minimal stress levels, FEA results indicated that stresses were extremely low, well below the design threshold, suggesting an opportunity to optimise material utilisation without sacrificing strength. The first iteration improved material efficiency by reducing wall thickness and joint diameters while maintaining structural performance. Although there was a slight increase in joint stress, it remained within acceptable limits. Topology optimisation highlighted further material savings through a more efficient geometric configuration.

Figure 39 shows that in the second iteration, the base was redesigned in a '+' shape, improving stress distribution and enabling better integration with the arm supports. Average stress levels were around 180 MPa, with no localised failures observed. Topological analysis confirmed that this configuration was nearly optimal for achieving mass efficiency and structural reliability.

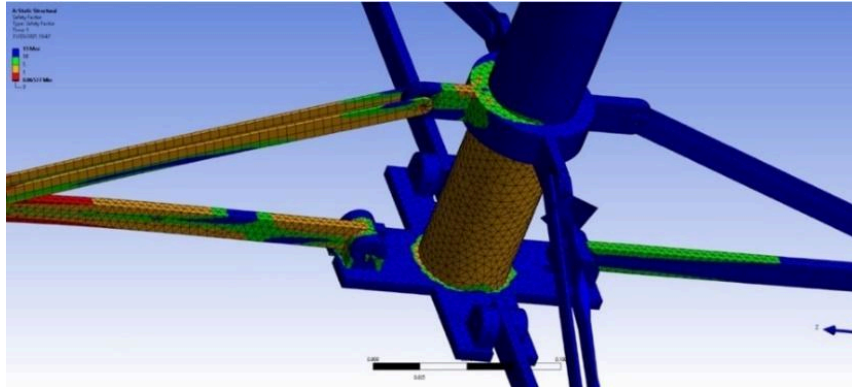


Figure 39 – Safety factor distribution focused on the base section.

Slider

The initial design of the slider was characterised by an oversized and bulky configuration, resulting in uneven stress concentrations around the connection holes. Although the component maintained a high overall safety factor, the stress distribution was inefficient, prompting the need for optimisation.

The first iteration focused on reducing the diameter of the joint holes, leading to a significant decrease in localised stresses around the connections without compromising overall stiffness. Additionally, minor reductions in low-stress regions were made to enhance weight efficiency.

The second iteration refined the slider by streamlining its overall profile and removing material from non-load-bearing areas. These modifications lowered the maximum stress to approximately 190 MPa while maintaining consistent safety factors and deflection within acceptable limits. The optimised geometry of the slider achieved a balance between weight savings and functional performance, displayed in Figure 40.

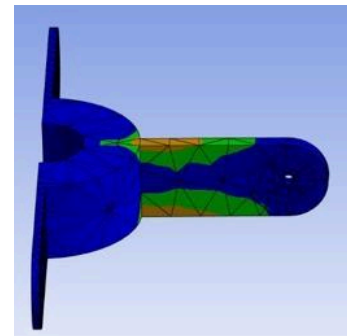


Figure 40 – Safety factor distribution of slider.

Central Column

The initial design of the central column featured a solid, thick, cylindrical shape that prioritised stiffness and negligible deformation under load. However, this rigidity resulted in excessive weight, as stress levels remained significantly below the material's yield threshold, revealing areas for optimisation.

During the first iteration, the external dimensions of the column were reduced while preserving structural safety margins. This approach achieved initial weight savings and opened the door for further optimisation by removing internal material.

The second iteration, shown in Figure 39, introduced a hollow core design, due to the optimisation in Section 4.5, which significantly reduced weight while maintaining uniform stress distribution under operational conditions. Stress levels peaked at approximately 90 MPa, remaining well within safety limits. This hollow design represented an effective balance between load distribution and material savings, substantially enhancing the agility and reliability of the capture mechanism.

Summary

To conclude, all components meet the objectives set, the safety factor is equal or greater than 1.5 for all sections, which means no plastic deformation occurs theoretically under the worst-case scenario (500 N). Furthermore, the deflection for the arm is lower than the limit set at 10 cm, and all other components have a deflection value much lower. Through the aid of topological optimisation, the total mass of the capture mechanism is 2 kg, which is 80 % lower than the maximum payload of the carrier drone, and 75 % lower than the payload objective set (8 kg).

5. Modelling and Validation

5.1 Flight Validation – Simulink

5.1.1 Introduction

This section of the report details the development and application of a Simulink flight model, built specifically to replicate the dynamic performance of the SG 15L FPV drone used in the project. The primary purpose of the flight model is to accurately simulate the drone's flight characteristics, enabling predictions and evaluations of how the addition of the designed capture mechanism, introducing extra mass and altered inertia, will affect overall flight performance. By assessing factors such as roll, pitch, and yaw stability thresholds and identifying potential points of flight failure, the flight model provides essential insights into the viability and limitations of the drone-capture system under realistic operational conditions.

5.1.2 Flight Model Development

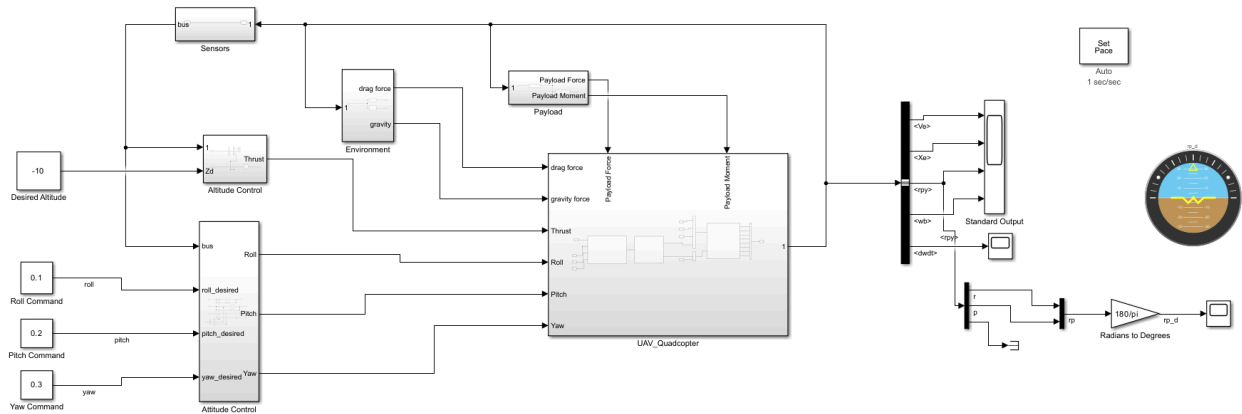


Figure 41 – Complete flight model architecture integrating attitude and altitude control systems, environmental effects (gravity and drag), and payload forces. Real-time feedback from sensors informs the UAV dynamics block, with outputs monitored via scopes and the artificial horizon instrument. [Simulink]

The main airframe flight model was developed in Simulink (MathWorks, 2023), guided primarily by a comprehensive MATLAB tutorial available on MATLAB Central (Thorburn, 2025). This tutorial provided foundational guidance in structuring key components such as the drone's parameters, propulsion dynamics, control systems, and sensor integration. Some additional advanced details (e.g. modelling aerodynamic drag) were informed by the methods illustrated in MathWorks' example of a Quadcopter model for a Parrot Minidrone (MathWorks, 2023). To ensure the flight model accurately represented the project-specific SG 15L FPV drone (SG UAV, 2024), personalised features were introduced, including customised input commands consistent with FPV drone operations, PID control, feed-forward features and detailed payload modelling. Together, these combined resources and customised additions enabled the creation of a highly representative flight simulation environment to predict drone performance and stability under payload conditions.

UAV Parameters

The SG 15L FPV drone was modelled as a quadcopter with specified dimensions (647 mm wheelbase, 517 mm x 517 mm x 80 mm body), mass (2.75 kg), and a detailed inertia profile derived from its datasheet (SG UAV, 2024) (see Appendix A.1). Carbon fibre (3K T300 carbon fibre) construction was assumed for mass distribution, accounting for the drone's geometry and structural layout.

Propulsion & Motor Mixing Model

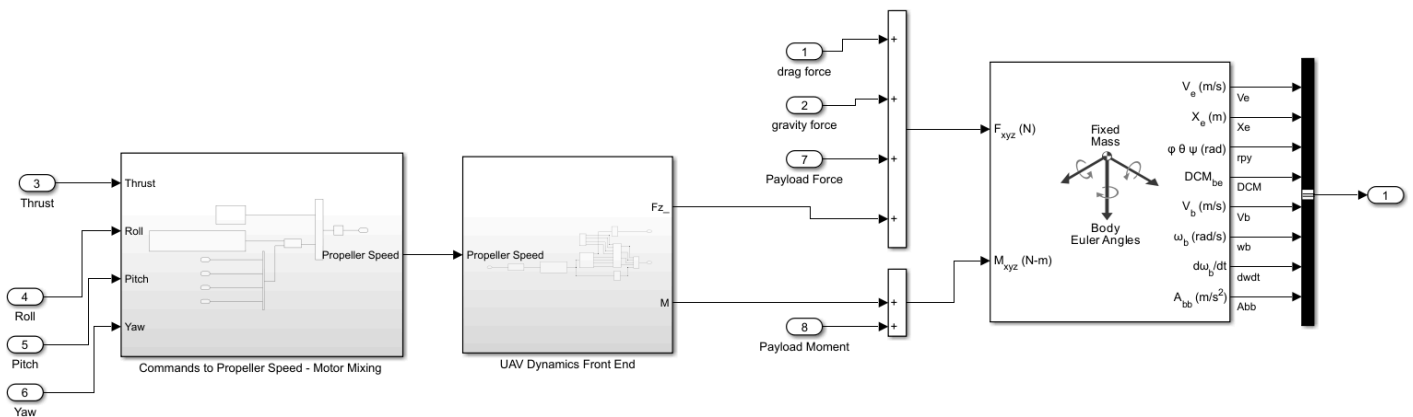


Figure 42 – UAV Quadcopter subsystem, illustrating the conversion of input commands into individual propeller speeds via motor mixing. The UAV dynamics front end computes resulting forces and moments, which are combined with environmental and payload effects before being fed into the flight dynamics model to generate key state outputs. [Simulink]

Propulsion dynamics were modelled based on experimentally derived motor constants for thrust and torque, obtained through controlled flight tests (detailed in Section 7.1 of the wider report). The thrust which the drone experiences is $F = \sum_{n=1}^4 F_{z,n}$ and the torque produced by the drone is $M = \sum_{n=1}^4 M_{z,n} + \sum_{n=1}^4 r_n \times F_n$. The thrust created by a single propeller in the z-direction (body frame of reference) is $F_z = k_t \omega^2$. The torque that is created by the motor and a single propeller spinning in the z-direction is $M_z = k_m \omega^2$ (Förster, 2015) (Gitbook, n.d.). Here, k_t [7.3070×10^{-5}] and k_m [2.1783×10^{-6}] represent thrust and torque coefficients respectively, r is the displacement of the n th propeller from the drone's centre of mass and ω is the angular velocity of each propeller. The coefficients are calculated experimentally in Section 7.1.

The drone employs a standard quadcopter motor-mixing matrix to translate pilot inputs: roll, pitch, yaw, and thrust into specific propeller speeds. Thrust is inputted in the form of desired altitude as its response is easier to track. Thrust is always input as a negative value because z corresponds to a direction downwards from the drone's centre of gravity. Input commands reflect those from an FPV drone controller, incorporating a feed-forward weight counter to ensure hover stability without continuous pilot input. Each propeller has an individual speed limiter matching the drone's physical constraints [4947 rpm]. This is also to avoid unrealistic output speeds. Forces and moments generated by propellers were calculated considering their spatial arrangement around the drone's centre of gravity, utilising their coordinates (using Cartesian coordinate system in Lagrangian frame of reference from the centre of gravity of the drone): [± 276.54 mm, ± 276.54 mm, -51 mm] from the drone's centre. Summation of these local forces and moments provided resultant dynamics that fed into the drone's 6-degree-of-freedom (6DOF) equations of motion.

Flight Dynamics and Environmental Factors

The dynamics of the flight validation system considers key environmental forces: gravitational force and aerodynamic drag, alongside payload influences from the capture mechanism.

Gravity Modelling: Gravity is modelled as a vertical force directed towards Earth, computed using standard gravitational acceleration. This force is continuously referenced to the drone's body frame using a Direction Cosine Matrix (DCM), aligning gravitational force according to the drone's orientation and position coordinates. The weight equation used is: $F_m = mg$ where m [2.3 kg] is the mass of the drone and g is the acceleration of gravity [9.81 ms^{-2}].

Aerodynamic Drag: Aerodynamic drag is modelled to realistically represent resistance against the drone's motion. Drag force is directly opposite to the drone's velocity vector across all three degrees of freedom (x, y, z). The drag force equation used is: $F_D = \frac{1}{2} \rho u^2 A C_d$ where ρ is the density of air [1.225 kg/m^3], u is drone velocity, A is the cross-sectional area perpendicular to the flight direction [$A_x = 5581.195 \text{ mm}^2$, $A_y = 8084.836 \text{ mm}^2$, $A_z = 22258.893 \text{ mm}^2$] (computationally derived in Section 7.1) and C_d is the experimentally derived coefficient of drag [37.4] (see Section 7.1). Each directional drag component (x, y, z) is calculated separately, corresponding to the projected area of the drone in each axis based on its current velocity vector.

Payload Effects: The inclusion of the capture mechanism introduces additional forces and moments to the drone system. Payload weight is treated analogously to drone mass by multiplying payload mass by gravitational acceleration, providing an additional vertical force acting downward.

A significant factor considered in payload modelling is the induced moment due to off-centre loading. This moment is determined using the cross-product of the payload's gravitational force and its offset position relative to the drone's centre of mass, represented by: $M_{\text{payload}} = r_{\text{payload}} \times F_{\text{payload}}$ where r_{payload} is the positional offset from the drone's centre [0, 0, 0.24 m] and F_{payload} is the gravitational force acting on the payload. This model effectively captures the influence of payload-induced changes to the drone's flight stability and performance, crucial for accurately predicting drone handling characteristics following the addition of the capture mechanism.

5.1.3 Control Algorithm and Stability Monitoring

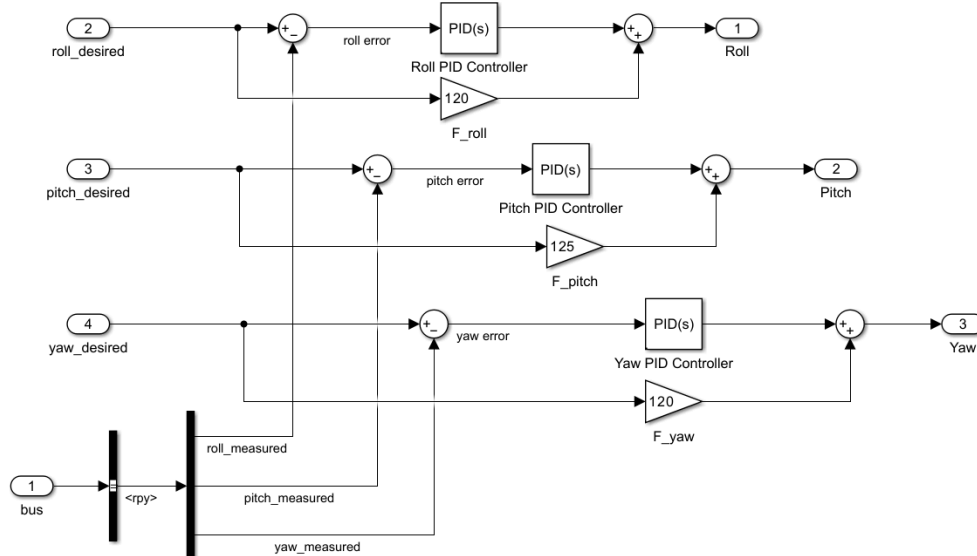


Figure 43 – Attitude Control subsystem. This block regulates the drone's roll, pitch, and yaw using three independent PID controllers, each receiving error signals between the desired and measured angles. Feed-forward gains are incorporated to enhance response sharpness, ensuring that control outputs closely track pilot stick inputs. [Simulink]

PID controllers were implemented for precise control of the drone's attitude and altitude. For attitude control, individual PID controllers managed the roll [$P_{\text{roll}} = 42$, $I_{\text{roll}} = 23$, $D_{\text{roll}} = 48$, $FF_{\text{roll}} = 120$], pitch [$P_{\text{pitch}} = 44$, $I_{\text{pitch}} = 25$, $D_{\text{pitch}} = 53$, $FF_{\text{pitch}} = 125$], and yaw [$P_{\text{yaw}} = 42$, $I_{\text{yaw}} = 28$, $D_{\text{yaw}} = 48$, $FF_{\text{yaw}} = 125$] axes by employing negative feedback loops. Real-time drone orientation data (roll, pitch, yaw) was continuously compared with the desired orientation setpoints corresponding to input commands from the FPV controller. PID gain values and filter coefficients for each of these attitude controllers were extracted directly from the drone's blackbox data, ensuring the Simulink model accurately replicated the real drone's response characteristics. A feed-forward (FF) gain term was also added to each PID channel, providing an immediate, latency-free component that mirrors the pilot's stick inputs and gives the simulated attitude response the same crisp feel observed on the real quad.

The altitude control subsystem also employed a PID controller, managing the vertical position of the drone. Due to the absence of recorded PID parameters in the drone's blackbox for altitude, the PID controller parameters were tuned using the Transfer Function-Based PID Tuner within Simulink. This approach facilitated automatic tuning, optimising altitude response to input commands. All tuning was incorporated before the payload effects were added.

Stability and performance during simulation runs were monitored using a real-time scope, visualising essential flight data, including the drone's position, velocity, angular velocity, angular acceleration, and roll/pitch/yaw angles. Additionally, the Aerospace Blockset's artificial horizon instrument provided an intuitive, real-time visualisation of the drone's stability, clearly indicating deviations from level flight. These monitoring tools ensured the drone's flight model accurately reflected stability and dynamic performance changes, particularly following the addition of the payload capture mechanism.

5.1.4 Experimental Validation

The motor constants (k_t and k_m) and the drag coefficient (C_d), essential parameters for accurately modelling the drone's performance and environmental interactions could not be extracted directly from the drone's flight code dump. Therefore, experimental testing was necessary to determine these parameters empirically. The specific experimental methodologies, testing procedures, and corresponding results are comprehensively detailed in Section 7.1.

5.1.5 Flight Capabilities and Outputs

The developed Simulink flight validation model enables comprehensive analysis and predictive capabilities crucial for safe and efficient drone operation. Specifically, the model provides the following key analytical insights:

Flight Stability Limits: By running simulations across varying roll, pitch, and yaw setpoints, the model determines the maximum allowable orientation angles that the drone can sustain without losing control. Exceeding these thresholds in simulation clearly highlights points at which instability or control loss may occur.

Payload Impact Assessment: The model accurately simulates drone responses to payload additions, predicting how increased mass affects handling, manoeuvrability, and overall stability. Through iterative simulations, critical payload limits can be identified beyond which the drone experiences significant degradation in control effectiveness, potentially leading to unstable flight conditions or crashes.

Stability Threshold Conditions: The model aids in identifying precise threshold conditions under which drone stability is compromised. It achieves this by analysing deviations in drone outputs from the commanded inputs, flagging scenarios where the outputs significantly diverge from desired flight paths or setpoints. This capability is instrumental in pre-emptively determining and avoiding conditions that could lead to flight failures or loss of the drone.

Overall, the Simulink flight model serves as a powerful tool for pre-flight validation and risk assessment, substantially reducing the likelihood of encountering unexpected flight behaviour during practical missions.

5.1.6 Discussion & Implications

The flight validation model developed showcases significant strengths, most notably, its versatility and accuracy in dynamic modelling of the drone's flight characteristics. Through precise tuning and validation against real-time data extracted from the drone's flight controller, the model effectively replicates essential flight behaviours and stability conditions. This allows thorough prediction and assessment of drone stability limits, payload handling effects, and potential flight failure scenarios.

However, the model presents certain limitations due to necessary simplifications and practical constraints. Not every detailed feature available in the drone's Betaflight flight code was included, primarily to avoid excessive complexity and computational demand. Specific features omitted include D-term dynamic reduction, gyro low-pass filtering, anti-gravity bumps, throttle-related shaping mechanisms, and potentially other nuanced control algorithms not explicitly listed in the provided Betaflight data dump.

Additionally, certain significant real-world factors were inherently challenging to incorporate. For example, modelling turbulent wind conditions, which significantly affect drone performance through increased and variable drag, was beyond the scope of this simulation. The inertia of payloads was another aspect difficult to accurately model, primarily due to uncertainty regarding the precise size, shape, and mass distribution of the payload capture mechanism.

These limitations suggest that while the developed model is suitable for predicting fundamental drone dynamics and response to control inputs, careful interpretation of the results is necessary when translating findings to practical scenarios. Real-world drone

behaviour under severe environmental disturbances or with dynamically changing payload characteristics may differ significantly from model predictions. Nevertheless, the Simulink model provides a robust foundational tool for preliminary assessment and informed decision-making in drone design and control strategy development, ensuring significant value despite its noted simplifications.

5.1.7 Summary

The developed Simulink flight model proved essential for predicting drone performance changes following the integration of the capture mechanism. By enabling detailed analysis of stability, control response, and payload impact, the model provided a reliable platform for anticipating and addressing flight behaviour deviations prior to physical deployment. The full developed flight model can be explored using the GitHub link in Appendix A.6.

5.2 Manoeuvre Limitations

5.2.1 Objective

The objective of this analysis is to determine the effect of the payload (capture mechanism) on the drone's ability to perform basic flight manoeuvres, specifically changes in altitude, roll, pitch, and yaw commands. This is evaluated using a comparative study between the flight responses of the drone alone versus the drone carrying the payload, through simulation using the 5.1 Flight Validation Model - Simulink. This analysis will inform whether the designed capture mechanism is light enough to be safely carried by the drone in a real flight, whether the inertia of the capture mechanism will make the drone unstable and if any control algorithms will need to be tuned to ensure that the capture mechanism can be flown as desired.

5.2.2 Simulation Setup

Simulations were run with the nonlinear 6-DOF model for two representative command sets.

- **Set A (moderate):** 10 m altitude, roll 0.1 rad, pitch 0.2 rad, yaw 0.3 rad
- **Set B (aggressive):** 10 m altitude, roll 0.2 rad, pitch 0.4 rad, yaw 0.6 rad

Each set was executed twice: first with the bare airframe/drone ($m_{\text{drone}} = 2.3 \text{ kg}$) and then with the capture-mechanism payload ($m_{\text{payload}} = 2.53 \text{ kg}$). A further altitude-only sweep to 50 m was added to test vertical authority.

The comparison of these scenarios allows for clear identification of the drone's performance constraints and the proportional command adjustments necessary when operating with the payload.

5.2.3 Results

The comparative analysis is primarily focused on drone position (X_e) and attitude (rpy), alongside secondary comparisons for velocity (V_e) and angular velocity (ω_b). The results are summarised using graphical comparisons as shown in Appendix A.7.1-A.7.2:

Under Set A, the unloaded vehicle reached the 10 m target in $\approx 15 \text{ s}$ with a peak undershoot of -1 m and a lightly damped single oscillation in the vertical velocity trace. Roll, pitch and yaw converged to their demands within 12 s, with maximal overshoots of 0.2 rad, 0.35 rad and 0.5 rad, respectively; angular-rate peaks stayed below 0.25 rad/s. When the payload was attached, the same command also settled after $\approx 15 \text{ s}$. Initial altitude error deepened to -5 m ($\approx 25 \%$ larger), the first two oscillations were clearly under-damped, and roll/pitch/yaw overshoots were the same.

For the 50 m climb, the bare airframe displayed a smooth, critically damped rise, settling in $\approx 15 \text{ s}$ again with $< 5 \%$ overshoot. With the payload, the vertical loop required $\approx 25 \text{ s}$ to settle and exhibited an 8 m error with similar overshoot. Lateral attitude channels remained stable but showed the same factor-of-two increase in peak magnitudes noted in Set A.

The more demanding Set B accentuated the trend. Without the payload, peak altitude deviation was 3 m (30 %) and the vehicle settled in $\approx 12 \text{ s}$; attitude overshoots boosted to around 250 % of the large, commanded values, and angular-rate excursions remained under 0.3 rad/s. Once loaded, the altitude error deepened to 80 %, and pitch overshoot reached 1.1 rad, which is almost 90 % above the 0.6 rad demand. Roll and yaw also displayed similar spikes. Angular-rate spikes of 0.5 rad/s appeared during the first two seconds before damping out.

Max roll = 0.6, max pitch = 0.7, max yaw = 1.2.

5.2.4 Discussion – Effect of the Capture Mechanism Payload

The payload doubles the airframe mass ($\approx +100 \%$) while leaving the rotor/ESC combination unchanged, cutting the thrust-to-weight ratio in half. The Simulink runs show that the controller still stabilises the vehicle, but thrust authority is no longer sufficient to reproduce the crisp transients of the unloaded case.

Table 10 – Summary of the impact of payload on altitude and attitude performance across different command sets.

Command set	Largest initial altitude sag (unloaded $\rightarrow$ loaded)	% of command height	Attitude peak increase
A (10 m climb, 0.1/0.2/0.3 rad)	-1 m $\rightarrow$ -5 m	10 % $\rightarrow$ 50 % (+40 p.p.)	none $\rightarrow$ +100 %
A' (50 m climb, 0.1/0.2/0.3 rad)	$\approx 0 \rightarrow$ -8 m	0 % $\rightarrow$ 16 % (+16 p.p.)	none $\rightarrow$ +100 %
B (10 m, 0.2/0.4/0.6 rad)	-3 m $\rightarrow$ -8 m	30 % $\rightarrow$ 80 % (+50 p.p.)	+150 % $\rightarrow$ +250 %

Altitude loop

The altitude sag grows almost linearly with payload weight but is nearly constant in absolute terms ($\approx 5\text{--}8\text{ m}$); consequently, the relative error falls from 50 % at a 10 m demand to 16 % at a 50 m demand ($\Delta h/h \approx 8\text{ m} / h$). Put differently, the loaded craft needs $\sim +20\%$ more thrust (or a 15–25 % higher altitude command) to reach the same height within the original 15 s settling target.

Settling time stretches from 12–15 s to 20–25 s, confirming that the integral term in the altitude PID saturates early and recovers slowly once headroom appears.

Attitude loop

Due to the capture mechanism's mass being centred on the column axis whilst mounted, no static pitching or rolling moment is introduced, so long-term attitude accuracy is retained. Transient peaks, however, are double (roll/pitch) or almost double (yaw), and the first two cycles are markedly under-damped. The payload's 60 % increase in pitch/roll inertia explains the sluggish response; the derivative term in the inner loops is no longer high enough to arrest the rates quickly.

Velocity-state observations

Vertical-speed traces show a 30–40 % greater first-cycle amplitude, and angular-rate spikes rise from $< 0.3\text{ rad/s}$ to $\approx 0.5\text{ rad/s}$ before converging in $< 6\text{ s}$. Although these rates are still well inside the ESC limit, they point to higher motor loading and thermal stress.

5.2.5 Implications & Recommended Tuning

- Altitude gain: increase integral gain $k_{i,z}$ by $\approx 50\%$ or add a feed-forward term proportional to measured mass to recover the original 12-s settling time and keep sag $< 10\%$ of demand
- Attitude gains: raise derivative gains in roll & pitch by $\approx 80\%$ to restore a 5 % overshoot and remove the first under-damped cycle
- Command scaling: for mission planning, budget an extra 2 m of climb per 10 m demanded ($\approx +20\%$) whenever the capture mechanism is armed; equivalently, reserve 25 % extra throttle margin

Implementing these modest changes is expected to return overshoot and settling metrics to within $\pm 10\%$ of the unloaded benchmarks, re-establishing the manoeuvre envelope required for reliable and safe net-deployment flights. Overall, there is no obvious, dangerously unstable flight behaviour displayed via the flight model; thus, the current capture mechanism's design is ready to be mounted to the drone for testing.

5.2.6 Stability Threshold Determination (Roll, Pitch, Yaw)

Further iterative simulations were conducted to identify the maximum allowable roll, pitch, and yaw commands before instability occurs for the drone attached with the payload. These tests systematically increased each attitude command in increments of 0.1 rad, while the other two commands were held constant (roll tested at pitch = 0.2 rad, yaw = 0.3 rad; pitch tested at roll = 0.1 rad, yaw = 0.3 rad; yaw tested at roll = 0.1 rad, pitch = 0.2 rad). Altitude was kept constant at 10 m. Instability was classified as clear signs of uncontrollable oscillations in the simulated attitude response (rpy plots of unstable behaviour once limits have been reached are shown in Appendix A.7.3), characterised by exponential growth and divergence from the desired commands. Unstable flight will lead to the drone-interceptor crashing and becoming destroyed.

The tests revealed the following stability limits:

- Roll: 0.6 rad
- Pitch: 0.7 rad
- Yaw: 1.2 rad

These findings define the operational limits for manoeuvring the drone safely with the payload attached, providing critical thresholds for practical flight operations and ensuring safe limits for capturing drone interception missions. Any input commands whilst flying the complete drone must not exceed the stability limits above. There are no thrust (altitude) thresholds as the propeller speed limiters limit the maximum allowable propeller speeds which would make the drone inherently unstable.

5.3 Mechanism Validation – Gazebo

5.3.1 Hardware and Operating Environment

The simulation framework for this project was developed within a virtualised setup using Parallels Desktop for Mac. The virtual machine runs Ubuntu 22.04 LTS on an ARM64 architecture, which introduced certain compatibility challenges, particularly with prebuilt packages for simulation and robotic control. To address this, key components such as Gazebo Classic 11.15.1 and its associated ROS 2 integration packages (`gazebo_ros_pkgs`) were built from source. The simulation stack is based on the ROS 2 Humble distribution. A dedicated ROS 2 workspace was created to house the required packages, including custom ones for the drone model. The environment was tailored to support real-time simulation and control of a drone system within Gazebo, with further technical details on package configuration, plugin integration, and workspace structure discussed in the next section. This setup allows the drone to be spawned and controlled in a physics-based environment with full access to Gazebo's simulation capabilities. It also enables integration with Simulink-based PID control logic, bridging model-based design with real-time robotic behaviour.

5.3.2 ROS 2 Workspace and Build Process

The simulation was structured around a custom ROS 2 workspace using the colcon build system. This workspace was organised under the typical `src/`, `build/`, and `install/` hierarchy, containing three core packages: `drone_description`, `drone_bringup`, `thrust_plugin`, and `joint_effort_plugin`. Each package served a specific role in the simulation pipeline and was designed for modular development, enabling isolated testing and incremental integration. The `drone_description` package holds the full URDF/Xacro model of the system, including all physical properties, joint definitions, and visual meshes. The configuration files defining joint controllers and PID gains were also kept in this package under `config/`, ensuring a self-contained structure. The URDF was constructed using a macro-based approach to reduce duplication and manage naming consistency, which became essential when resolving joint connection errors in Gazebo. The `drone_bringup` package provides the launch infrastructure and runtime scripts. It includes ROS 2 launch files for spawning the drone in Gazebo (`control_and_spawn.launch.py`), initialising joint positions (`init_joint_positions.py`), launching the visualisation in RViz2, and running the PID control node. The initial joint positions script was critical for resetting the mechanism into a neutral configuration on simulation startup and was carefully sequenced to avoid race conditions with the spawn process. The `thrust_plugin` package contains a custom C++ plugin, `ThrustPlugin.cpp`, compiled into `libthrust_plugin.so`. This plugin bypasses `ros2_control` and instead listens directly to ROS topics for each rotor's thrust command. It applies both vertical thrust and torque to the rotor links, based on predefined coefficients of thrust and torque. This was essential for simulating aerial dynamics with sufficient flexibility and removing the need for complicated controller setup which often introduced compatibility issues in an ARM-based environment.

Several challenges were encountered during the build and integration process, particularly involving shared library paths and plugin loading. To ensure proper runtime linking of all compiled `.so` files, environment variables such as `GAZEBO_PLUGIN_PATH` and `LD_LIBRARY_PATH` were configured in `.bashrc`. These were critical for Gazebo to detect and load both standard and custom plugins, such as the thrust and IMU sensor plugins. Additionally, issues related to duplicate joint names, arising from macro expansion, were resolved by carefully prefixing joint and link names and introducing intermediary dummy links to isolate repeated structures. All source code, plugin implementations, controller settings, and launch instructions are maintained in a dedicated GitHub repository whose link can be found in Appendix A.8. The repository includes documentation and setup scripts to allow reproducible builds, and local simulation runs on other machines using similar environments.

5.3.3 Mesh Preparation and URDF

The mechanical structure of the drone and its attached umbrella-style capture mechanism was designed in FUSION360 and prepared for simulation in Blender. The goal was to produce a lightweight yet accurate representation suitable for real-time physics simulation. All components were exported from CAD models in STL format, then imported into Blender for scaling, axis correction, and conversion into DAE (Collada) format, which is required for use with Gazebo and ROS 2 URDF models. To ensure physical plausibility, the geometry was simplified where appropriate without compromising essential shape or dimensions. Origin points were aligned to facilitate correct joint articulation within the robot description. Each part was also assigned its correct material and volume, allowing the estimation of mass and inertia properties. The 3D Print Toolbox add-on in Blender was used to calculate volumes, which, when combined with the known aluminium density, enabled the calculation of approximate inertias. For further refinement and verification, the inertial tensors were validated using MeshLab, and the final values were integrated directly into the `<inertial>` blocks within the URDF. The model's mechanical structure includes fixed, prismatic, and revolute joints. A macro-based Xacro approach was adopted to define repeated structures like the four-bar linkage system, reducing code duplication while avoiding joint naming conflicts. Dummy intermediary links were introduced where needed to isolate scopes and resolve duplication issues encountered during simulation in Gazebo.

5.3.4 Simulation Parameters

All dynamic parameters used in the simulation were selected to reflect the physical behaviour of the system modelled earlier in Simulink. The full set of inertial properties, including masses and tensors defined in the URDF, were interpreted by Gazebo's physics engine to simulate flight dynamics under the influence of gravity, rotor-generated thrust, and interaction forces within the mechanism. Aerodynamic actuation was handled via a custom thrust plugin. Each rotor was assigned its own topic for force commands and configured with a direction of rotation (clockwise or counterclockwise) and specific aerodynamic coefficients: thrust coefficient $k_t = 7.3070 \times 10^{-5}$ and torque coefficient $k_m = 2.1783 \times 10^{-6}$ (derived in Section 7.1). Additionally, damping and linear drag effects were approximated via URDF-defined joint dynamics and rotor-level damping. While a precise drag coefficient (C_d) was not assigned at the body level, the current model allows future extension through Gazebo's aerodynamic plugins or custom drag components if more accurate airflow interaction modelling is required. These parameters were defined in the URDF `<plugin>` block and allowed the plugin to apply force and torque directly to the rotor links in real time. Joint-level constraints, including limits, damping, and friction, were configured to reflect realistic mechanical boundaries, for such restricting the slider to its operating range and preventing overextension of the brace or arm. The control loop was designed to run at 100 Hz, aligned with the sampling rate used in Simulink during PID tuning.

An IMU sensor was attached to `base_link` and configured to publish angular velocity and acceleration data to the `/imu_plugin/out` topic, enabling closed-loop control and future use in stability monitoring. In addition to rotor thrust simulation, the mechanism actuation was handled using a custom C++ plugin (`JointEffortPlugin`). This plugin subscribed to the topic `/column_to_slider/effort_command` and directly applied joint effort to the slider, which is connected to the vertical column via a prismatic joint. This approach removed the need for `ros2_control` and enabled smoother and more direct control over the capture mechanism's actuation.

To enable thrust-based control in Gazebo, each rotor required a dedicated physical link and joint within the URDF/Xacro model. Gazebo's physics engine mandates that any force or torque must be applied to a defined link with mass and inertia. As a result, simple circular-shaped links were added above each motor as rotor instances. These are visually represented as white cylindrical discs in the simulation. Each disc was instantiated via a reusable macro (`rotor_link`) that defined a small inertial component (0.01 kg), a cylindrical geometry for basic collision and visual reference, and a continuous joint connecting it to the drone base. These joints are not dynamically rotated but serve as anchors for applying vertical thrust and yaw torque through the custom ThrustPlugin. Each rotor is addressed individually through topics such as `/rotor1/thrust`, allowing the plugin to deliver force commands in real time. These placeholder discs are structural necessities rather than visual representations of propellers and could be substituted or visually suppressed in future iterations if required.

Together, these simulation parameters provided the foundation for replicating real-world behaviour in scenarios such as flight with shifting inertia or mid-air mechanism deployment, allowing analysis of controller performance under non-ideal and dynamic conditions.

5.3.5 System Architecture

The system architecture for the drone simulation is composed of several interacting ROS 2 nodes and Gazebo plugins, each responsible for a specific layer of functionality: physical modelling, control, visualisation, and actuation. The structure reflects a separation of concerns between description, deployment, control logic, and visualisation, enabling modular testing and clear traceability of the simulation's behaviour.

At the core of the architecture is the URDF/Xacro model defined in the `drone_description` package. This model encodes all links and joints of the drone and its attached capture mechanism, including the inertial and visual properties needed by Gazebo to simulate rigid-body dynamics. The system includes a hierarchical assembly of components with nested joints representing the base, column, slider, four-bar linkages, and rotors. Fixed, prismatic, and revolute joints are used in combination to simulate realistic mechanical behaviour, especially for the deployable arms driven by the vertically sliding ring. The control layer is implemented through a Python-based PID node (`pid_control_node.py`), which interfaces with the custom thrust plugin via ROS topics. Each rotor is addressed individually through a topic (e.g. `/rotor1/thrust`, `/rotor2/thrust`), allowing the PID node to adjust vertical thrust and torque dynamically. This arrangement replaces the default `ros2_control` mechanism, offering a more direct and lightweight approach that avoids known compatibility issues on ARM64 systems. The deployment and initialisation logic is managed by the `drone_bringup` package. It includes a launch file that sequentially brings up all required components: the Gazebo server, the spawn entity service, the joint initialisation script (`init_joint_positions.py`), and the PID controller. The joint initialisation node ensures that the net mechanism starts in a known configuration and avoids physical instability during the early simulation steps.

Mechanism control was implemented through a GUI node (`mechanism_gui.py`) that published toggle commands to a dedicated ROS 2 service (`/toggle_mechanism_service`). This service triggered direct effort commands to the prismatic slider joint via the `JointEffortPlugin`, allowing actuation of the net mechanism. The system architecture supported both manual deployment (via GUI buttons) and scripted actuation through command-line ROS 2 service calls.

Visualisation can also be provided through RViz2, which allows monitoring of link positions, coordinate frames, and joint states. The robot description is published via `robot_state_publisher`, and all transforms are made available through the `/tf` and `/tf_static` topics. This setup is essential for observing changes in the drone's attitude, centre of mass, and joint articulations during mid-air deployment. This architecture allows coordinated interaction between physical simulation, control logic, and observation tools. It is designed to be extensible, with clearly defined communication interfaces between nodes, and supports both manual actuation and automated control testing.

5.3.6 Visual Staging and Plugin-Based Mechanism Actuation

A targeted simulation scenario was developed to validate the dynamic behaviour, control stability, and mechanical reliability of the quadrotor drone system equipped with the umbrella-style net capture mechanism. All simulations were conducted in Gazebo Classic, with RViz2 used for transforming and joint state visualisation, and `rqt` tools for control signal analysis.

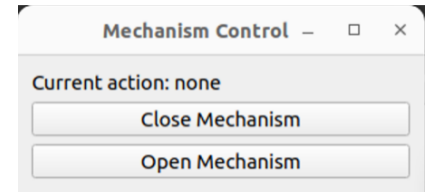
It is important to note that achieving full aerial stability and autonomous hovering in Gazebo represents a particularly demanding challenge, especially in a custom-built simulation environment developed entirely from the ground up. The simulation stack involved the design and integration of multiple custom plugins, including effort-based joint actuation and per-rotor thrust control, all developed manually without reliance on pre-existing templates or controller packages. This level of modularity and depth makes the system highly extensible but also introduces significant complexity in achieving stable flight. Despite rigorous efforts, including the implementation of a functioning ThrustPlugin for rotor force application and a separate JointEffortPlugin for mechanism actuation, the drone has not yet achieved sustained hover within the simulation. Preliminary debugging suggests that this is likely due to unresolved issues in data flow between the control node and thrust topics (such as inconsistent message publishing or lack of actuator feedback on `/rotorX/thrust` topics). Further investigation is required to refine the real-time control loop, assess thrust vector alignment, and ensure correct force application under the influence of simulated inertial parameters. This simulation framework remains an active area of development. The current state of the system demonstrates successful deployment mechanics and plugin-based actuation and provides a strong foundation for future work.

While full aerial control and autonomous hovering remain ongoing challenges, the simulation system has successfully demonstrated robust mechanical behaviour and plugin-based actuation under realistic constraints. In particular, the umbrella-style capture mechanism was fully integrated, tested, and visualised across multiple deployment stages, providing a valuable basis for validating joint-level motion, effort-based actuation, and visual consistency in a physics-based environment. The following results focus on the deployment dynamics of the mechanism, evaluated through a custom plugin-actuated control pipeline. Despite the broader complexity

of real-time thrust balancing and full 6-DOF flight, the capture system itself performed reliably across a range of configurations, as detailed below.

The primary focus was on mechanism deployment under realistic physical constraints, with the system evaluated in three distinct configurations: fully retracted: slider fully raised; arms vertical, aligned with the column; semi-deployed: slider halfway extended; arms at approximately 45° from vertical; and fully deployed: slider fully extended; arms horizontal, achieving the maximum net span.

The deployment sequence was actuated using JointEffortPlugin responsible for applying force directly to the prismatic joint that connects the slider to the central column. This plugin subscribes to the topic `/column_to_slider/effort_command` and interprets incoming float values as joint efforts in Newtons. These values are applied to the joint at each simulation timestep, enabling continuous movement until an effort of 0.0 is received or a joint limit is reached. Actuation was triggered through a ROS 2 service-based control architecture. A GUI node (`mechanism_gui.py`) published toggle requests to a service interface (`/toggle_mechanism_service`). Upon receiving a request, the service issued either a positive or negative effort command (+10 to deploy, -10 to retract) to the plugin's topic. The GUI offered a minimal interface with two user actions: Open Mechanism and Close Mechanism, allowing for manual staging and repeatable tests (Figure 44).



Figures 45-47 illustrate the deployment stages as rendered in Gazebo. The actuation was visually consistent and symmetric across all arms, reflecting the correct behaviour of the four-bar linkage. In the Appendix A.8, there is terminal feedback from the JointEffortPlugin, confirming successful subscription to the effort command topic and real-time response to GUI-triggered actuation commands. The mechanism stayed strictly within defined joint limits across all configurations. No oscillations, unintended collisions, or joint state violations were observed during simulation runs. The prismatic joint's response to effort commands was smooth and predictable under the current damping and timestep settings. For instance, with a sustained effort of +10, the slider reached full extension in approximately 2-3 seconds, after which the effort was set to 0.0 to stabilise the pose. Similarly, retracting the mechanism followed the same behaviour in reverse. The actuation could be interrupted at any stage by issuing a neutral effort command, effectively freezing the system at an intermediate configuration, useful for visual verification and staged validation.

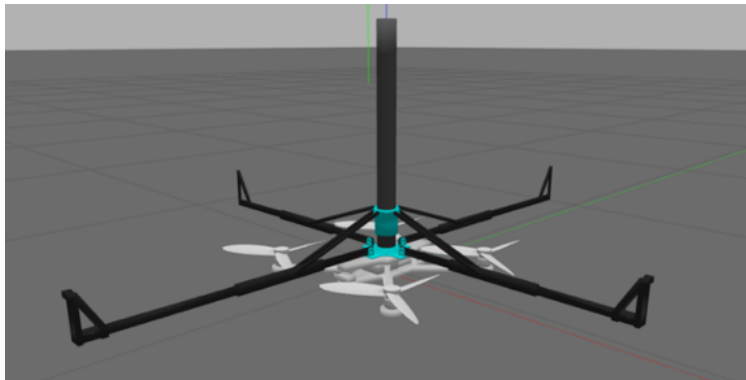


Figure 45 – Fully deployed

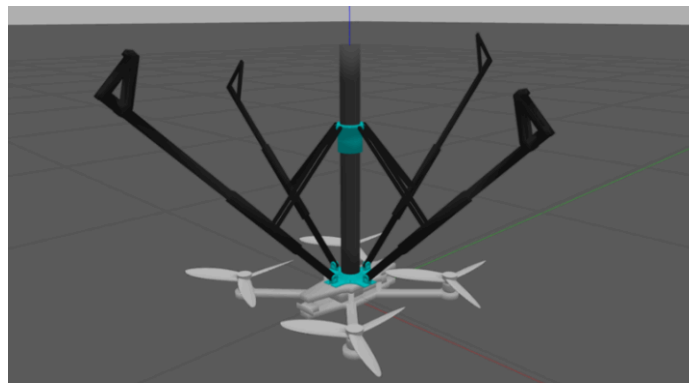


Figure 46 – Semi-deployed

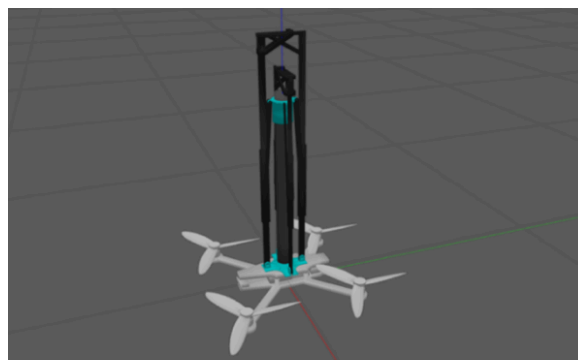


Figure 47 – Fully retracted

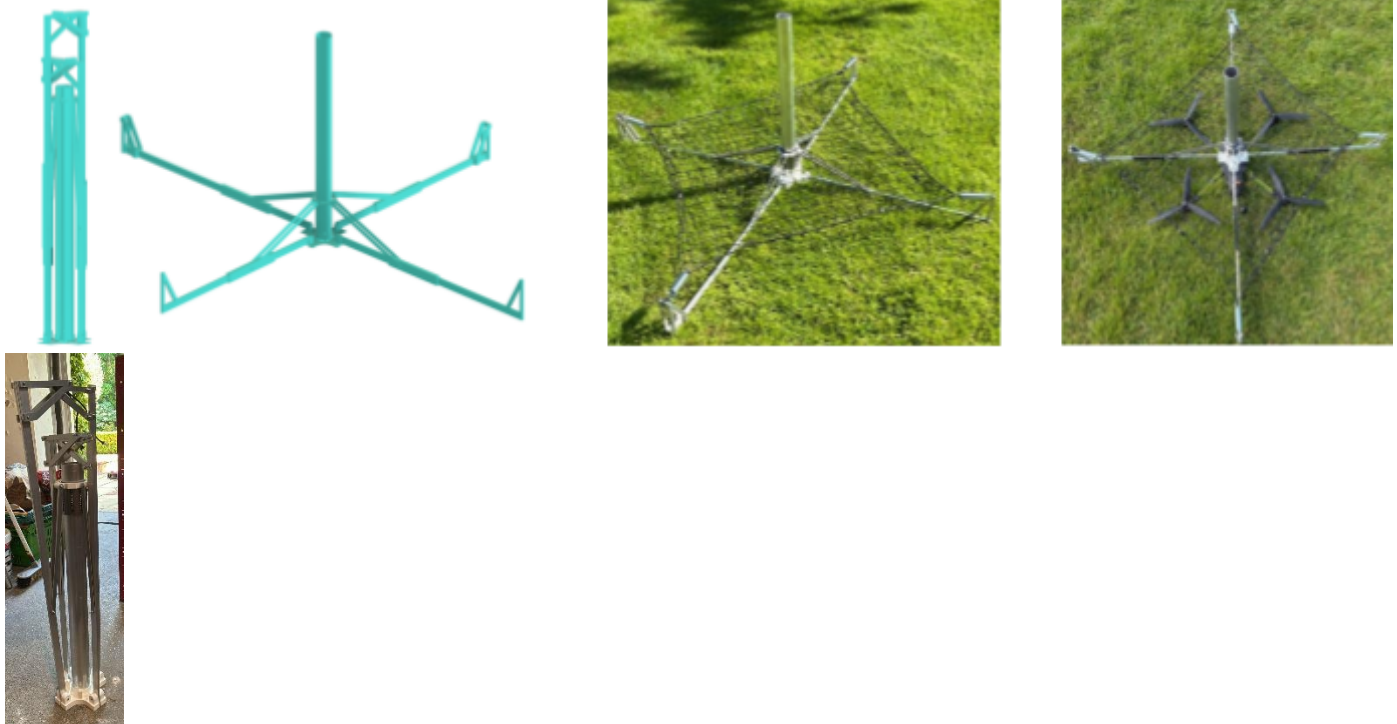
5.3.7 Summary

The simulation work presented here offers a critical step in validating the mechanical feasibility and dynamic integration of the umbrella-style drone capture mechanism under realistic physics-based conditions. Despite the inherent complexity of simulating aerial dynamics in Gazebo Classic, particularly with a fully custom-built plugin architecture and from-scratch URDF/Xacro modelling, the mechanism performed reliably across all tested deployment states. The successful implementation of the JointEffortPlugin and GUI-triggered deployment control establishes a functioning pipeline for repeatable and configurable mechanism actuation. Unresolved thrust control and feedback problems prevented sustained autonomous flight, but the foundation this simulation lays directly supports the more general project goals. It specifically confirms that the deployment system can remain structurally stable, run within specified physical limitations, and react consistently to actuation driven by plugins. This strengthens the feasibility of the suggested concept for securely capturing hostile drones in civilian surroundings and offers a strong basis for next control integration, including closed-loop flight stability and live aerial interception scenarios (Civil Aviation Authority, 2019).

6. Manufacturing & Assembly

6.1 Manufacturing Process

All the constituent parts which make up the capture mechanism were sourced from UK retailers, with all the major components being manufactured in aluminium 6061. The arms were sourced from rectangular aluminium stock, with only the length of the bar being specified to the retailer. This was because the thickness and width dimensions of the readily available solid bars followed the desired dimensions extremely closely, expediting the purchase. Ideally, hollow bars, as determined, would be used instead to minimise mass. The sleeves on the arms were 3D printed and hammered onto the arms due to the necessity of a tight press fit. The base and slider were custom-ordered and manufactured via the use of a CNC mill. CNC milling was chosen due to its ability to manufacture complex shapes out of a singular block of material, limiting the necessity of stress-concentrating fixtures. For the test explored before, the base was a 3D printed prototype. This was due to time constraints, as the aluminium base required further sanding to achieve the desired fit. The endcaps and braces were also custom-ordered and were waterjet cut after providing the .dxf file. Nuts and bolts were sourced from an existing variety kit, with any brackets having to be manufactured manually out of scrap aluminium 6061.



A small chamfer is to be created on some edges for aesthetics. All the holes in the arms, endcaps and brackets were drilled. The slider was pressed onto the bearing, which then slides over the top of the central column and into position. The central column also achieves a tight press fit with the base. A small hole was drilled into the central column, of which a bolt has placed into. This prevents the slider from exceeding a certain travel down the column, stopping the arms from extending below the horizontal.

The pieces were then able to be easily assembled, with nuts and bolts being the only fixtures required to complete the frame of the capture mechanism. A length of paracord was wrapped around the springs attached to the four endcaps to act as a frame for the netting. This was then able to be placed over the central column and tied around the paracord frame with cable ties. The excess of net was then cut off, leaving the final design assembled with a final mass of 2 kg. The entire process took one to two people around 10 labour hours, with the prototype costing around £900, with spare parts bringing the total cost up to £1200.

6.2 Net and Spring

The net selected for this drone capture mechanism is a 2 mm thick, UV-stabilised polyethene net from Net World Sports, with a 50 mm mesh size. This material was chosen after evaluating several options, including Dyneema, Kevlar, and nylon, balancing strength, weight, flexibility, cost, and ease of integration. While Dyneema and Kevlar offer superior strength-to-weight ratios, they were considered unnecessarily expensive and difficult to source or modify. The chosen net, originally designed for drone enclosure systems, is lightweight and durable enough to withstand the forces generated during rotor entanglement, estimated at up to 500-900 N per thread, based on a total impact energy of 450J spread over 5 cm of deformation, with 10-20 net strands absorbing the load. The net area is approximately $0.3 - 0.4 \text{ m}^2$, with a rhombus shape of about 700 by 800 mm, suitable for the capture zone defined by the mechanism. The 50mm mesh size, although larger than some alternatives, was found to be ideal: smaller meshes (10–25 mm) would increase air resistance and hinder deployment speed, while larger meshes might fail to catch rotors. Since the system targets small consumer drones, the mesh allows reliable rotor entrapment without compromising flexibility or actuation speed. This makes the net a practical and effective choice for a safe, mid-air, reusable capture system.

The springs used to support the net were the LE48J 06 M model. This spring was chosen for its K-value, which matches the target stiffness of 10,000 N/m as derived in Section 4.1. Additionally, it has a rated maximum applied force of 502 newtons, ensuring it can withstand the highest modelled stress. Finally, it was selected for its small wire diameter of 5 mm, which is significantly smaller than that of alternative springs with similar properties. This smaller diameter helped minimise the size of the holes in the endcaps, thereby preserving their structural integrity.

7. Testing & Analysis

This section consists of experiments completed. The purpose of the tests is to analyse the performance of the capture mechanism, which includes the performance of the carrier (SG 15L FPV drone) with the capture mechanism attached to it. To support this analysis, a video in Appendix A.9 is provided via a shared online folder. The folder includes footage from key experiments: the full drone test flight, a live interception scenario, and three net deployment cases. Each test is presented from both a drone-mounted camera perspective and an external observer viewpoint. The link is accessible in Appendix A.9 and is intended to supplement the written findings with direct visual evidence.

7.1 Drone Flight Experiment – Drag & Motor Coefficients

7.1.1 Objective

To determine the thrust constant k_t , torque constant k_m and aerodynamic drag coefficient C_d of the SG 15L FPV drone, controlled flight tests using the drone are required as these parameters cannot be extracted otherwise. These experimentally derived parameters will be fed into the Simulink flight model to replace nominal estimates, ensuring the model predicts attitude response, power demand, and payload-induced performance limits with engineering accuracy.

7.1.2 Experimental Set-up

Instrumentation

Table 11 – Summary of instrumentation and onboard sensor configuration used in flight testing.

Component / Sensor	Function in the test campaign	Connection / Location on the airframe
SG 15 L FPV quadcopter (SG UAV, 2024)	Test platform; supplies ESC telemetry, battery voltage and current, and RPM data to the flight controller	N/A
Matek M9N-5883 GNSS + Compass GPS (Unmanned Tech Shop, 2024)	Logs ground-referenced vertical & horizontal velocity and absolute position (GNSS lock ≥ 4 sats)	Mounted on the top of the drone, data streamed to OSD
On-board barometric altimeter	Provides high-rate altitude (ALT) for thrust-constant (k_t , k_m) tests	Integrated on the F405 FC; data streamed to OSD
6-DoF IMU (ICM 42688-P)	Measures rotational attitude: pitch (Pit) channel recorded for drag-coefficient runs	FC-integrated; 8 kHz gyro / 1 kHz filtered log
Betaflight Blackbox recorder (32 MB flash)	Captures motor PWM, D-shot RPM, pack voltage, current and FC sensor buses at 1 kHz	On-board flash memory, downloaded post-flight
Skyzone SKY04X PRO digital FPV goggles w/ OSD overlay (Skyzone FPV, 2023)	Real-time display of GPS speed, altitude, pitch angle, elapsed flight time, propeller speeds and power	5.8 GHz recorded video linked from VTX; eliminates the need for a handheld stopwatch
Ground station laptop (Betaflight Configurator)	Arming/disarming, log retrieval, and live console monitoring	2.4 GHz ELRS link + USB post-flight

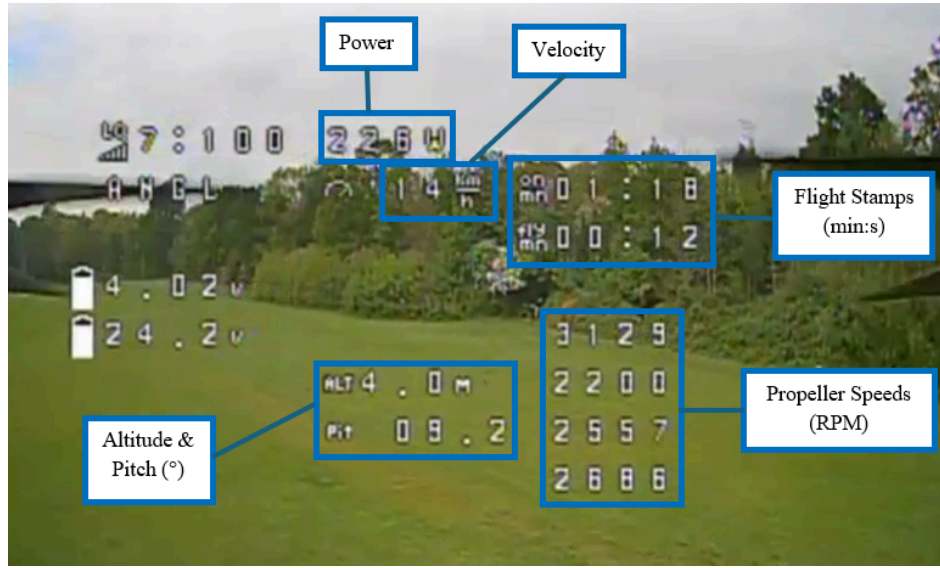


Figure 49 – OSD overlay from FPV goggles showing key flight telemetry.

Test Environment

Experiments were flown in calm weather ($\leq 3 \text{ m s}^{-1}$ wind) without any precipitation to minimise unaccountable environmental effects and to prevent water damage to sensitive electrical components of the drone. All experiments took place on a wide, quiet and unobstructed grass field to remove ground effect from tall objects and to guarantee GNSS lock. Flights were restricted to a minimum altitude of 15 m and a maximum altitude of 50 m (much below the legal restriction of 120 m) to remain in a visual line-of-sight. The drone was always flown a minimum distance of 50 m away from any public persons to ensure the safety of bystanders, and due to legal restrictions (Civil Aviation Authority, 2019).

Recorded Display

All required variables: velocity u (m/s), pitch angle θ ($^\circ$), altitude (m), power P (W), propeller speeds ω (RPM) were overlaid on the goggles' OSD, and concurrently, written to Blackbox flash (see sample frame, Figure 49). Consequently, no external chronometer or tracking system was necessary; flight-time stamps came directly from the drone's timer.

Baseline Configuration

No payload was attached during these characterisation flights. Establishing the airframe's intrinsic coefficients first allows the Simulink flight validation model to add artificial payload masses parametrically and predict their influence.

7.1.3 Experimental Methodology

The experimental methodology adopted the assumption that the drone's mechanical power input P is entirely converted into overcoming aerodynamic drag (F_d) or generating thrust (F_z), expressed through the fundamental power relation: $P = Fu$ where u represents the drone's velocity. For the determination of motor constants (thrust constant k_t and torque constant k_m), vertical flight experiments were performed, carefully maintaining a constant pitch angle of 0° , which was continuously monitored using the drone's OSD. This ensured that measurements exclusively reflected thrust and torque without including power consumption for pitch, roll, or yaw manoeuvres. The motor thrust (F_z) and torque (M_z) were computed from experimental measurements using the power relation:

$F_z = \frac{P - P_{\text{hover}}}{4u}$ (where P_{hover} is the power required for hovering) and $M_z = \frac{P}{4\omega}$. They were then related to the motor constants by the equations introduced in Section 5.1.2: $F_z = k_t \omega^2$, $M_z = k_m \omega^2$.

For the determination of the aerodynamic drag coefficient C_d , two separate flight scenarios, vertical and forward flight, were executed to reflect the variation in drag forces arising from changes in cross-sectional area exposure and wind conditions during different drone manoeuvres. Drag forces were calculated from measured data by: $F_d = \frac{P - P_{\text{hover}}}{u}$. The drag coefficient can then be derived using the aerodynamic drag equation: $C_d = \frac{F_d}{\frac{1}{2}\rho u^2}$.

An average of the vertical and forward drag coefficients was taken for a conservative estimate. All experiments were repeated multiple times to ensure the consistency and reliability of the results, with the drone's onboard sensors and OSD continuously recording necessary parameters. From all the parameters for several different velocities gathered, aerodynamic coefficients and motor constants could be derived by constructing graphs and performing linear regression, where the gradient of the resulting trend lines provided the coefficient of drag (C_d), the motor thrust constant (k_t), and the motor torque constant (k_m).

7.1.4 Assumptions & Limitations

Assumptions

- Still air conditions ($\rho = 1.225 \text{ kg/m}^3$) (Ahmed, 2019) were assumed throughout the testing process, neglecting any external wind influence.
- Net tension and payload mass were considered constant.
- Thrust produced by the drone was assumed to be evenly distributed across all four motors (validated by matching propeller speeds in the OSD).
- The angle of attack (direction of wind flow relative to the drone's orientation) was assumed to match the drone's measured body tilt.
- All electrical power input to the drone was assumed to be converted into mechanical power for producing thrust and performing manoeuvres (i.e., no power loss through video recording or heat).
- The power of the drone is required to hover P_{hover} including any non-manoeuve drone operations is assumed to be constant, at all altitudes, at 50W (verified by flying the drone to a hover experimentally).
- Motor efficiency is unknown, so it was estimated to 1, leading to potential inaccuracies in power consumption calculations.

Limitations

- GPS module accuracy is limited to $\pm 1 \text{ km/h}$, introducing potential variability in measured velocity
- External environmental factors such as wind gusts and ground effects were not controllable, introducing variability and turbulence into the results
- Drone components are incredibly sensitive, so readings may vary between tests
- Human error could influence the drone's ability to maintain a consistent speed during testing

7.1.5 Results & Discussion

Drag Coefficient (C_d) Determination

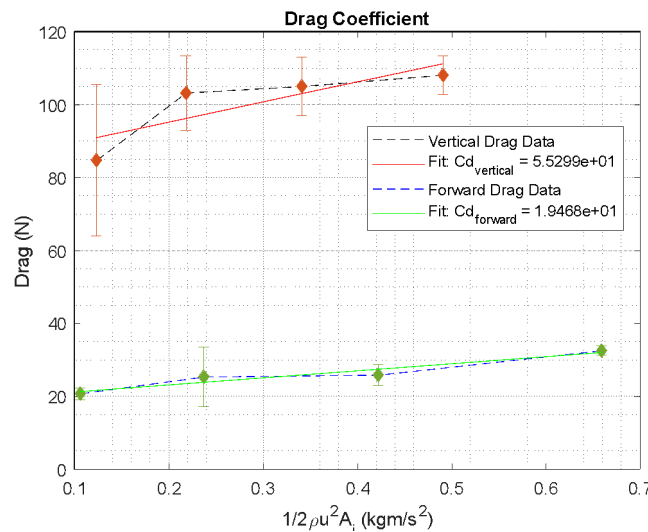


Figure 50 – Drag force vs dynamic pressure for vertical and forward flight.

The drag coefficient was experimentally determined to inform and enhance the accuracy of the flight validation model. A linear relationship was found between drag forces and their respective dynamic pressure-area terms ($1/2 \rho u^2 A_i$). The vertical drag coefficient ($C_{d_{\text{vertical}}} = 55.3$) significantly exceeds the forward drag coefficient ($C_{d_{\text{forward}}} = 19.5$), indicating notably higher resistance during vertical ascent, likely due to the greater cross-sectional area and induced aerodynamic effects such as propeller wash and turbulent airflow interacting with the drone body.

Notably, vertical drag measurements exhibit significantly greater standard deviations. This variability stems primarily from two factors. It is inherently challenging to precisely control vertical ascents using manual remote piloting, causing irregularities in ascent velocities. In addition, the drone's GPS unit does not reliably measure vertical velocity, necessitating manual estimation through altitude and time readings from the OSD. This introduces additional human observational error.

Given these limitations and the inherent uncertainties involved, the average drag coefficient ($C_{d_{\text{avg}}} = 37.4$) used in the flight model represents a conservative estimate. The large standard deviation associated with vertical measurements further ensures that the model incorporates an appropriate safety margin. Therefore, the adopted drag coefficient is suitable and conservative enough for integration into the flight model, ensuring realistic predictions and robust safety considerations for drone performance with the intended payload and capture mechanism.

Motor Constants (k_t & k_m) Determination

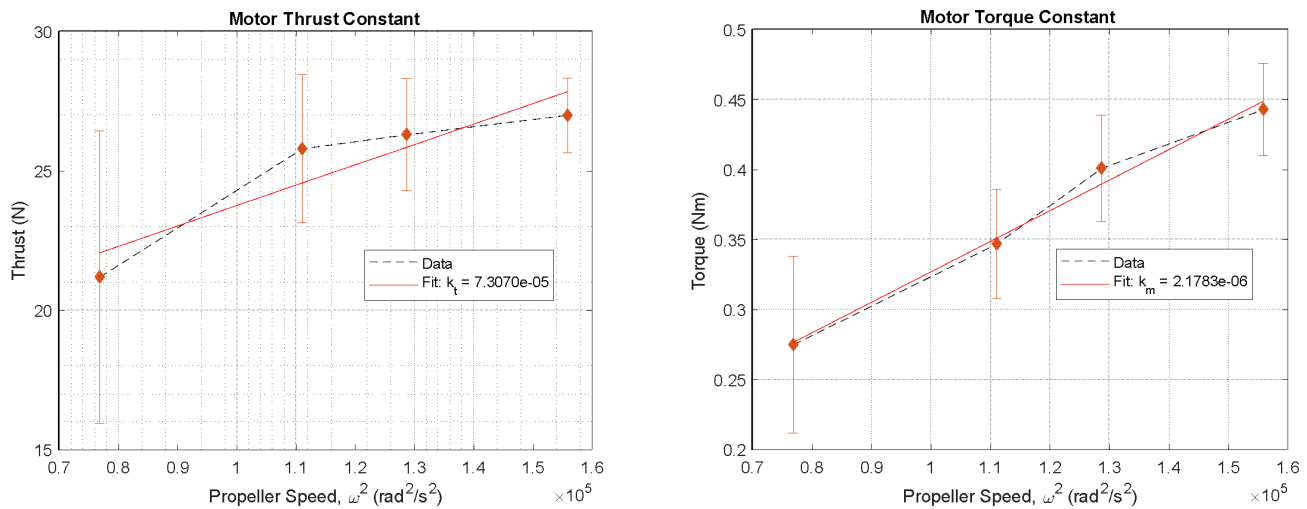


Figure 51 – Motor thrust (left) and torque (right) constant determination.

The motor thrust constant ($k_t = 7.3070 \times 10^{-5}$) and torque constant ($k_m = 2.1783 \times 10^{-6}$) were obtained by fitting linear relationships between measured thrust and torque values against the square of the propeller angular velocity (ω^2). As with the vertical drag experiment, uncertainties arose primarily due to the difficulty in maintaining a consistent vertical climb using manual remote piloting. This introduced variability in both propeller speed and power draw, reflected in the relatively large standard deviations in the data.

Despite these limitations, the resulting trends were sufficiently linear to yield reliable estimates for k_t and k_m . These values are now suitable for integration into the 5.1 Flight Validation – Simulink model and the 5.3 Mechanism Validation – Gazebo model, allowing simulations of accurate predictions of thrust and rotational dynamics of the drone-interceptor under varying flight conditions.

7.2 Net Test

7.2.1 Experimental Setup and Assumptions

The experiment assessed three net configurations to evaluate their effectiveness in capturing and entangling an incoming drone. Each trial involved launching the drone from a set distance at a 20° angle while maintaining a constant flight speed of 30 km/h, which helped isolate the impact of net tension. Once entangled, the drone's high-speed, low-torque FPV motors lacked the necessary force to break free, and the onboard system was programmed to shut down the motors to prevent additional instability immediately. The configurations varied in tension: Case 1 featured a loosely anchored net with two fixed corners, allowing for 15 cm of movement at the centre; Case 2 added a third anchor point, decreasing the movement to 10 cm; and Case 3 secured all corners, restricting movement to just 5 cm. Each configuration was tested five times to evaluate repeatability. Table 14 (Appendix A.12) includes before-and-after images of the net, illustrating how variations in tension influenced both deformation and success in entangling the drone. Although the controlled conditions and fixed flight path do not fully replicate the unpredictability of real-world drone behaviour, this approach is suitable for comparative analysis, as it provides consistency across trials and isolates the impact of net design on capture effectiveness. The table below summarises the success rates observed during the trials:



Figure 52 – Image of the drone being caught during testing.

Table 12 – Success Rates for different net configurations.

Case	Number of Trials	Successful Captures	Success Rate
Case 1	5	4	80%
Case 2	5	5	100%
Case 3	5	3	60%

7.2.2 Observations and Analysis

The net exhibited impressive durability throughout all the trials, remaining intact and fully functional without visible damage. The drone, too, sustained minimal harm, with all components securely in place. Its motors consistently stopped upon entanglement, ensuring safe and controlled captures. Case 1 performed admirably, capturing the drone four times out of five. The net's flexible design allowed it to wrap around the drone's rotors and arms; however, the larger gaps occasionally let it slip through or bounce off. Case 2 emerged as the most successful setup, achieving a perfect capture rate in all five trials. The net provided a stable yet responsive surface with moderate tension that absorbed impacts and deformed just enough to ensnare the drone securely. This combination of firmness and flexibility maximised contact and minimised the chances of rebounds or escapes. Conversely, Case 3 demonstrated the least effectiveness, with only three successful captures out of five. The net's rigidity often caused the drone to bounce away upon

impact, limiting entanglement opportunities. While some rotor interference was noted, the net's lack of flexibility hindered its ability to wrap around the drone's structure, reducing its effectiveness. Furthermore, the area of the net is 0.78 m^2 , which meets the objective of covering 0.5 m^2 of area to capture the malicious drone and clearly shows the net captures and secures the drone. In conclusion, Case 2 is the preferred configuration for future applications, demonstrating an impressive 100% capture success rate.

7.3 Arm Static Test

7.3.1 Experimental Setup

A static load test was performed using a 50 kg weight, roughly equivalent to the 500 N load simulated in the FEA, to assess the arm's structural integrity. The arm was positioned vertically against a wooden backboard firmly secured to a brick wall for stability. A spring was attached to a bracket above the arm, with a cord connecting it to the weight; this arrangement redirected the vertical load into a horizontal force, creating a lateral load path that induced bending stress on the arm. In this scenario, the lateral force represents the sideways load applied perpendicular to the arm's orientation, while lateral deflection refers to the resulting sideways bending. Given the fixed and rigid nature of the setup, any observed deflection can be directly linked to the arm's structural response. Although the FEA simulated a 500 N load at a 45° angle to mimic realistic drone impact conditions, replicating this in the physical test would have required a more complex setup. Instead, a purely lateral load allowed for a more straightforward, controlled approach to evaluate the arm's resistance to bending. While this physical configuration does not ideally mimic the conditions of the FEA, it still provides valuable insights by effectively isolating a key direction of loading and demonstrating the arm's stiffness and potential real-world performance.



Figure 53 – Arm static test set up.

7.3.2 Observations and Analysis

As depicted in Figure 53, the arm exhibited 5 cm of lateral deflection when subjected to a load of 500 N. This deformation was entirely elastic, allowing the arm to revert to its original shape once the load was lifted, which indicates that no permanent or plastic deformation occurred. To replicate boundary conditions similar to those defined in the FEA, wooden blocks were positioned at both the base and midpoint of the arm. The base block mimics the base connection to the arm and is a simple support, preventing translation and rotation ($U_x, U_y, U_z, R_x, R_y = 0$). In contrast, the midpoint block, which mimics the brace connection to the arm, functioned as a roller support, permitting vertical movement but restricting lateral translation. Although the applied force was purely horizontal, differing from the 45° angled load used in the simulation, the test still generated significant bending, effectively evaluating the arm's stiffness and structural response. The absence of permanent deformation under these conditions reaffirms the arm's capacity to withstand considerable lateral loads and endorses the overall durability of the design in practical applications.

7.4 Structural Integrity Test

7.4.1 Experimental Setup



Figure 54 – Drone entangled in net.

To evaluate the structural durability of the capture mechanism in semi-realistic conditions, a controlled ground test was conducted to mimic the drone interception process. As depicted in Figure 55, the mechanism was affixed to a wooden board set at an angle of about 45 degrees, chosen to simulate its intended deployment angle, albeit more extreme than what was experienced during the final live interception (refer to Section 7.5). This test setup allowed for consistent and repeatable impacts; however, the mechanism itself was firmly mounted and did not have the ability to move or react as it would in an actual aerial situation. A 1000 g FPV drone was manually piloted toward the apparatus at a speed of approximately 35 km/h from a distance of 80 meters and at an altitude of 4 to 5 meters, creating



Figure 55 – Structural integrity test set up.

a realistic forward momentum while the pilot maintained control. Telemetry data confirmed stable flight and consistent power delivery at impact. Battery voltage and signal strength remained steady, verifying that the drone struck the net at full velocity without deceleration or control loss. This ensured the structure absorbed the drone's complete kinetic energy.

7.4.2 Observations and Analysis

All test runs resulted in successful captures. As illustrated in Figure 54, the net effectively entangled the drone's propellers and frame, halting its motion completely. The net remained semi-taut (as shown in section 7.3) and intact, and all mounting points withstood the impact without failure. No tearing or disconnections were observed, confirming the reliability of the net under repeated impacts.

The structural frame exhibited varied responses based on the impact point of the drone. When the drone hit the centre of the net, the load was evenly distributed, leading to no



permanent deformation. However, direct impacts with the braces resulted in localised plastic deformation, as illustrated in Figure 56. These braces were not intended to absorb direct forces; instead, their primary role was to support the arms once a load was applied through the net. The observed plastic deformation under direct impact highlights the need to redesign these components to withstand such forces better, even if they are not part of the primary load path. The initial FEA did not consider this scenario, which concentrated on distributed loads across the frame rather than localised impacts. While one could view this as a limitation of the simulation, the actual implications were minimal. The slight bending didn't affect the mechanism's functionality and could be corrected manually. This indicates that the braces absorbed some impact energy, protecting more critical components from damage. In this sense, the deformation served as a controlled failure mechanism, helping to maintain the overall system's integrity. However, repeated impacts in the same area could threaten the structure's long-term durability, an essential consideration for future design improvements. On a positive note, the main support arms remained rigid and undamaged throughout the testing process, suggesting a degree of overengineering that could allow for material or weight savings without sacrificing structural strength. While the FEA did not capture all the intricacies of real-world behaviour, the system performed well under the test conditions. The minor, recoverable deformations in non-critical areas represent an acceptable compromise in a design focused on resilience and low maintenance.

Figure 56 – Brace bending due to direct impact.

7.5 Live Interception

7.5.1 Experimental Setup and Conditions

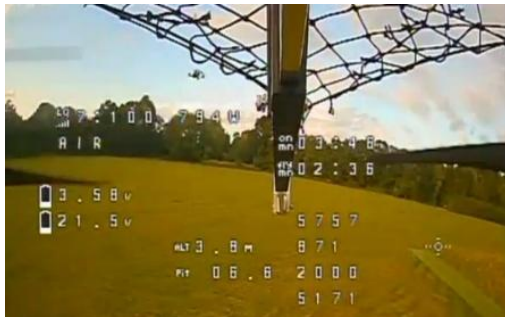


Figure 57 – FPV of experiment.

The interception tests took place in a designated open-field area measuring 30×30×40 meters, specifically selected to mimic the spatial constraints found in urban and suburban environments. Moderate wind during the trials added natural disturbances, such as drift and shifts in position, which helped to validate the interceptor's stability in realistic conditions. For safety reasons, a minimum distance of 30 meters was kept between observers and the testing area, with continuous visual monitoring throughout the process. The capture mechanism was mounted on a quadcopter drone as the interceptor. Despite the extra weight, the drone maintained stable and predictable flight, remaining agile during and after interception attempts, which facilitated consistently safe landings.

7.5.2 Observations and Analysis of Results

Five interception attempts were made consistently at a fixed altitude of 7.5 meters, with the target drone travelling at a steady speed of approximately 25 km/h. Of these attempts, only one was successful, resulting in a 20% success rate. The unsuccessful attempts were mainly due to manual piloting errors and the challenges of visually tracking the moving drone while simultaneously operating the capture drone. The situation was further complicated by limited visibility from the front-mounted arm of the capture mechanism, which hindered the pilot's line of sight, as illustrated in Figure 57. Notably, despite the failures, neither the capture mechanism nor the target drone suffered any damage. This outcome highlights a significant safety advantage over conventional interception methods, which often involve kinetic or signal-jamming techniques that can result in drone destruction or falling debris, posing risks in civilian environments.

As shown in Figures 58 and 59, the successful interception process involved effective net entanglement that secured the drone without causing any damage. Following the capture, the interceptor drone exhibited stable and controlled flight, enabling a safe landing. Inspections conducted post-test revealed no structural damage or deformation in the capture mechanism, which underscores its robustness and reliability under realistic conditions. However, some mechanical issues were noted, particularly regarding the looseness of the nuts securing the braces to the ribs. This emphasises the necessity for self-locking nuts in future designs to enhance joint reliability.

The moderate winds during the trials significantly impacted the precision and repeatability of the interceptions. Drift from the drone made it challenging to maintain precise alignment during manual operation, and limitations in human reaction time and spatial awareness further hindered interception accuracy. These observations indicate that relying solely on manual control may not yield consistent results in real-world scenarios. Additionally, the onboard camera positioned behind the front structural arm limited visibility and affected targeting precision. Moving the camera to the top of the central column could provide a clear forward view,

greatly enhancing piloting accuracy. Compared to traditional kinetic interception methods, the net-based system demonstrated here presents a safer and more controlled approach. Avoiding destructive collisions reduces the risk of falling debris, a crucial aspect in populated areas. Even during failed attempts, there was no damage to the intercepted drones, further supporting the system's appropriateness for civilian applications where safety is paramount.



Figure 58 – Drone in the air after catching a malicious .

Future designs should embrace autonomous flight control through computer vision and trajectory prediction algorithms. Automation would mitigate human error and enhance interception accuracy, consistency, and operational reliability, while also reducing costs by negating the need for skilled pilot training. Upgrades such as self-locking nuts will bolster durability, and repositioning the camera will solve current visibility challenges. Finally, conducting more tests under various environmental conditions, including higher wind speeds and reduced visibility, will be vital to thoroughly validate the capture mechanism's readiness for real-world application.

Figure 59 – Successful interception.

8. Conclusion

In conclusion, the drone capture mechanism successfully achieved its goal of offering a safe, reusable, and structurally robust solution for intercepting harmful drones in civilian settings. Key features such as non-destructive capture, stable flight integration, and reusability were effectively demonstrated in realistic scenarios. The mechanism design was also verified through simulations, mechanical testing, and live trials. The final mechanism weighed 2.53 kilograms, comfortably under the 8-kilogram target. It exceeded the required coverage of 0.5 square meters and could fold compactly within the drone's dimensions. FEA confirmed that the structural components could handle impacts up to 500 N without yielding, and the overall structural integrity confirmed during testing. Aluminium 6061 T6, which was chosen for its excellent strength-to-weight ratio, corrosion resistance, and recyclability proved itself as an excellent choice for the construction of this mechanism.

Live flight tests demonstrated stable operation and safe landings post-engagement. This allowed for a live interception trial during which a one successful capture of a fast FPV drone resulted in a 20% success rate. All trials were conducted without causing any damage to the mechanism, satisfying the requirements for safety, robustness, and repeatability. The design prioritised sustainability and reusability by utilising recyclable aluminium and a modular approach allowing easy component replacement. The mechanism is straightforward to disassemble for recycling purposes. Efficient manufacturing techniques minimise material waste, such as CNC machining and waterjet cutting. More than 15 rapid prototyping iterations using 3D printing facilitated quick and environmentally friendly development, with early components reused in the final product. The mechanism operates on battery-powered actuation, which produces no emissions during operation. A net-based capture method avoids creating debris, making it a safer alternative to destructive interception techniques and suitable for deployment in populated areas. Its safe functionality also enhances opportunities for broader applications in logistics, inspection, and environmental monitoring. However, the manual piloting and the position of the rear-mounted camera limited visibility, contributing to the 20% success rate. Incorporating onboard visual tracking and autonomous flight controls would enhance targeting accuracy and consistency. Additionally, using self-locking nuts would address the issue of bolt slack, thereby improving structural integrity under repetitive loading. Overall, the mechanism met all technical, environmental, and operational criteria. It successfully captured a fast FPV drone, proved reusable, and maintained control throughout the trials. These findings confirm its effectiveness as a reliable, regulations-compliant interception system for safeguarding civilian airspace.

9. Future Work

9.1 In-flight deployment mechanisms

A way in which the design can be further improved, which could not be done due to time constraints, is the ability to deploy mid-flight. Despite the fact that the design was demonstrated as being able to open and close manually, a fully autonomous opening system was not introduced. This development will have several advantages, such as full integration with a rocket body, as described in 9.3, as well as a smaller profile area of the interceptor during flight, makes it harder for the malicious drone to detect and avoid.

The deployment mechanism needs to transition from a fully closed state, with arms aligned parallel to the central column, to a fully open state, with arms parallel to the top plate of the drone, activated by the push of a button. The system would be required to be reliable, reusable, and lightweight. Ideally, it would include a single electronic component, thereby streamlining the design and minimise the number of moving parts.

Several methods for applying force were evaluated:

- A large solenoid at the base (heavy and slow)
- A compression spring at the top of the column (simple and fast, but very heavy due to a long column)
- Tension springs at the base, which are attached to the slider (simple, fast, and lightweight)

However, an issue with tension springs is that conventional springs would be inadequate in this application due to the long travel distance (from 565 mm to 110 mm). To solve this, it was decided that a spring-loaded retractable pulley offered the best solution, as they provide consistent tension over a wide range of lengths. To release the slider, causing the pulleys to move it down, a 24-volt solenoid with a 10 mm stroke is mounted below the slider. It in turn, is connected to a switch drive module powered by the drones, battery to activate the mechanism. Therefore, when the flight controller sends a PINIO signal, the solenoid contracts, allowing the pulleys, which are attached at the base, to draw the slider downward. Thereby the capture arms are deployed, spreading the net and allowing for the target drone to be captured.

9.2 Targeting and Guidance System

The existing prototype operates exclusively through manual control and does not include integrated targeting or autonomous guidance systems. While this setup is adequate for initial testing, it limits the system's effectiveness in complex real-world scenarios where drones can exhibit unpredictable or high-speed movements. A significant focus for future development will be the integration of autonomous targeting and navigation capabilities, enabling the platform to identify, track, and intercept rogue drones with minimal

operator input. The system would employ sensor fusion to enhance targeting, utilising radar, LiDAR, infrared, and optical camera data. Information from these sensors will be processed with algorithms like Kalman filters or machine learning (ML) models, such as neural networks, created using Python. Real-time trajectory estimation will empower the system to anticipate movements rather than merely respond, significantly improving the likelihood of successful captures. The guidance system will need control algorithms capable of adjusting to external factors like wind and quick evasive manoeuvres, allowing it to make autonomous corrections to optimise its interception trajectory. While these automated features will reduce the burden on operators and enhance performance, they also add technical complexity and associated risks. The system must be resilient against threats such as jamming, spoofing, and signal loss, which necessitates including redundant sensors and robust cybersecurity measures. These enhancements will impact the overall weight and cost of the system, so careful consideration of trade-offs is essential during the development phase.

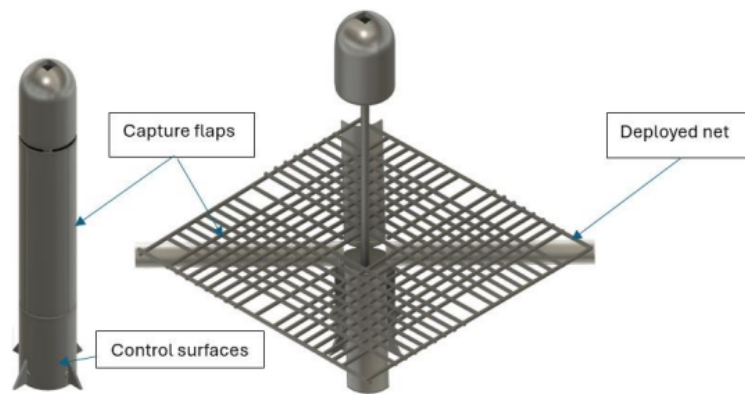
9.3 Drone-Rocket Hybrid

This project primarily focused on the physical capturing aspect of catching a drone, and due to time constraints, there was not enough time to work on a carrier. Ideally, to catch a target you need a carrier platform that is an order of magnitude faster than the it. Therefore, it has been suggested that a rocket motor should be incorporated into the design. The resultant drone-rocket hybrid platform would improve pursuit speed and enhance closing distances. This innovative approach merges the vertical take-off and stability of a multirotor drone with the acceleration capabilities of a solid-fuel rocket.



Figure 60 – Drone-rocket hybrid concept.

Thereby, it enables quick, stable positioning and a powerful final boost for engaging high-speed or distant targets. An example of a similar concept is illustrated in Figure 60, showcasing the potential of integrating drone and rocket technology into a unified aerodynamic structure.



The proposed hybrid system emphasises modularity and compact design. The drone is housed entirely within the rocket fuselage, which minimises aerodynamic drag while enclosing propulsion, control mechanisms, and the capture system. Upon reaching an appropriate range from the target, the platform initiates a controlled deployment procedure and fires the rocket motor. This allows for the interceptor to close with the target before it can react. Following a successful capture, the system utilises the stability and manageability provided by the drone rotors to descend to a designated safe recovery area.

Figure 61 – CAD hybrid concept with net.

solid-fuel rocket, and a parabolic nose cone to minimise drag. The aerodynamic configuration was influenced by principles of The Handbook of Model Rocketry (Stine & Stine, 2007) deploying clipped delta fins and optimised length-to-diameter ratio.

Early flight performance assessments were conducted using OpenRocket simulations on this initial design. The results in supplementary figures indicate that with a K550W-6 motor, the hybrid platform can achieve up to 113 m/s, altitudes nearing 680 m, and accelerations above 60 m/s², varying with body material and design. Stability margins were within acceptable thresholds across most configurations, confirming the system's capacity to reach targets while quickly maintaining control during high-speed engagements. Further technical advancements will focus on enhanced aerodynamic modelling through CFD to analyse airflow around the structure at various speeds and angles of attack. FEA will evaluate the structural integrity of the net deployment mechanism and flaps under launch and deployment stress. A priority will be re-examining deployment methods; alternatives such as spring-loaded, pneumatic, and mechanical linkages will be assessed based on their size, responsiveness, and repeatability. Mass balancing and stability assessments will also optimise internal layouts for payload, electronics, and recovery components.

In addition to engineering hurdles, integrating rocket propulsion entails stringent regulatory requirements. Rocket motors exceeding 160 Ns necessitate prior authorisation from the Civil Aviation Authority (CAA) (Civil Aviation Authority, 2016), and testing protocols must adhere to established safety standards. Ethical considerations also arise when deploying in urban or civilian airspace, including risks to bystanders, the potential for misuse, and data privacy concerns stemming from onboard sensors. These issues will be addressed by ensuring compliance with legal requirements, promoting responsible use-case scenarios, and establishing strong data governance practices. Despite these challenges, the hybrid system provides a viable and scalable solution to the increasing problem of unauthorised drones in restricted airspace. Its unique potential for rapid, reliable engagement and non-destructive capture aligns with operational and ethical benchmarks for civilian applications.

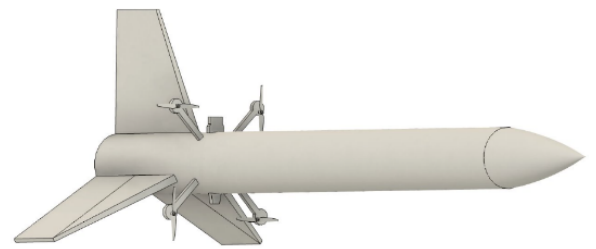


Figure 62 showcases an initial CAD design of the drone-rocket hybrid design, which features four electric motors affixed to stabilising fins, a centrally housed

Figure SEQ Figure * ARABIC 62 – CAD Design made to show hybrid concept.

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Appendix

A.1 Appendix for SG 15L Drone Datasheet

SG 15L Product Images

SG 15L Size

SG 13L Parameters

Frame	15 inches
Wheelbase	647mm
Product size	517*517*80mm
Material	3K T300 carbon fiber
Propeller	1507 (glass fiber nylon)
Max. take-off altitude	5000m
Max. flight speed	130km/h
Max. flight distance	20km
Rated load	7kg
Max. load	8kg
Max. take-off weight	13kg
No load endurance	42 mins (8s16000mah,4.9kg)
Full load endurance	12 mins (8s16000mah,11.8kg)
Net weight (with weight)	2.08kg
Product weight	2.75kg
Color box size	500*330*90mm
Recommended battery	8S 16000 50C LCB50 plug

Image Transmission Parameters

Model	TCMM 3W image transmission (optional)
Frequency	5.8G
Channels	48CH
Input Voltage	7-26V
Output Power	25mw/1600mw/2000mw/2500mW/3000mW
Protocol	Smart Audio
Dimensions	41*25*13.35MM

ESC Parameters

Model	TCMM 80A single ESC
Continuous current	80A
Peak current	85A (10S)
Lithium battery	3-8S
Weight	22g
Firmware	AM32

Flight Control Parameters

Model	F405ICM85MINIOSD flight control (optional)
BEC	10V2A/5V3A/3.3V
MCU	STM32F405/STM32F722
Gyroscope	MPU6000/ICM42688-P
OSD	AT7456E
Firmware	TCMMF405/TCMMF722
Weight	8g
Mounting holes	M3*30.5*30.5

Motor Parameters

Model	TCMM 4214 380KV
Weight	208g
Maximum continuous current	60A
Maximum continuous power	1450W
Screw hole spacing	19mm
Screw hole	M3

Camera Parameters

Model	TCMM1800TVL night vision FPV camera (optional)
Sensor	1/3.cmos
Horizontal resolution	1800TVL
Horizontal field of view	165°
Lens	2.1mm lens
Video format	NTSC & PAL (adjustable)
Image ratio	16:9
Low illumination	0.0001LUX
Video output	CVBS
Day and night	External/internal/black and white/color
Operating temperature	-20°C~+60°C

Receiver Parameters

Model	ELRS915M receiver
MCU	ESP32 PICO D4, dual SX1276
Antenna	IPEX MHF 1/U.FL
RF frequency	915MHz&868MHz
Remote control power	17dBm
Receiver protocol	CRSF
Input voltage	+5V DC @ "+" pad
PCB size	16mm x 16mm
Protocol version	ELRS V3.3.0

Figure 63 – SG 15L FPV drone full comprehensive datasheet.

A.2 Appendix for 4.1 Force Calculations

A.2.1 Force Calculations

Maximum extension calculations:

$$\begin{aligned}\sin \sin (\theta) &= \frac{y_{max}}{a} \\ a &= \frac{y_{max}}{\sin(\theta)} \\ b^2 &= a^2 + L^2 - 2aL\cos(\theta) \\ b^2 &= \frac{y_{max}^2}{\sin^2(\theta)} + L^2 - 2 * \frac{y_{max}}{\sin(\theta)} * L\cos(\theta) \\ L &= x + \frac{y_{max}}{\sin(\theta)} + b \\ x &= k \left(L - \frac{y_{max}}{\sin(\theta)} - \left(\frac{y_{max}^2}{\sin^2(\theta)} + L^2 - 2 * \frac{y_{max}}{\sin(\theta)} * L\cos(\theta) \right)^{0.5} \right)\end{aligned}$$

Based on the initial design dimensions:

$$\theta = 45 \text{ degrees}, y_{max} = 0.4m, L = 1m$$

So $x = 0.3m$ (1sf)

Energy calculations:

$$KE = \frac{1}{2}mv^2 = 450J$$

$$PE = \frac{1}{2}kx^2$$

$$k = 10\,000 \text{ Nm}^{-1}$$

Therefore, the force on the system was found to be:

$$F = kx = 500N$$

A.3 Appendix for 4.2 Material Selection Analysis

A.3.1 MPI for yield strength vs density

$$\frac{\sigma_y}{\rho} = C \text{ (constant for a given design target)}$$

$$\sigma_y = C\rho$$

$$\log \log (\sigma_y) = \log(C\rho)$$

$$\log \log (\sigma_y) = C + \log(\rho)$$

A.3.2 MPI for Young's Modulus vs Fracture Toughness

$$EK_{IC}^2 = C \text{ (constant for a given design target)}$$

$$E = \frac{C}{K_{IC}^2}$$

$$\log(E) = \log\left(\frac{C}{K_{IC}^2}\right) = \log(C) - \log(K_{IC}^2)$$

$$\log(E) = C - 2\log(K_{IC})$$

A.3.3 MPI for Machinability vs CO2 footprint

$$\text{Machinability} = C \times \text{CO}_2 \text{ footprint (constant for a given design target)}$$

$$\log(\text{Machinability}) = \log(C \times \text{CO}_2 \text{ footprint})$$

$$\log(\text{Machinability}) = C + \log(\text{CO}_2 \text{ footprint})$$

A.4 Appendix for 4.4 Net Arms & Brace

A.4.1 Mass Minimisation

$$m = \rho V \quad m = \rho AV$$

Mass of Beam 1:

$$m_1 = \rho r L (H_1^2 - h_1^2)$$

Mass of Beam 2:

$$m_2 = \rho r c (H_2^2 - h_2^2) = \rho r \sqrt{a^2 + b^2} (H_2^2 - h_2^2)$$

Mass of Beam 3:

$$m_3 = \rho r b H_3^2$$

Mass of Beam 4:

$$m_4 = \rho r \frac{b}{\cos \cos(45^\circ)} (H_4^2 - h_4^2)$$

Total mass:

$$m = m_1 + m_2 + m_3 + m_4$$

A.4.2 Stress Minimisation (Maximum Stress in Net Arm)

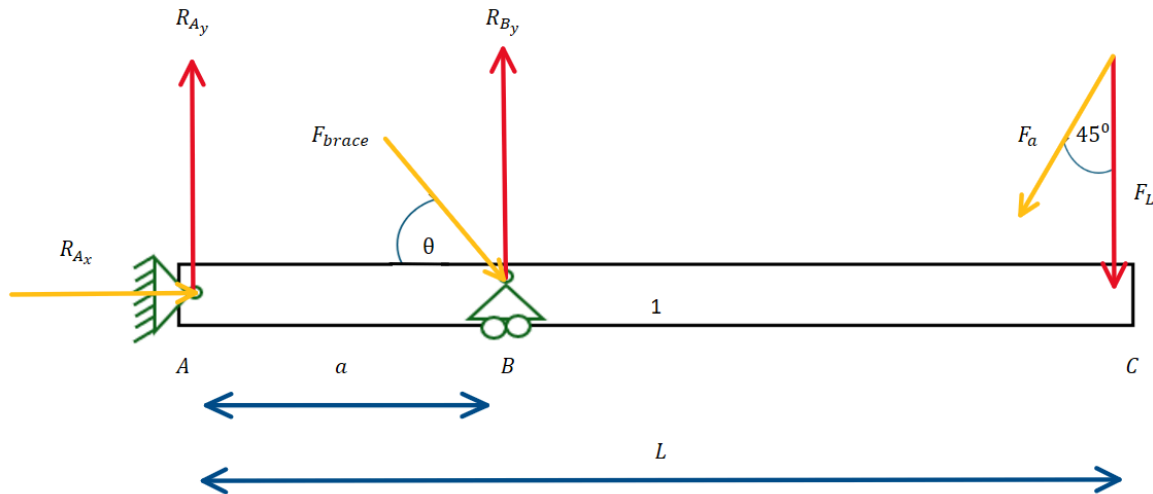


Figure 64 – Free-body diagram of the net arm (Beam 1) showing all external forces and support reactions.

Reaction forces

Sum of vertical forces:

$$\sum F_y = 0: R_{A_y} + R_{B_y} - F_L = 0 \Rightarrow R_{A_y} + R_{B_y} = F_L$$

Where:

$$F_L = F_a \cos \cos(45^\circ) \text{ (Vertical component of } F_a \text{ acting on Beam 1)}$$

Sum of moments about A:

$$Mom(A) = 0 \therefore R_{B_y} a - F_L L + M_0 = 0 \Rightarrow R_{B_y} = \frac{F_L L - M_0}{a}$$

Where:

$$M_0 = F_a b \sin \sin(45^\circ) \text{ (Moment induced at the end of Beam 1 due to the horizontal component of } F_a \text{)}$$

$$\Rightarrow R_{A_y} = F_L - R_{B_y} = F_L - \frac{F_L L - M_0}{a} \Rightarrow R_{A_y} = \frac{F_L(a-L) + M_0}{a}$$

Internal bending moments

Region 1 ($0 \leq x < a$):

$$M_y(x) = R_{A_y} x$$

$$M_y(x) = \frac{F_L(a-L) + M_0}{a} x$$

At $x = a$, the internal bending moment is:

$$M_y(x) = F_L(a - L) + M_0$$

Region 2 ($a \leq x < L$):

$$M_y(x) = R_{A_y} x + R_{B_y}(x - a)$$

$$M_y(x) = \frac{F_L(a-L) + M_0}{a} x + \frac{F_L L - M_0}{a} (x - a)$$

At $x = L^-$ (just inside the right boundary), the internal bending moment is:

$$M_y(L) = \frac{F_L(a-L) + M_0}{a} L + \frac{F_L L - M_0}{a} (L - a)$$

$$M_y(L) = M_0$$

Thus, the maximum bending moment in the beam is:

$$M_{y_{max}} = M_y(L^-) = M_0$$

Maximum bending stress

Stress in arm (Beam 1), σ_{arm} :

$$\sigma_{arm} = \frac{M_{y_{max}} c}{I}$$

Where:

$$c = \frac{H}{2} \text{ (Distance from the neutral axis to the extreme fibre of Beam 1)}$$

$$I = \frac{r}{12} (H_1^4 - h_1^4) \text{ (Second moment of area of Beam 1)}$$

$$\therefore \sigma_{arm} = \frac{M_0 \frac{H}{2}}{\frac{r}{12} (H_1^4 - h_1^4)} \Rightarrow \sigma_{arm} = \frac{6M_0 H_1}{r(H_1^4 - h_1^4)}$$

A.3 Stress Minimisation (Maximum Stress in Brace)

Stress in brace (Beam 2), σ_{brace} :

$$\sigma_{brace} = \frac{F_{brace}}{A_2}$$

Where:

$$F_{brace} = \frac{R_{B_y}}{\sin \sin(\theta)} \text{ (Force exerted on Beam 2 by Beam 1)}$$

$A_2 = r(H_2^2 - h_2^2)$ (Area of Beam 2 in contact with Beam 1 where the reaction force is exerted)

$$F_{brace} = \frac{R_{B_y}}{\sin \sin(\theta)} \therefore F_{brace} = \frac{F_L L - M_0}{a \sin \sin(\theta)}$$

$$\sigma_{brace} = \frac{\frac{F_L L - M_0}{a \sin \sin(\theta)}}{r(H_2^2 - h_2^2)} \Rightarrow \sigma_{brace} = \frac{F_L L - M_0}{a \sin \sin(\theta) r(H_2^2 - h_2^2)}$$

A.4 Deflection Minimisation (Maximum Deflection of Net Arm)

Bending moment

Macaulay's method:

$$M_y(x) = R_A x + R_{B_y} \langle x - a \rangle - F_L \langle x - L \rangle - M_0 \langle x - L \rangle^0$$

Deflection

Euler-Bernoulli beam theory:

$$EI \frac{d^2 y}{dx^2} = M_y(x)$$

$$EI y'(x) = \int M_y(x) dx = \int \left[R_A x + R_{B_y} \langle x - a \rangle - F_L \langle x - L \rangle - M_0 \langle x - L \rangle^0 \right] dx + C_1$$

$$EI y'(x) = \frac{R_A}{2} x^2 + \frac{R_{B_y}}{2} \langle x - a \rangle^2 - \frac{F_L}{2} \langle x - L \rangle^2 - M_0 \langle x - L \rangle + C_1$$

$$EI y(x) = \int \left[\frac{R_A}{2} x^2 + \frac{R_{B_y}}{2} \langle x - a \rangle^2 - \frac{F_L}{2} \langle x - L \rangle^2 - M_0 \langle x - L \rangle \right] dx + C_1 x + C_2$$

$$EI y(x) = \frac{R_A}{6} x^3 + \frac{R_{B_y}}{6} \langle x - a \rangle^3 - \frac{F_L}{6} \langle x - L \rangle^3 - \frac{M_0}{2} \langle x - L \rangle^2 + C_1 x + C_2$$

$$y(x) = \frac{1}{EI} \left[\frac{R_A}{6} x^3 + \frac{R_{B_y}}{6} \langle x - a \rangle^3 - \frac{F_L}{6} \langle x - L \rangle^3 - \frac{M_0}{2} \langle x - L \rangle^2 \right] + \frac{C_1 x}{EI} + \frac{C_2}{EI}$$

Apply boundary conditions to find unknown constants:

$$y(0) = 0$$

$$\therefore y(0) = \frac{1}{EI} \left[\frac{R_A}{6} \times 0 + 0 - 0 - 0 \right] + \frac{C_1 \times 0}{EI} + \frac{C_2}{EI} = 0 \Rightarrow C_2 = 0$$

$$y(a) = 0$$

$$y(a) = \frac{1}{EI} \left[\frac{R_A}{6} a^3 + \frac{R_{B_y}}{6} \langle a - a \rangle^3 - 0 - 0 \right] + \frac{C_1 a}{EI} = 0$$

$$\frac{1}{EI} \left[\frac{R_A a^3}{6} + C_1 a \right] = 0 \Rightarrow C_1 = -\frac{R_A a^2}{6}$$

$$y(x) = \frac{1}{EI} \left[\frac{R_A}{6} x^3 + \frac{R_{B_y}}{6} \langle x - a \rangle^3 - \frac{F_L}{6} \langle x - L \rangle^3 - \frac{M_0}{2} \langle x - L \rangle^2 - \frac{R_A a^2}{6} x \right]$$

$$\Rightarrow y(x) = \frac{12}{Er(H_1^4 - H_1^4)} \left[\frac{F_L(a-L) + M_0}{6a} x^3 + \frac{F_L L - M_0}{6a} \langle x - a \rangle^3 - \frac{F_L}{6} \langle x - L \rangle^3 - \frac{M_0}{2} \langle x - L \rangle^2 - \frac{a(F_L(a-L) + M_0)}{6} x \right]$$

Maximum deflection occurs at the free end at C:

$$\therefore y_{max}(x) = y(L)$$

$$\Rightarrow y(L) = \frac{12}{Er(H_1^4 - H_1^4)} \left[\frac{F_L(a-L) + M_0}{6a} L^3 + \frac{F_L L - M_0}{6a} (L - a)^3 - \frac{F_L}{6} (L - L)^3 - \frac{M_0}{2} (L - L)^2 - \frac{a(F_L(a-L) + M_0)}{6} L \right]$$

$$\Rightarrow y_{max} = \frac{12}{Er(H_1^4 - H_1^4)} \left[\frac{F_L(a-L) + M_0}{6a} L^3 + \frac{F_L L - M_0}{6a} (L - a)^3 - \frac{a(F_L(a-L) + M_0)}{6} L \right]$$

A.5 Appendix for 4.5 Central Column

A.5.1 Mass

$$A_i = \pi r_i^2 \quad A_o = \pi r_o^2 \quad A = A_o - A_i \Rightarrow A = \pi(r_o^2 - r_i^2)$$

$$V = AL \Rightarrow V = L\pi(r_o^2 - r_i^2)$$

$$m = \rho V \Rightarrow m = \rho L\pi(r_o^2 - r_i^2)$$

A.5.2 Maximum Bending Stress

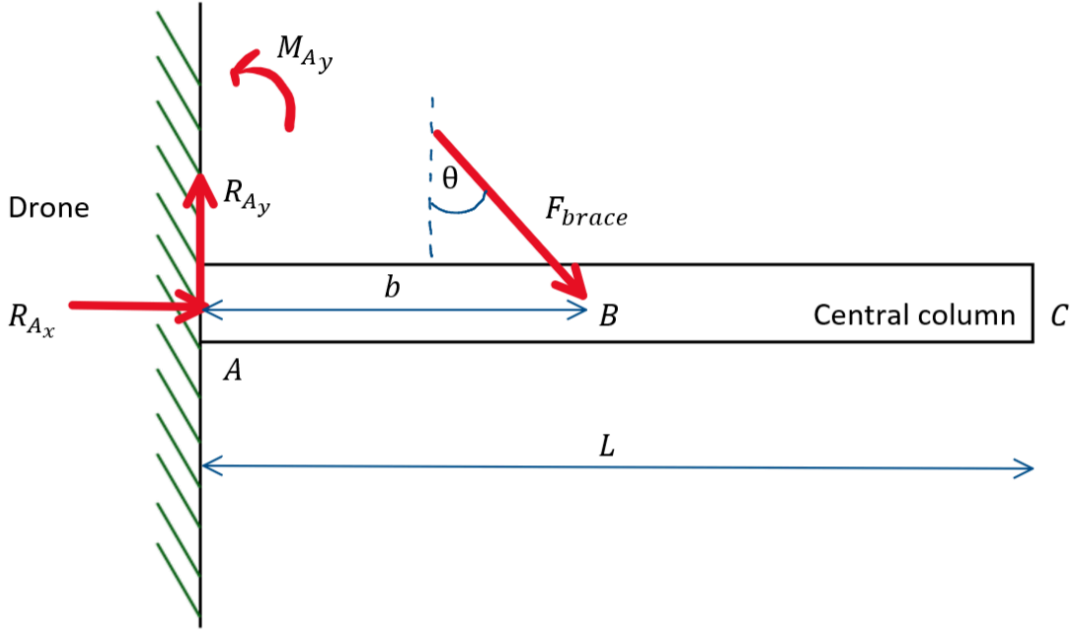


Figure 65 – Free body diagram of the central column model showing applied loads and reaction forces.

Force in brace, F_{brace}

From Net Arms & Brace Optimisation:

$$F_{brace} = \frac{F_L L - M_0}{a \sin(\theta)}$$

$$F_{brace_y} = F_{brace} \cos(\theta) \quad F_{brace_x} = F_{brace} \sin(\theta)$$

Reaction forces

Sum of vertical forces:

$$\sum F_y = 0 \therefore R_{Ay} - F_{brace_y} = 0 \Rightarrow R_{Ay} = F_{brace_y}$$

Sum of horizontal forces:

$$\sum F_x = 0 \therefore R_{Ax} - F_{brace_x} = 0 \Rightarrow R_{Ax} = F_{brace_x}$$

Sum of moments about A:

$$Mom(A) = 0 \therefore M_{Ay} - F_{brace_y} b = 0 \Rightarrow M_{Ay} = F_{brace_y} b$$

Internal bending moments

Region 1 ($0 \leq x < b$):

$$M_y(x) = R_{Ay} - M_{Ay} \Rightarrow M_y(x) = F_{brace_y} x - F_{brace_y} b \Rightarrow M_y(x) = F_{brace_y} (x - b)$$

$$M_y(x) = F_{brace} \cos(\theta) (x - b)$$

Region 2 ($b \leq x < L$):

$$M_y(x) = R_{A_y} x - M_{A_y} - F_{brace_y}(x - b)$$

$$M_y(x) = F_{brace_y} x - F_{brace_y} b - F_{brace_y}(x - b)$$

$$M_y(x) = F_{brace_y}(x - b) - F_{brace_y}(x - b) \Rightarrow M_y(x) = 0$$

At $x = 0$, the internal bending moment is:

$$M_y(0) = F_{brace} \cos \cos(\theta) (0 - b) \Rightarrow M_y(0) = -F_{brace} \cos \cos(\theta) b$$

Thus, the maximum bending moment in the beam is:

$$M_{y_{max}} = M_y(0) = -F_{brace} \cos \cos(\theta) b$$

Maximum bending stress

Stress in central column, σ_{max} :

$$\sigma_{max} = \frac{M_{y_{max}} c}{I}$$

Where:

$c = r_o$ (Distance from the neutral axis to the extreme fibre of Beam 1)

$I = \frac{\pi}{4}(r_o^4 - r_i^4)$ (Second moment of area for a hollow cylinder)

$$\sigma_{max} = -\frac{F_{brace} \cos \cos(\theta) b r_o}{\frac{\pi}{4}(r_o^4 - r_i^4)} \Rightarrow \sigma_{max} = -\frac{4F_{brace} \cos \cos(\theta) b r_o}{\pi(r_o^4 - r_i^4)}$$

Bi-directional loading (from 2 braces)

Stress in central column, σ_{max} :

$$\sigma_{max, 2 brace} = \sigma_{max,y} + \sigma_{max,z}$$

Where:

$$\sigma_{max,y} = \frac{M_{y_{max}} c_y}{I_y} \text{ (Bending about y)}$$

$$\sigma_{max,z} = \frac{M_{z_{max}} c_z}{I_z} \text{ (Bending about z)}$$

Assuming that the drone is captured in one of the net quadrants so that it is an equal distance between 2 arms, force from one brace, $F_{1/2}$:

$$F_{1/2} = \frac{F_{brace}}{2}$$

Due to symmetry:

$$I_y = I_z \quad c_y = c_z$$

$$\sigma_{max,y} = \sigma_{max,z} = -\frac{4\left(\frac{F_{brace}}{2}\right) \cos \cos(\theta) b r_o}{\pi(r_o^4 - r_i^4)}$$

$$\therefore \sigma_{max, 2 brace} = -\frac{4\left(\frac{F_{brace}}{2}\right) \cos \cos(\theta) b r_o}{\pi(r_o^4 - r_i^4)} \pm \frac{4\left(\frac{F_{brace}}{2}\right) \cos \cos(\theta) b r_o}{\pi(r_o^4 - r_i^4)}$$

$$\sigma_{max, 2 brace} = \sigma_{max} = -\frac{4F_{brace} \cos \cos(\theta) b r_o}{\pi(r_o^4 - r_i^4)}$$

A.5.3 Maximum Shear Stress

Assumption of uniform shear flow

Shear flow, q (force per unit length along the wall):

$$q = \tau t$$

Total shear flow

Transverse shear force, V :

$$V = qC_o \Rightarrow V = q(2\pi r_o) \therefore q = \frac{V}{2\pi r_o}$$

Maximum shear stress

Shear stress, τ :

$$\tau = \frac{q}{t} \Rightarrow \tau = \frac{V}{2\pi r_o t}$$

Where:

$V = F_{brace} \sin \sin(\theta)$ (Transverse force applied by brace against the central column)

$t = r_o - r_i$ (Thickness of central column)

$$\therefore \tau_{max} = \frac{F \sin \sin(\theta)}{2\pi r_o (r_o - r_i)}$$

A.5.4 Graphical Analyses

Outer diameter $D_o = 40 \text{ mm}$

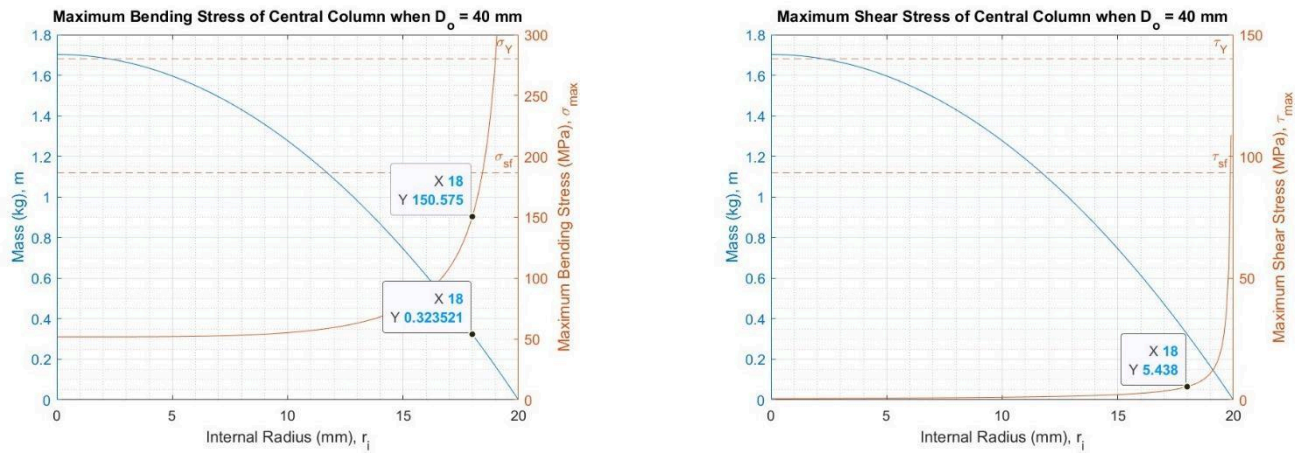


Figure 66 – Variation of mass, maximum bending (left) and shear stress (right) with internal radius for a central column of outer diameter 40 mm.

When $D_o = 40 \text{ mm}$, the optimal internal radius was $r_i = 18 \text{ mm}$, yielding $m = 0.324 \text{ kg}$ mass, $\sigma_{max} = 150.575 \text{ MPa}$ and $\tau_{max} = 5.438 \text{ MPa}$.

Outer diameter $D_o = 30 \text{ mm}$:

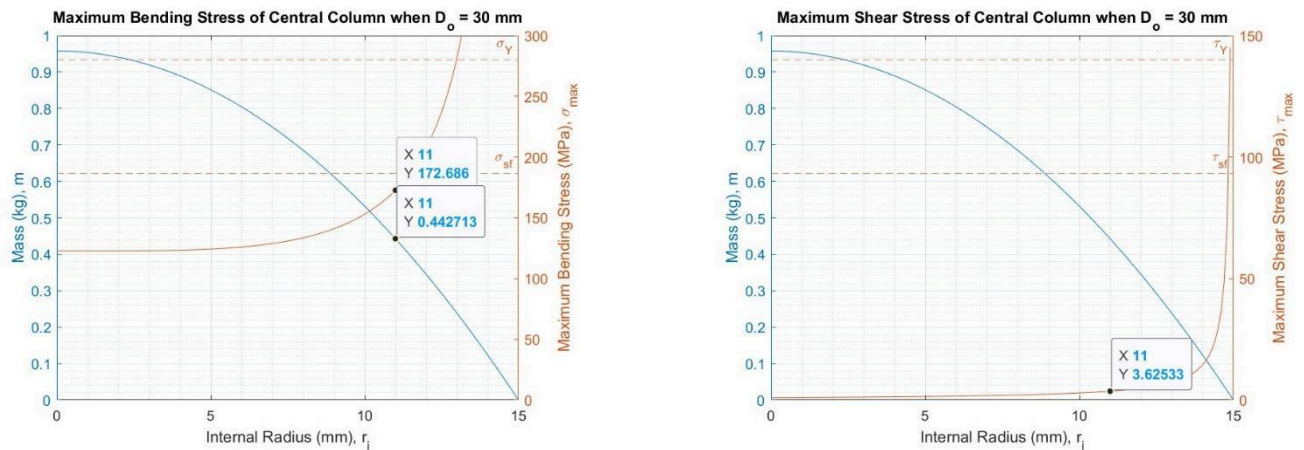


Figure 67 – Variation of mass, maximum bending (left) and shear stress (right) with internal radius for a central column of outer diameter 30 mm.

For $D_o = 30 \text{ mm}$, the feasible internal radius was $r_i = 11 \text{ mm}$, with mass at $m = 0.443 \text{ kg}$, $\sigma_{\max} = 172.686 \text{ MPa}$ and $\tau_{\max} = 3.625 \text{ MPa}$.

Outer diameter $D_o = 20 \text{ mm}$:

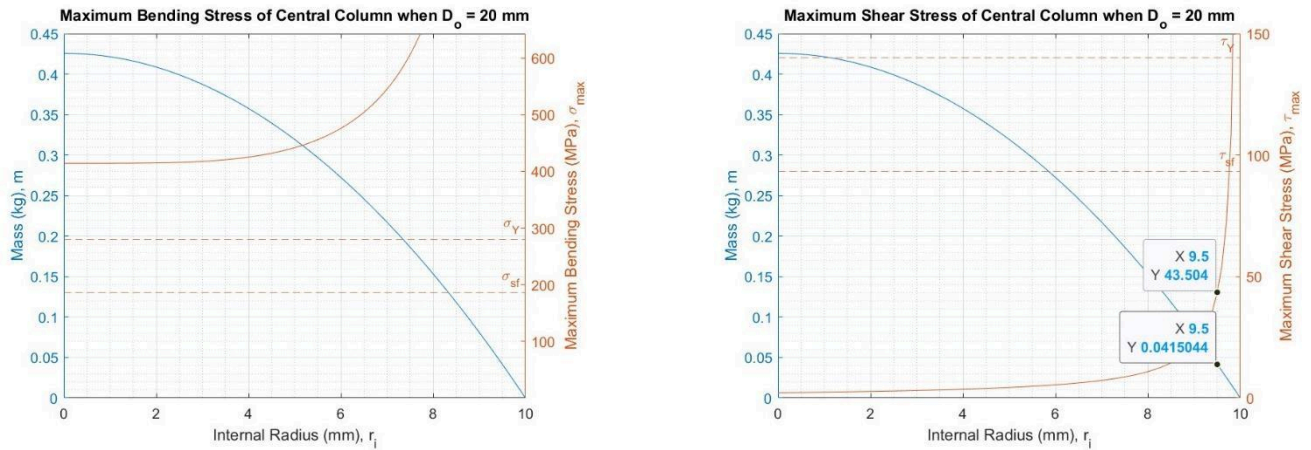


Figure 68 – Variation of mass, maximum bending (left) and shear stress (right) with internal radius for a central column of outer diameter 10 mm.

For $D_o = 20 \text{ mm}$, no feasible internal radius satisfied the maximum bending stress constraints under the safety factor. Only the maximum shear stress constraint could be satisfied such that $r_i = 9.5 \text{ mm}$ and $\tau_{\max} = 43.504 \text{ MPa}$ where $m = 0.042 \text{ kg}$.

A.6 Appendix for 5.1 Flight Validation – Simulink

The complete source code for the drone flight simulation with the payload, including all initialisation scripts, is available in the following GitHub repository:

<https://github.com/umayr5/flight-validation-simulink>

A.7 Appendix for 5.2 Manoeuvre Limitations

A.7.1 Set A (altitude: 10m, roll: 0.1 rad, pitch: 0.2 rad, yaw: 0.3 rad) Results

Base airframe (drone), on the following page:

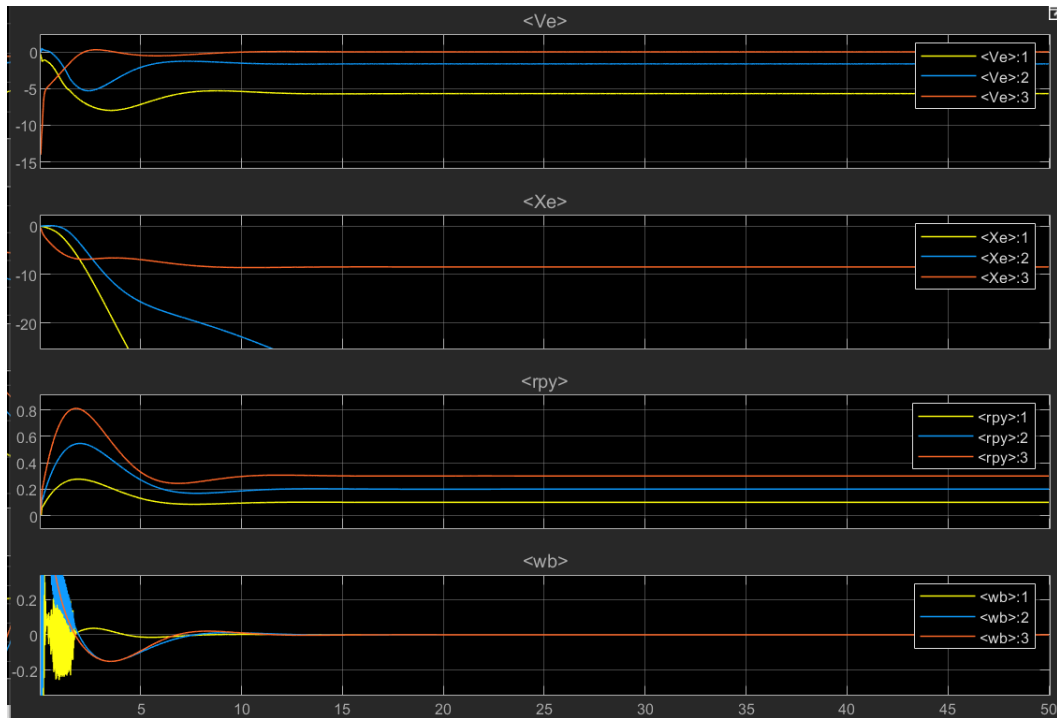


Figure 69 – Drone response (Set A).

With payload (capture mechanism):

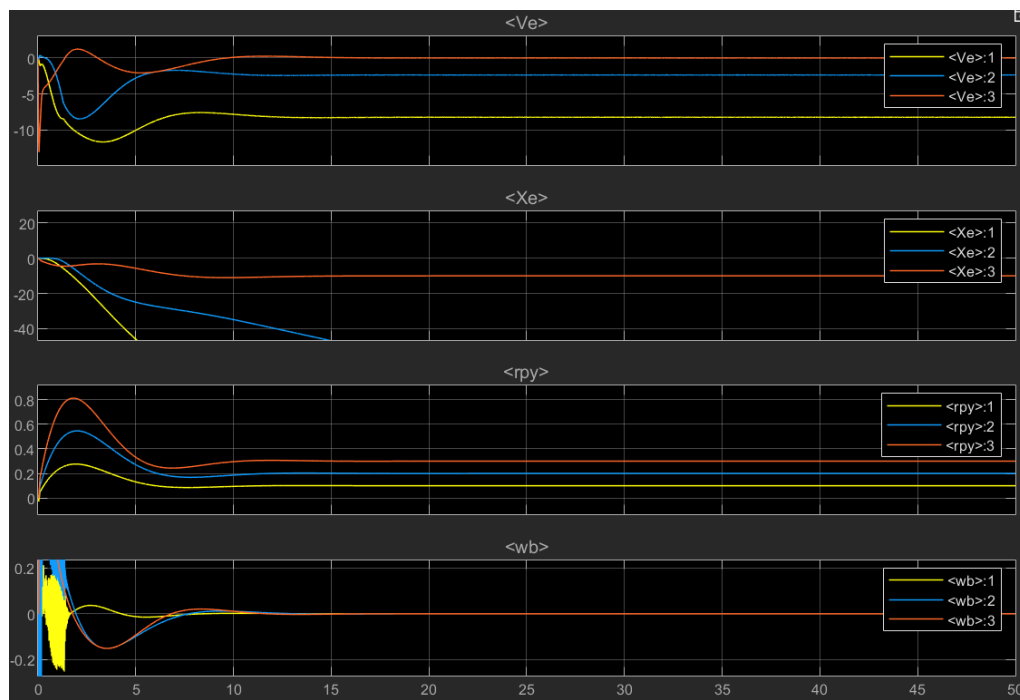


Figure 70 – Drone with payload response (Set A).

Altitude-sweep (50 m):

Base airframe (drone) on the following page:

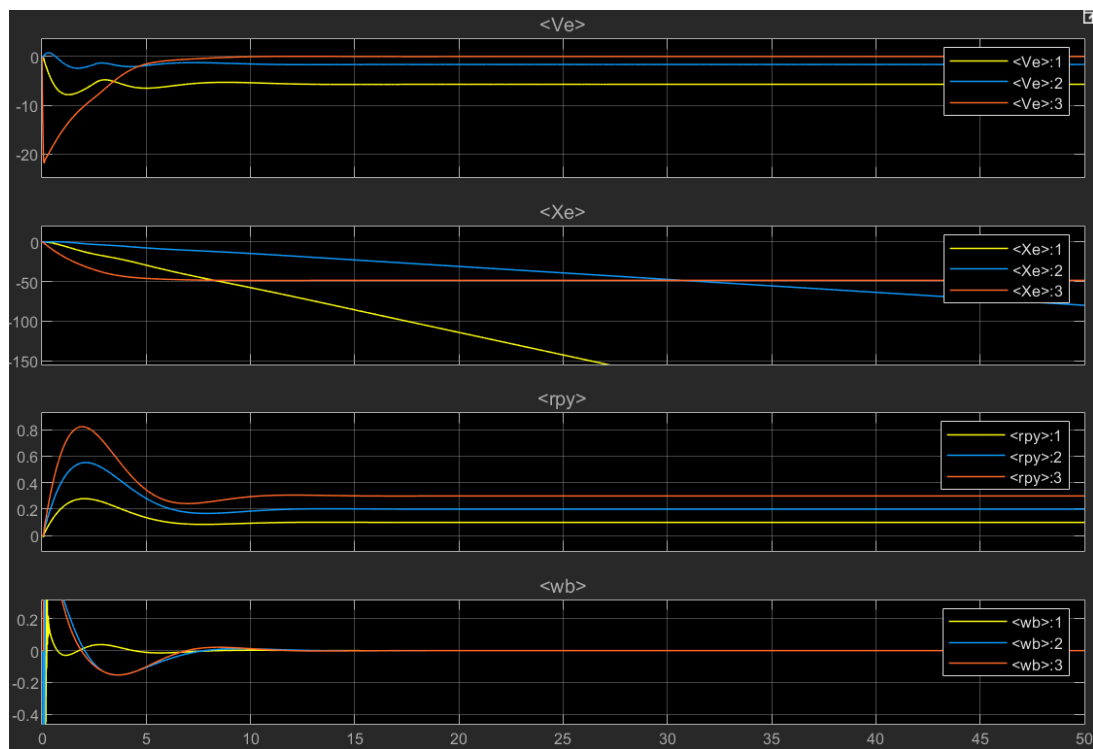


Figure 71 – Drone response for 50 m sweep (Set A).

With payload (capture mechanism):

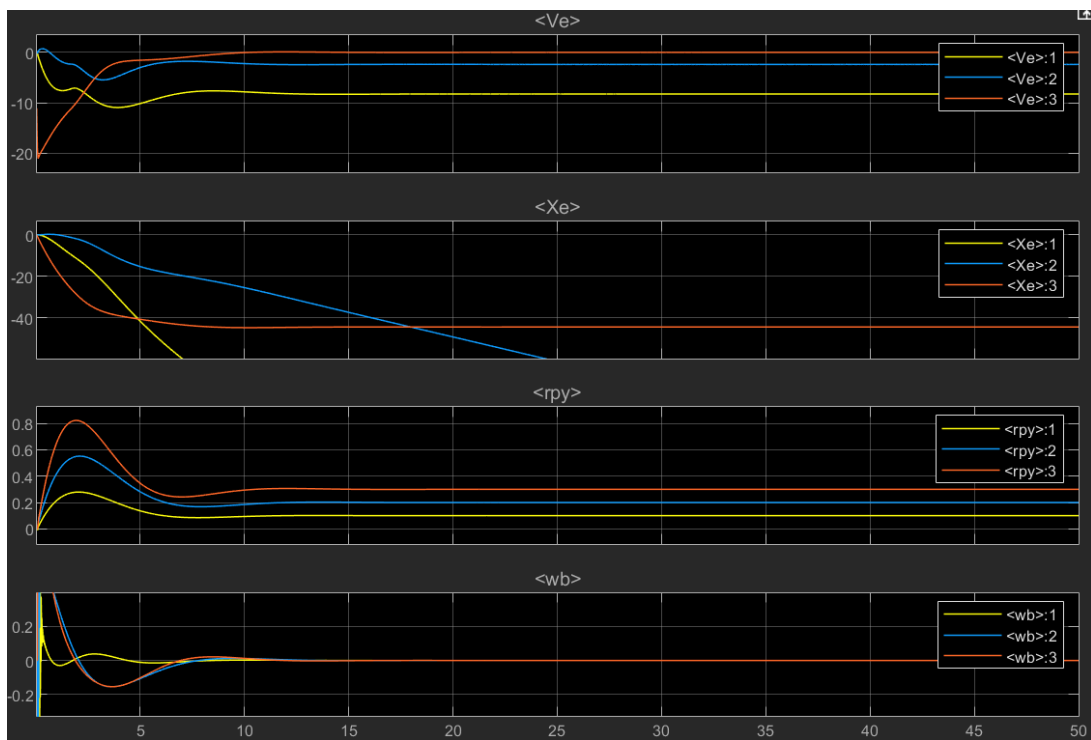


Figure 72 – Drone with payload response for 50 m sweep (Set A).

A.7.2 Set B (altitude: 10m, roll: 0.2 rad, pitch: 0.4 rad, yaw: 0.6 rad) Results

Base airframe (drone) on the following page:

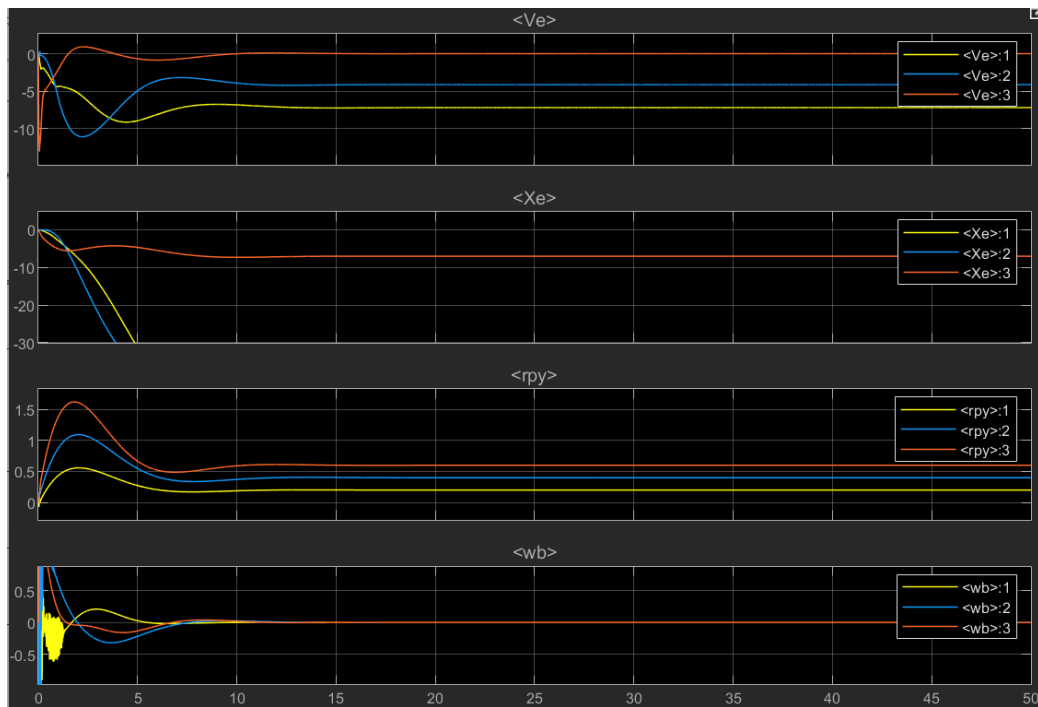


Figure 73 – Drone response (Set B).

With payload (capture mechanism):

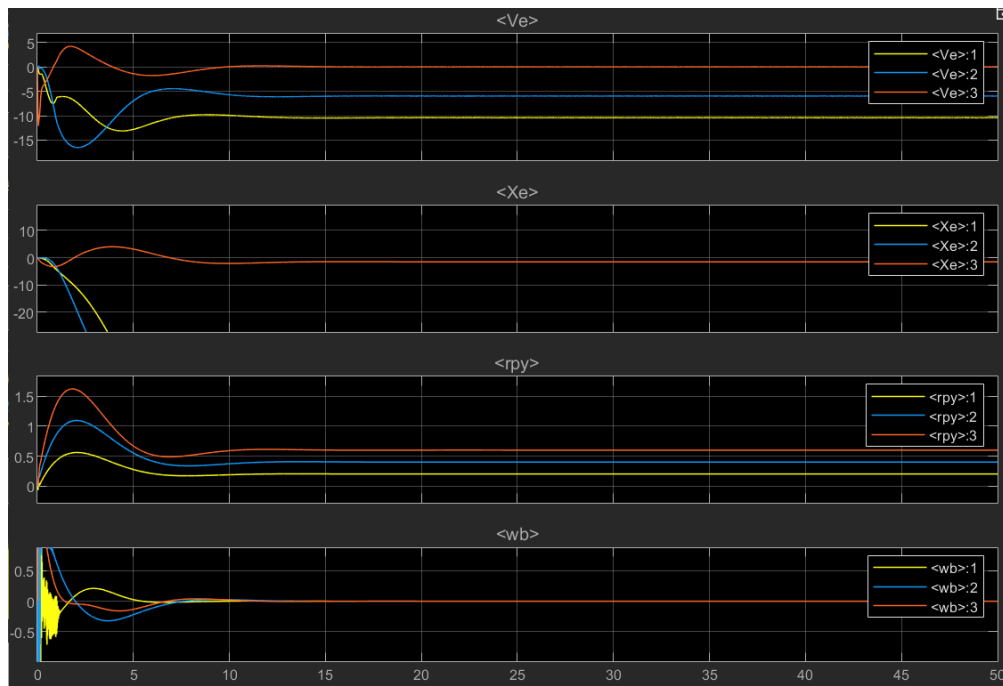


Figure 74 – Drone with payload response (Set B).

A.7.3 Attitude Stability Threshold Determination Results

Roll = 0.6 (pitch = 0.2, yaw = 0.3):

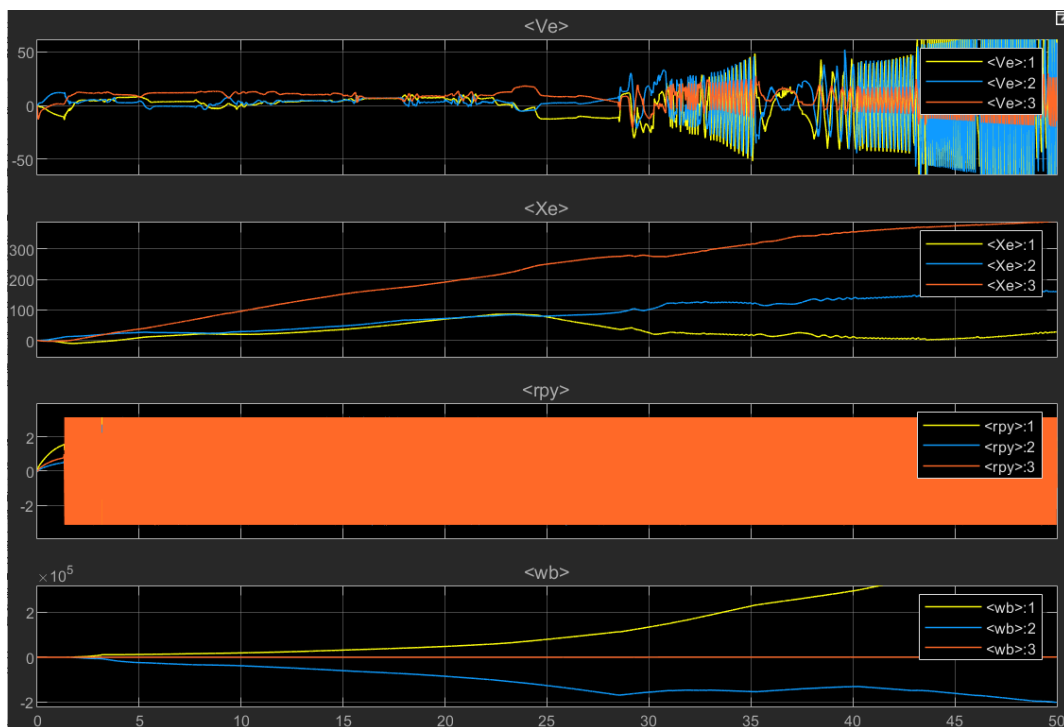


Figure 75 – Unstable flight behaviour at maximum roll limit.

Pitch = 0.7 (roll = 0.1, yaw = 0.3):

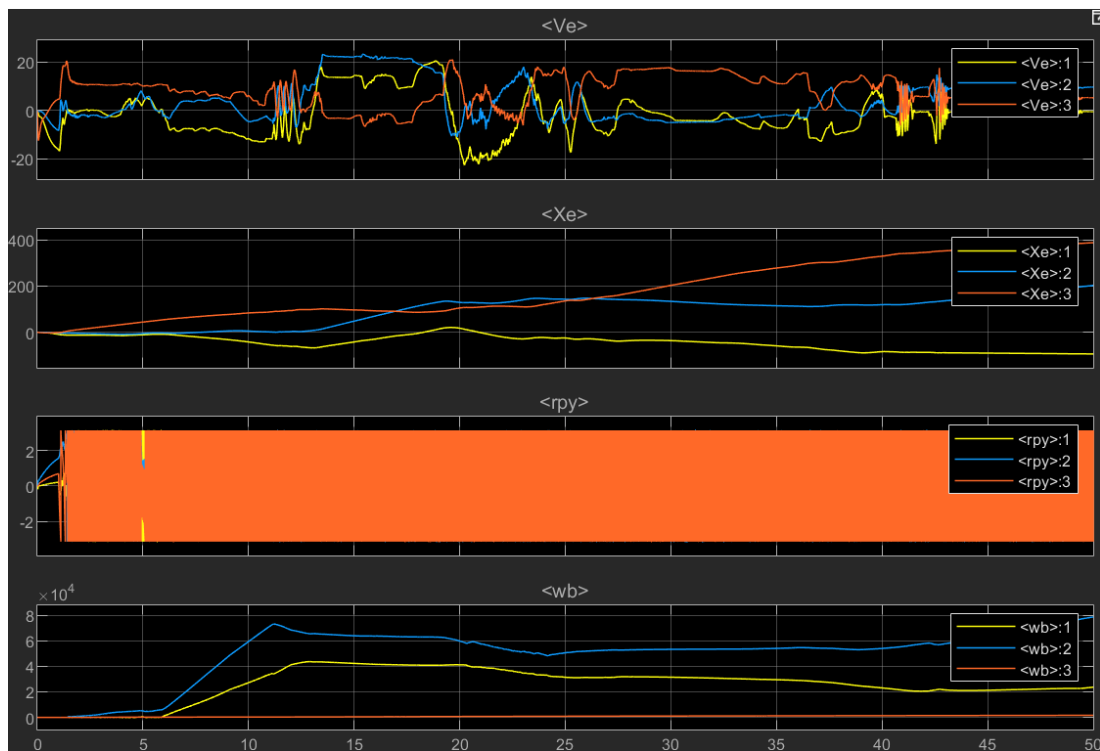


Figure 76 – Unstable flight behaviour at maximum pitch limit.

Yaw = 1.2 (roll = 0.1, pitch = 0.2):

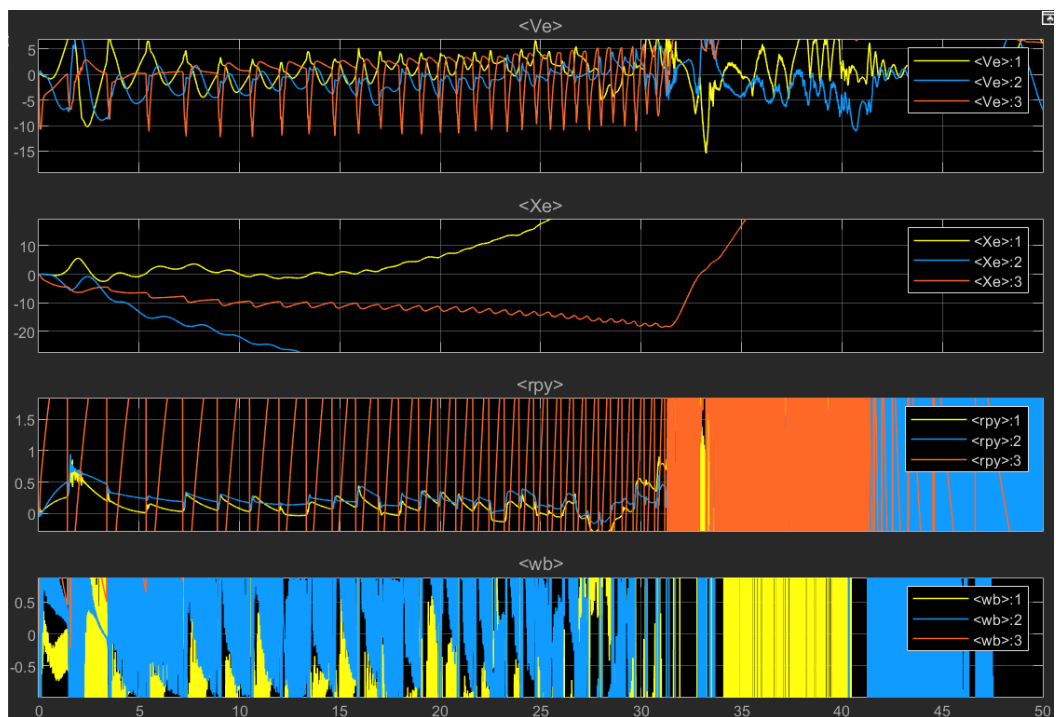
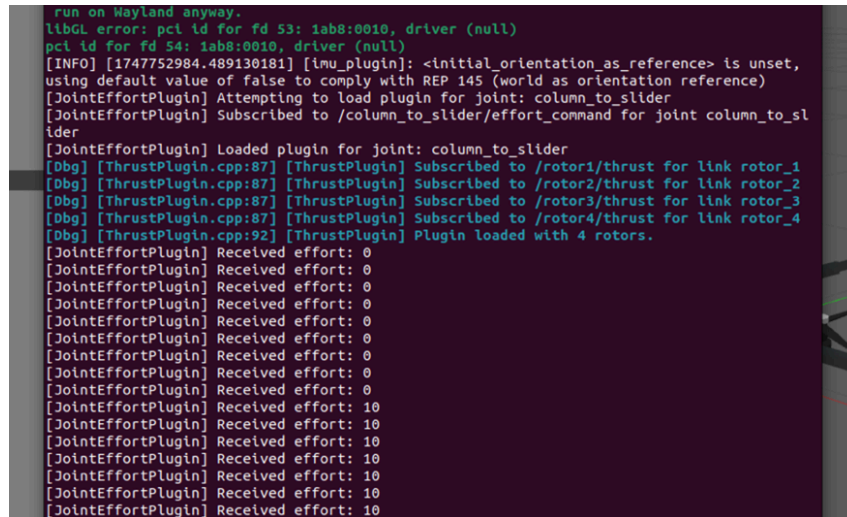


Figure 77 – Unstable flight behaviour at maximum yaw limit.

A.8 Appendix for 5.3 Mechanism Validation – Gazebo

A.8.1 Terminal feedback from the JointEffortPlugin



```

run on Wayland anyway.
libGL error: pct id for fd 53: 1ab8:0010, driver (null)
pct id for fd 54: 1ab8:0010, driver (null)
[INFO] [1747752984.489130181] [imu_plugin]: <initial_orientation_as_reference> is unset,
using default value of false to comply with REP 145 (world as orientation reference)
[JointEffortPlugin] Attempting to load plugin for joint: column_to_slider
[JointEffortPlugin] Subscribed to /column_to_slider/effort_command for joint column_to_sl
ider
[JointEffortPlugin] Loaded plugin for joint: column_to_slider
[Dbg] [ThrustPlugin.cpp:87] [ThrustPlugin] Subscribed to /rotor1/thrust for link rotor_1
[Dbg] [ThrustPlugin.cpp:87] [ThrustPlugin] Subscribed to /rotor2/thrust for link rotor_2
[Dbg] [ThrustPlugin.cpp:87] [ThrustPlugin] Subscribed to /rotor3/thrust for link rotor_3
[Dbg] [ThrustPlugin.cpp:87] [ThrustPlugin] Subscribed to /rotor4/thrust for link rotor_4
[Dbg] [ThrustPlugin.cpp:92] [ThrustPlugin] Plugin loaded with 4 rotors.
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 0
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10
[JointEffortPlugin] Received effort: 10

```

Figure 78 – Ubuntu Terminal.

A.8.2 GitHub repository for Gazebo

The complete source code for the drone and mechanism simulation, including all ROS 2 packages, plugins, configuration files, and launch scripts, is available in the following GitHub repository:

<https://github.com/01711-01711/drone-simulation>

A.9 Appendix for 7. Testing & Analysis

A folder containing all recorded test videos is available at the following link:

<https://drive.google.com/drive/u/0/folders/1xkR3W3RiZiV9nqUZVLoO7WdGSU2O7DCf>

It includes: drone flight test with capture mechanism, 7.2 three net deployment cases (open, partial, and closed) and 7.5 Live interception attempt.

A.10 Appendix for 7.1 Drone Flight Experiment – Drag & Motor Coefficients

A.10.1 Experimental Procedure

Drag Coefficient (C_d) Experiment:

1. Power on and arm the drone on a level take-off board.
2. Fly the drone to a fixed altitude of 15 m.
3. Gradually increase the throttle to 3 m/s and hold for 5 seconds.
4. Confirm minimal forward/lateral movement using GPS; pitch angle should be 0 to ensure thrust is mostly vertical.
5. Record velocity, altitude, time and power P using the recorded OSD video during this period.
6. Repeat steps 1 – 5 two more times for reliability.
7. Repeat steps 1 – 6 but for throttle speeds of 4, 5 and 6 m/s.
8. After each flight, extract the data.
9. Repeat steps 1 – 7 but fly the drone forward for velocities of 20, 30, 40 and 50 km/h. Confirm minimal vertical movement using GPS; altitude should be constant. Also, use the GPS to confirm the direction of flight, minimising lateral movement.
10. Calculate the aerodynamic drag force F_D in each direction of flight using $F_{D_i} = \frac{P - P_{\text{hover}}}{u_i}$.
11. Plot a graph of F_D (y-axis) - $\frac{1}{2}\rho u^2 A$ (x-axis) for the 2 different orientations of flight. The cross-sectional area of the drone in each plane was determine via measuring tools in Autodesk FUSION360 for a 3D CAD replica of the SG 15L drone ($A_x = 5581.195 \text{ mm}^2$, $A_y = 8084.836 \text{ mm}^2$, $A_z = 22258.893 \text{ mm}^2$).
12. Calculate the aerodynamic drag coefficient C_d by finding the gradient of the slope as $C_d = \frac{F_D}{\frac{1}{2}\rho u^2 A}$.
13. Calculate the average of the 2 different aerodynamic drag coefficients.

Motor Constants (k_t and k_m):

1. Power on and arm the drone on a level take-off board.
2. Fly the drone to a fixed altitude of 15 m.
3. Gradually increase the throttle to 3 m/s and hold for 5 seconds.
4. Confirm minimal forward/lateral movement using GPS; pitch angle should be 0 to ensure thrust is mostly vertical.
5. Record velocity, propeller speeds ω , altitude, time and power P using the recorded OSD video during this time.
6. Repeat steps 1 – 5 two more times for reliability.
7. Repeat steps 1 – 6 but for throttle speeds of 4, 5 and 6 m/s.
8. After each flight, extract the data.
9. Convert propeller RPM to angular speed ω (rad/s).
10. Calculate the thrust force F_z produced by the propellers in the z-direction (body frame of reference) using $F_z = \frac{P - P_{\text{hover}}}{4u_z}$.
11. Calculate the torque M_z generated by a motor and propeller spinning in the z-direction (body frame of reference) using $M_z = \frac{P}{4\omega}$.
12. Plot a graph of F_z (y-axis) - ω^2 (x-axis).
13. Plot a graph of M_z (y-axis) - ω^2 (x-axis).
14. Calculate the motor constant k_t by finding the gradient of the slope as $F_z = k_t \omega^2$.
15. Calculate the motor constant k_m by finding the gradient of the slope as $M_z = k_m \omega^2$.

A.10.2 Raw Data Tables

Drag force:

Table 12 – Drag force data.

Direction of Flight	Velocity (km/h)	Velocity (m/s)	Trial	Total Power Input (W)	Total Power Input - Hover Power (W)	Drag Force F_d (N)	$\frac{1}{2}\rho u^2 A$
Vertical	10.8	3	1	239	189	84.8	0.123
			2	363	313		
			3	311	261		
			Avg.	304	254		
	14.4	4	1	510	460	103.3	0.218
			2	444	394		
			3	435	385		
			Avg.	463	413		
	18	5	1	581	531	105.1	0.341
			2	533	483		
			3	612	562		
			Avg.	575	525		
	21.6	6	1	731	681	108.2	0.491
			2	699	649		
			3	667	617		
			Avg.	699	649		
Forward	20	5.6	1	162	112	20.8	0.106
			2	176	126		
			3	159	109		
			Avg.	166	116		
	30	8.3	1	260	210	25.4	0.237
			2	195	145		
			3	330	280		

	40	11.1	Avg.	262	212	25.9	0.422
			1	351	301		
			2	360	310		
			3	301	251		
	50	13.9	Avg.	337	287	32.6	0.659
			1	483	433		
			2	512	462		
			3	514	464		
			Avg.	503	453		

Motor thrust force and torque:*Table 13 – Motor constant data.*

Velocity (m/s)	Avg. Power (W)	Total Avg. Power Input - Hover Power (W)	Propeller No.	Propeller Speeds (RPM)	Propeller Speeds (rad/s)	Motor Thrust Fz (N)	Motor Torque (Nm)
3	304	254	1	2914	305	21.2	0.275
			2	2329	244		
			3	2843	298		
			4	2500	262		
			Avg.	2647	277		
4	463	413	1	3271	343	25.8	0.347
			2	2886	302		
			3	3471	363		
			4	3100	325		
			Avg.	3182	333		
5	575	525	1	3714	389	26.3	0.401
			2	3143	329		
			3	3543	371		
			4	3300	346		
			Avg.	3425	359		
6	699	649	1	3966	415	27.0	0.443
			2	3391	355		
			3	3844	403		
			4	3877	406		
			Avg.	3770	395		

A.11 Appendix for 7.2 Net Test

Table 14 – Net configurations and before and after results.

Case	Side	Before Collision	After Collision
1			
2			
3			